

Strategic Multi-User Offloading and Service Provider Pricing in Mobile Edge Computing

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Abstract—Mobile Edge Computing (MEC) is a promising paradigm for supporting computation-intensive mobile applications. It enables users to offload tasks to the network edge. In practice, the MEC ecosystem involves multiple self-interested entities. In particular, an Edge Service Provider (ESP) and a Network Service Provider (NSP) offer complementary computation and network resources, respectively. They strategically price their resources to maximize revenue. Meanwhile, the users selfishly compete for these limited resources to minimize costs. To capture such strategic interactions, we formulate the system as a two-stage multi-leader multi-follower Stackelberg game. In Stage I, the two Service Providers (SPs) selfishly price their respective resources. In Stage II, given the SPs' prices, users jointly decide whether to offload and how much computation and network resources to purchase under coupled capacity constraints. We show that the user offloading game in Stage II is an exact generalized potential game, and further prove that computing its generalized Nash equilibrium is NP-hard. To address this challenge, we develop a distributed primal-dual resource allocation algorithm and a distributed offloading-user selection algorithm with a theoretical approximation bound. In Stage I, the SP pricing game is analytically intractable because each SP's revenue is implicit and non-smooth due to the NP-hardness of Stage II. To overcome this difficulty, we first characterize the piecewise monotonicity of the SPs' revenues and reveal that the Nash equilibrium in Stage I with non-empty equilibrium offloading user set in Stage II must lie in a set of switching prices. We then estimate the SPs' best responses using a set of boundary prices. Finally, we develop an iterative pricing algorithm to approximate a Nash equilibrium of the SP pricing game in Stage I and the corresponding Stackelberg equilibrium. Extensive simulations validate the efficiency and effectiveness of the proposed framework.

Index Terms—Mobile Edge Computing, Task Offloading, Pricing Strategy, Strategic Interaction, Stackelberg Game

I. INTRODUCTION

A. Background and Motivations

Recent advances in AI applications (e.g., LLM-based mobile agents [1], [2], embodied robots [3], [4], world models for autonomous vehicles [5]) highlight the growing computational intensity of mobile applications. However, most user devices cannot support such applications due to limited computation

capacity. To resolve this conflict, Mobile Edge Computing (MEC) [6], [7] is a promising computing paradigm. It places computation resources at the network edge close to user devices, supporting users' delay-sensitive and computation-intensive applications. Users can access these resources by offloading their tasks to nearby edge servers via wireless networks, thereby reducing task delay and saving energy.

Users' joint access to computation and network resources in MEC has two fundamental characteristics. First, edge environments offer limited computation and network resources [8], [9], compared with centralized cloud platforms. Second, edge infrastructures are often deployed and operated by multiple self-interested Service Providers (SPs) [9]–[11], rather than by a single SP. In practice, the computation resources at edge servers are provided and managed by an Edge Service Provider (ESP) (e.g., AWS Wavelength or Huawei Intelligent EdgeCloud), while the network resources are managed by a Network Service Provider (NSP) (e.g., China Telecom or AT&T). These two resources are both essential in user task offloading, yet they are limited and separately controlled. Therefore, resource access in MEC inherently involves complex strategic interactions.

Such strategic interactions arise at two levels: user level and SP level. At the user level, selfish users compete for limited computation and network resources, making both offloading and resource-purchasing decisions. Moreover, these decisions are impacted by the SPs' prices. Each user needs to balance the potential reduction in delay and energy consumption against the payment for the acquired resources. The user-level strategic interaction motivates us to explore the following key question:

Key Question 1: *Given the ESP's and NSP's prices, how do users compete for computation and network resources?*

At the SP level, both the ESP and the NSP selfishly price their own resources to maximize their own revenues. As their resources are complementary, the prices of the two resources affect the users' resource purchasing behaviors, thereby affecting both SPs' revenues. This SP-level strategic interaction drives us to ask the following key question:

Key Question 2: *How should the ESP and the NSP price their resources under complex strategic interdependence?*

The ESP–NSP–user interaction poses a practically relevant and highly complex strategic problem in MEC. However, such a problem has not yet been fully investigated in the MEC literature. Existing studies generally approach task offloading through centralized optimization [12]–[14] or model user competition under simplified settings [15]–[17]. While a separate line of work investigates pricing mechanisms [10], [11], [18]–

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[23], these studies have distinct limitations. Specifically, they typically assume a single centralized provider [18], [19], [21], [23], focus on competition over identical resources [10], [22], or overlook user-level competition for shared resources [9], [11], [20]. Therefore, the joint problem of ESP–NSP pricing and competitive user offloading remains underexplored.

B. Methodology and Contributions

We model the strategic interaction among an ESP, an NSP, and multiple users as a two-stage Multi-Leader Multi-Follower (MLMF) Stackelberg game. In Stage I, the ESP and the NSP strategically set prices for computation and network resources, respectively. In Stage II, users decide whether to offload and how much computation and network resources to purchase under shared resource capacity constraints.

It is challenging to analyze this Stackelberg game. Specifically, the user offloading game in Stage II is a Generalized Nash Equilibrium Problem (GNEP) with mixed-integer variables [24], [25]. Consequently, it is difficult to solve the SPs' Nash equilibrium in Stage I. Each SP's revenue is difficult to characterize analytically because it is implicit and non-smooth due to the mixed-integer GNEP in Stage II. This makes a direct analytical derivation of the Nash equilibrium in Stage I intractable. To the best of our knowledge, Stackelberg games with such a mixed-integer GNEP follower subgame remain largely unaddressed in the existing literature, and no tractable solution frameworks are available.

We use backward induction to solve the considered Stackelberg game. For Stage II, we exploit the potential game structure [24], [26] of the user game to identify its Generalized Nash Equilibrium (GNE). For Stage I, we observe that, given the opponent's price, each SP's revenue is piecewise monotone in its own price. Moreover, any Nash equilibrium in Stage I that induces a nonempty equilibrium offloading user set in Stage II must be located at switching prices where the offloading users in Stage II change. Based on this structure, we estimate each SP's best response using a set of boundary prices to approximate the switching prices. Finally, we develop an iterative pricing algorithm to approximate the Nash equilibrium in Stage I and the corresponding Stackelberg equilibrium.

We summarize our main contributions as follows.

- We formulate the strategic interaction among an ESP, an NSP, and multiple users as a two-stage MLMF Stackelberg game. The formulation captures two independent SPs selling complementary computation and network resources, as well as users' competition for these resources, thereby reflecting practical MEC settings.
- For Stage II, we model the user competition for computation and network resources as a GNEP and prove the existence of a GNE. To compute a GNE, we propose a distributed primal-dual algorithm for resource allocation and a distributed offloading-user selection algorithm with a theoretical approximation bound.
- For Stage I, we model the SP pricing as a non-cooperative game. We identify a key structural property: given the opponent's price, each SP's revenue is piecewise monotone in its own price. Moreover, any Nash equilibrium in Stage

I with non-empty offloading user set in Stage II must occur at switching prices where the offloading users in Stage II change. Exploiting this property, we approximate the SPs' best responses using a set of boundary prices. Furthermore, we develop an iterative pricing algorithm to approximate a Nash equilibrium of the pricing game and the corresponding Stackelberg equilibrium.

- We validate the proposed framework through extensive simulations. The results show that the proposed algorithms efficiently achieve near-optimal performance in computing the Stackelberg equilibrium. We further quantify the impact of the ESP–NSP–user strategic interactions by comparing with baselines representing alternative strategic settings.

II. RELATED WORK

In this section, we review the related work from two perspectives: task offloading and resource allocation, and pricing mechanisms in MEC.

A. Task Offloading and Resource Allocation

Task offloading and resource allocation in MEC have been widely studied [12]–[17]. Many studies approach these problems from an optimization perspective. For example, Wang *et al.* [12] proposed a trust-based computation offloading framework for mobile cloud–edge computing networks to jointly optimize system cost and users' quality of experience while considering dishonest edge servers. Wei *et al.* [13] proposed a hybrid quantum–classical reinforcement learning model to maximize the number of completed tasks in the MEC task offloading scenario. Dai *et al.* [14] designed a Lyapunov-guided federated deep reinforcement learning online task offloading algorithm to optimize the long-term average system throughput with task queue and battery energy constraints. Such works often aim to optimize overall system performance and neglect the users' selfish behaviors.

Some studies explore users' strategic task offloading from a game-theoretic perspective. For example, Hu *et al.* [15] addressed users' congestion in heterogeneous task offloading with incomplete information using a minority game. Chen *et al.* [16] studied mobile user devices' competition for limited resources in unmanned aerial vehicle-assisted low earth orbit satellite edge computing networks, formulating the task offloading problem as a non-cooperative game. Chu *et al.* [17] investigated users' multi-task computation offloading game in MEC and proposed an efficient multi-task offloading scheme. Such studies typically assume that users make simplified offloading decisions: either choosing whether to offload with evenly allocated resources [17], determining the number of subtasks to offload [15], or selecting only one type of resource, such as wireless channels [16]. However, to the best of our knowledge, no prior work has studied users' offloading competition for computation and network resources under the selfish pricing of both the ESP and the NSP.

B. Pricing Mechanisms

Resource pricing mechanisms in MEC have been extensively studied in the existing literature [10], [11], [18]–[23]. Studies in [18], [19], [21], [23] investigated pricing mechanisms involving only a single SP. Tong *et al.* [18] analyzed the interaction between an MEC server and multiple end users in MEC systems using a Stackelberg game, and proposed a dynamic-programming–based resource allocation algorithm to jointly optimize bandwidth allocation and pricing. Wang *et al.* [19] conducted a business-model study of the mobile Internet ecosystem, jointly analyzed users’ content acquisition and task offloading decisions, and developed a pricing policy for the ESP to monetize edge computing service. Tütüncüoğlu and Dán [21] studied joint resource allocation, pricing, and application caching for latency-sensitive task offloading in serverless edge computing. They modeled the interaction between the operator and wireless devices as a Stackelberg game and developed approximation algorithms for the subgame perfect equilibrium. Wang *et al.* [23] proposed a marginal-value–based pricing mechanism for an edge computing platform to incentivize resource allocation for deadline-sensitive tasks with higher marginal value. The above studies either assumed a single SP managing one type of resource [18], [19] or a centralized operator jointly managing multiple resources [21], [23], and thus overlooked the strategic interactions among multiple SPs.

In contrast, research in [10], [11], [20], [22] studied pricing mechanisms that involve multiple SPs. Chen *et al.* [10] investigated edge servers’ price competition and users’ offloading competition in MEC networks using a two-stage game under different service models. Ding *et al.* [20] investigated optimal pricing strategies in IoT systems with multiple service providers under coordinated and uncoordinated interaction structures. Wang *et al.* [22] designed profit-maximization incentive mechanisms for edge clouds in MEC by analyzing both non-competitive and competitive settings. Liu *et al.* [11] developed a market model involving multiple edge clouds, access points, and smart mobile devices in MEC, where computation and radio resources are provided by the edge clouds and access points, respectively. They applied microeconomic theory to identify the market equilibrium prices of these resources. The above works either focused on SPs of the same resource type (e.g., computation) [10], [22], or neglected the selfishness of SPs [11], or failed to capture the strategic interactions between the ESP and the NSP [20].

Among recent studies, [9] is the closest to ours. The authors studied cooperative and competitive pricing between a Computation Service Provider (CSP) and an NSP while incorporating users’ offloading decisions through a two-stage Stackelberg game. However, [9] considered a simplified user model in which each user independently determines the offloading ratio and the amounts of computation and network resources to purchase. As a result, it did not capture the competition among multiple users for shared computation and network resources, nor the strategic SP pricing shaped by such user competition. In contrast, our work explicitly captures the strategic interaction among the ESP, the NSP, and the

users by jointly considering strategic pricing by the SPs and offloading competition among users. This joint interaction remains largely unexplored in the MEC literature.

III. SYSTEM MODEL

We consider an MEC system with an ESP, an NSP¹, and a set of users $\mathcal{I} = \{1, \dots, I\}$. The ESP manages computation resources at an edge server, while the NSP manages network resources at a base station. Each user has an indivisible computation task [27] that can either be computed locally or fully offloaded to the ESP’s edge server via the NSP’s base station.

In this section, we first introduce the user model in Section III-A and the service provider model in Section III-B. Then, we model their strategic interaction as a two-stage MLMF Stackelberg game in Section III-C.

A. The User Model

Each user $i \in \mathcal{I}$ has an indivisible task characterized by workload w_i (in CPU cycles) and input data size d_i (in bits). The user i makes a binary offloading decision $o_i \in \{0, 1\}$, where $o_i = 1$ corresponds to computation offloading and $o_i = 0$ corresponds to local computation. If choosing to offload, it needs to decide the amount of computation resource f_i (in CPU cycles/s) and bandwidth b_i (in Hz) to purchase from the ESP and the NSP, respectively.

1) *Local Computation Model:* Executing a task locally incurs both computation delay and energy consumption.

Local Computation Delay. Each user $i \in \mathcal{I}$ has a local computation resource f_i^l (in CPU cycles/s) on its local device. If it chooses to execute its task locally, the delay is as follows:

$$T_i^l = \frac{w_i}{f_i^l}. \quad (1)$$

Energy Consumption of Local Computation. Each user $i \in \mathcal{I}$ ’s local device consumes energy while processing its task. The energy consumption of the user i ’s local computation is computed according to a quadratic energy consumption model [28], [29]:

$$E_i^l = \kappa_i w_i (f_i^l)^2, \quad (2)$$

where κ_i is a constant energy coefficient related to the hardware architecture of the user i ’s local device.

2) *Computation Offloading Model:* If a user chooses to offload its task, the total offloading delay consists of transmission delay and remote computation delay. Meanwhile, the data transmission process incurs additional energy consumption.

Transmission Delay. To offload its task, each user $i \in \mathcal{I}$ needs to transmit d_i bits of data to the ESP’s edge server via the NSP’s base station using a bandwidth of b_i Hz. Here, we adopt the *Orthogonal Frequency Division Multiple Access* (OFDMA) model to compute the transmission delay, similar to previous works [18], [21], [30]. We define ρ_i as the transmission power of the user i ’s device, ς_i as the channel

¹Although there are multiple ESPs and NSPs in practical MEC environments, analyzing even a single pair of ESP and NSP is already highly complex. Hence, we focus on one ESP and one NSP, which sufficiently capture the essential strategic interactions in the environment.

gain between the user i and the base station, and N_0 as the background noise power. Let $\sigma_i \triangleq \log_2(1 + \rho_i \varsigma_i / N_0)$ denote the channel spectral efficiency (in bits/s/Hz) between the user i and the base station. The uplink transmission delay is:

$$T_i^u(b_i) = \frac{d_i}{b_i \sigma_i}. \quad (3)$$

We assume that the downlink transmission delay of the computation result is negligible [6], [21], as the size of the result is significantly smaller than the size of the input data.

Energy Consumption of Data Transmission. During data transmission, the energy consumption of user $i \in \mathcal{I}$'s local device depends on its transmission power and the duration of transmission, which can be expressed as follows:

$$E_i^u(b_i) = T_i^u(b_i) \rho_i \varpi_i, \quad (4)$$

where ϖ_i denotes the transmit antenna power efficiency parameter of user i 's device.

Remote Computation Delay. Upon completion of data transmission, the ESP's edge server computes user $i \in \mathcal{I}$'s task using f_i CPU cycles/s of computation resources. We assume that these resources are exclusively allocated to the user i without queueing delay, similar to previous works [17], [21]. This is consistent with real-world cloud-edge service models of renting a virtual machine instance with guaranteed CPU capacity, rather than submitting tasks to a shared queue. The remote computation delay of the user i 's task at the edge server is:

$$T_i^c(f_i) = \frac{w_i}{f_i}. \quad (5)$$

Therefore, the total delay of user i 's task via computation offloading is the sum of transmission delay and remote computation delay:

$$T_i^e(f_i, b_i) = T_i^u(b_i) + T_i^c(f_i). \quad (6)$$

3) The User's Costs: We define α_i, β_i as the user i 's marginal monetary costs incurred by delay and energy consumption, respectively. These two parameters reflect the user i 's personalized sensitivity to delay and energy consumption.

Under local computation, the user i 's cost is modeled as:

$$C_i^l = \alpha_i T_i^l + \beta_i E_i^l. \quad (7)$$

Under computation offloading, its cost includes the costs caused by delay and energy consumption, and the payments to the SPs:

$$C_i^e(f_i, b_i) = \alpha_i T_i^e(f_i, b_i) + \beta_i E_i^u(b_i) + p_E f_i + p_N b_i. \quad (8)$$

Here, p_E is the unit price of the computation resource set by the ESP, and p_N is the unit price of the bandwidth set by the NSP.

Recall that binary variable $o_i \in \{0, 1\}$ indicates the user i 's offloading decision. Accordingly, the user i 's overall cost is defined as a binary selection between the local computation cost and the computation offloading cost:

$$C_i(o_i, f_i, b_i) = o_i C_i^e(f_i, b_i) + (1 - o_i) C_i^l. \quad (9)$$

4) The User's Cost Minimization Problem: Given the SPs' prices p_E and p_N , each user $i \in \mathcal{I}$ selects the offloading decision o_i and resource usage (f_i, b_i) to minimize its overall cost by solving the following cost minimization problem:

$$\min_{o_i, f_i, b_i} C_i(o_i, f_i, b_i) = o_i C_i^e(f_i, b_i) + (1 - o_i) C_i^l \quad (10a)$$

$$s.t. f_i + \sum_{n \in \mathcal{I}, n \neq i} f_n \leq F, b_i + \sum_{n \in \mathcal{I}, n \neq i} b_n \leq B, \quad (10b)$$

$$o_i \in \{0, 1\}, f_i \geq 0, b_i \geq 0. \quad (10c)$$

The constraints in (10b) reflect the system-wide resource capacity limits imposed by the ESP and the NSP. They imply that each user's feasible space depends on other users' decisions, which we will further illustrate in Section IV-A.

B. Service Provider Model

The ESP Model. The ESP manages limited computation resources at the edge server compared with those available in the cloud [21], [27], [31]. We denote the capacity of computation resources at the edge server as F . The users' total computation resource usage should not exceed the capacity, i.e., $\sum_{i \in \mathcal{I}} f_i \leq F$. To offer these resources, the ESP incurs a unit cost $c_E > 0$, which captures the unit operational cost of computation resources [20], [32]. The ESP charges the users a unit price p_E for computation resource consumption, with $p_E \geq c_E$ to ensure economic rationality. The ESP's revenue is:

$$U_E = (p_E - c_E) \sum_{i \in \mathcal{I}} f_i. \quad (11)$$

The NSP Model. The NSP has a total wireless bandwidth B at the base station. The users' bandwidth usage should not exceed the capacity, i.e., $\sum_{i \in \mathcal{I}} b_i \leq B$. Similarly, the unit cost and unit price of bandwidth are denoted by c_N and p_N , respectively. We have $p_N \geq c_N$ to ensure economic rationality. The NSP's revenue is:

$$U_N = (p_N - c_N) \sum_{i \in \mathcal{I}} b_i. \quad (12)$$

The SPs' Revenue Maximization Problem. The ESP and the NSP selfishly set their prices p_E and p_N to maximize their revenue. We highlight that each user i 's resource usages f_i and b_i depend on the prices p_E and p_N , and thus we explicitly denote them as $f_i(p_E, p_N)$ and $b_i(p_E, p_N)$.

The ESP solves the following revenue maximization problem:

$$\max_{p_E} U_E(p_E, p_N) = (p_E - c_E) \sum_{i \in \mathcal{I}} f_i(p_E, p_N) \quad (13a)$$

$$s.t. p_E \geq c_E, \quad (13b)$$

$$\sum_{i \in \mathcal{I}} f_i(p_E, p_N) \leq F. \quad (13c)$$

Similarly, the NSP solves the following problem:

$$\max_{p_N} U_N(p_N, p_E) = (p_N - c_N) \sum_{i \in \mathcal{I}} b_i(p_E, p_N) \quad (14a)$$

$$s.t. p_N \geq c_N, \quad (14b)$$

$$\sum_{i \in \mathcal{I}} b_i(p_E, p_N) \leq B. \quad (14c)$$

C. Stackelberg Game Formulation

We model the interactions among the ESP, the NSP, and the users as a two-stage MLMF Stackelberg game. The ESP and the NSP first set the prices of computation resources and bandwidth, respectively, to maximize their own revenues. Given the SPs' prices, the users then make their offloading and resource-purchasing decisions to minimize their costs.

A Stackelberg equilibrium is a strategy profile that consists of the SPs' prices and the users' strategies such that:

- given the SPs' prices, none of the users can decrease its cost by unilaterally deviating to another strategy.
- given that users play the GNE in Stage II under any price profile, neither SP can increase its revenue by unilaterally changing its own price.

The Stackelberg equilibrium represents a stable outcome of the strategic interactions among the ESP, the NSP, and the users. We use backward induction to derive it. Specifically, given the ESP's and the NSP's prices in Stage I, we compute the user GNE in Stage II by addressing the social cost minimization problem. Then, we analyze the SP pricing game in Stage I based on the user GNE in Stage II.

IV. STAGE II: USER OFFLOADING GAME AND GENERALIZED NASH EQUILIBRIUM COMPUTATION

In this section, we study the user offloading competition via a user offloading game. First, in Section IV-A, we formulate the user offloading game as a GNEP and establish the existence of a GNE. Then, we focus on computing its GNE. We formulate a social cost minimization problem in Section IV-B, and develop its solution by addressing the inner resource allocation problem in Section IV-C and the outer offloading-user-set selection problem in Section IV-D.

A. User Offloading Game Formulation and Equilibrium Existence

Given the SPs' prices, we model the users' offloading competition as a user offloading game. As all users share common resource capacity constraints in (10b), each user's feasible strategy space depends on other users' strategies. We define $s_i \triangleq (o_i, f_i, b_i)$, $\mathbf{s} \triangleq (s_1, \dots, s_I)$, and let $\mathbf{s}_{-i}$ denote the strategy profile of all users except user i . Given a fixed $\mathbf{s}_{-i}$, we denote the strategy space of user i defined by (10c) and (10b) as $\mathcal{S}_i(\mathbf{s}_{-i})$. We denote the joint strategy space of all users as $\mathcal{S}$. Formally, the user offloading game is defined as the following GNEP.

Definition 1 (User Offloading Game). *The User Offloading Game in Stage II consists of*

- **Players:** All users $i \in \mathcal{I}$.
- **Strategies:** Each user i selects a strategy $s_i = (o_i, f_i, b_i) \in \mathcal{S}_i(\mathbf{s}_{-i})$. Its strategy space $\mathcal{S}_i(\mathbf{s}_{-i})$ depends on other users' strategies, where $\mathcal{S}_i(\mathbf{s}_{-i}) \triangleq \{0, 1\} \times [0, F - \sum_{j \in \mathcal{I}, j \neq i} f_j] \times [0, B - \sum_{j \in \mathcal{I}, j \neq i} b_j]$.
- **Cost Functions:** Each user i 's cost is given in (9).

We next define the GNE [24], [30], [33] of the user offloading game. It is a stable point where no user can lower its cost by unilateral deviation.

Definition 2 (GNE of User Offloading Game). *A feasible strategy profile $\mathbf{s}^* = (s_1^*, \dots, s_I^*) \in \mathcal{S}$ is a GNE of the user offloading game if the following condition holds:*

$$C_i(s_i^*) \leq C_i(s_i), \forall s_i \in \mathcal{S}_i(\mathbf{s}_{-i}^*), \forall i \in \mathcal{I}. \quad (15)$$

We now investigate the existence of a GNE of the user offloading game through exact generalized potential game characterization. To proceed, we first introduce the concept of exact generalized potential game [24], [26], which is an extension of the classical potential game [34].

Definition 3 (Exact Generalized Potential Game). *A GNEP is an exact generalized potential game if it admits an exact potential function $\Phi(\mathbf{s})$ such that for every player $i \in \mathcal{I}$ and any two feasible strategy profiles $(s_i, \mathbf{s}_{-i}), (s'_i, \mathbf{s}_{-i}) \in \mathcal{S}$, $C_i(s'_i, \mathbf{s}_{-i}) - C_i(s_i, \mathbf{s}_{-i}) = \Phi(s'_i, \mathbf{s}_{-i}) - \Phi(s_i, \mathbf{s}_{-i})$.*

The following lemma shows that the user offloading game is an exact generalized potential game.

Lemma 1. *The user offloading game is an exact generalized potential game with a potential function $\Phi(\mathbf{s}) = \sum_{i \in \mathcal{I}} C_i(s_i)$. The joint strategy space $\mathcal{S}$ is non-empty and compact.*

Proof. The proof is provided in Appendix A. $\square$

In an exact generalized potential game, the players are unknowingly minimizing the same potential function in the joint strategy space. According to Lemma 1, the exact potential function of the user offloading game coincides with the social cost of all users. Thus, whenever a user unilaterally changes its strategy to reduce its own cost, the social cost decreases by exactly the same amount.

Any global optimum of the exact potential function Φ over the joint strategy space $\mathcal{S}$ is a GNE of the exact generalized potential game [24], [26]. This property provides a convenient way to establish the existence of a GNE in the user offloading game. Based on Lemma 1, we establish the existence of a GNE in the user offloading game.

Theorem 1 (Existence of GNE). *The user offloading game admits at least one GNE.*

Proof. The key idea of the proof is that the exact potential function $\Phi(\mathbf{s}) = \sum_{i \in \mathcal{I}} C_i(s_i)$ admits a global minimizer over the joint feasible strategy space $\mathcal{S}$, and the global minimizer is a GNE of the user offloading game. The detailed proof is provided in Appendix B. $\square$

Theorem 1 guarantees the existence of a GNE under any prices set by the ESP and the NSP. More importantly, it implies that any global minimizer of the exact potential function is a GNE of the user offloading game. Since Lemma 1 shows that the exact potential function coincides with the users' social cost, we can obtain a GNE by minimizing the social cost. Therefore, in the following subsection, we formulate a social cost minimization problem to identify the GNE of the user offloading game.²

²Although best-response updates are natural for potential games, we do not adopt them here for two reasons. First, standard best-response dynamics may oscillate and have no convergence guarantee due to the users' mixed-integer strategies. Second, solving the social cost minimization problem provides structural information for the analysis of the SP pricing game in Section V.

B. Social Cost Minimization Problem Formulation

In this subsection, we aim to identify the GNE of the user offloading game by minimizing the social cost of all users, *i.e.*, the exact potential function $\Phi(\mathbf{s}) = \sum_{i \in \mathcal{I}} C_i(s_i)$ in Lemma 1. We equivalently rewrite the social cost minimization problem by first selecting an offloading user set $\mathcal{X} \subseteq \mathcal{I}$, and then optimizing the resource allocation of users in $\mathcal{X}$. This gives the following Social Cost Minimization (SCM) problem.

$$\text{SCM} : \min_{\mathcal{X} \subseteq \mathcal{I}} \min_{\{(f_i, b_i)\}_{i \in \mathcal{X}}} \sum_{i \in \mathcal{X}} C_i^e(f_i, b_i) + \sum_{j \in \mathcal{I} \setminus \mathcal{X}} C_j^l \quad (16a)$$

$$\text{s.t. } C_i^e(f_i, b_i) \leq C_i^l, \forall i \in \mathcal{X}, \quad (16b)$$

$$\sum_{i \in \mathcal{X}} f_i \leq F, \quad (16c)$$

$$\sum_{i \in \mathcal{X}} b_i \leq B, \quad (16d)$$

$$f_i > 0, b_i > 0, \forall i \in \mathcal{X}. \quad (16e)$$

The SCM problem is a bi-level problem: the inner problem allocates resources among a given offloading user set, while the outer problem selects the offloading user set that minimizes the social cost. Constraints (16b) ensure the users' individual rationality, that is, every offloading user in $\mathcal{X}$ incurs an offloading cost no greater than its local-computation cost. Constraints (16c) and (16d) limit the consumed computation resource and bandwidth to the SPs' capacities. Constraints (16e) require that each offloading user's resource allocations be strictly positive. We next demonstrate the complexity of the SCM problem in the following proposition.

Proposition 1. *The SCM problem in (16) is NP-hard.*

Proof. The proof is provided in Appendix C. $\square$

The NP-hardness of the SCM problem implies that finding the solution in polynomial time is computationally intractable. Thus, we are interested in designing a distributed approximation algorithm that can compute a near-optimal solution efficiently. To this end, we tackle the SCM problem in two steps. First, we develop a distributed primal-dual algorithm for the resource allocation in the inner problem (see Section IV-C). Second, we propose a distributed algorithm for the offloading-user-set selection in the outer problem (see Section IV-D).

C. Resource Allocation in the Inner Problem

Given a set $\mathcal{X} \subseteq \mathcal{I}$ of offloading users, we optimize the resource allocation $\{(f_i, b_i)\}_{i \in \mathcal{X}}$ to minimize the total computation offloading cost for offloading users in $\mathcal{X}$, by considering the inner problem of the SCM problem:

$$\min_{\{(f_i, b_i)\}_{i \in \mathcal{X}}} \sum_{i \in \mathcal{X}} C_i^e(f_i, b_i) \quad (17)$$

$$\text{s.t. } (16b) - (16e)$$

We first show the strict convexity of the inner problem in (17).

Lemma 2. *The inner problem in (17) is a strictly convex optimization problem with a unique optimal solution.*

Proof. The proof is provided in Appendix D. $\square$

Lemma 2 implies that there is a unique optimal resource allocation for the inner problem. Although such an allocation can be computed by a centralized solver, doing so would incur high computational overhead. We thus design a distributed primal-dual algorithm for solving the inner problem.

We introduce dual variables $\lambda_F \geq 0$ and $\lambda_B \geq 0$ for the capacity constraints in (16c) and (16d), respectively. We also introduce dual variables $\mu_i \geq 0$ for the individual rationality constraints in (16b) for all $i \in \mathcal{X}$. The Lagrangian of the inner problem in (17) is

$$\begin{aligned} \mathcal{L}(\{f_i, b_i\}_{i \in \mathcal{X}}, \lambda_F, \lambda_B, \{\mu_i\}_{i \in \mathcal{X}}) \\ = \sum_{i \in \mathcal{X}} C_i^e(f_i, b_i) + \lambda_F \left(\sum_{i \in \mathcal{X}} f_i - F \right) + \lambda_B \left(\sum_{i \in \mathcal{X}} b_i - B \right) \\ + \sum_{i \in \mathcal{X}} \mu_i (C_i^e(f_i, b_i) - C_i^l). \end{aligned} \quad (18)$$

Here, dual variables λ_F and λ_B can be interpreted as the shadow prices for computation resource and bandwidth, respectively. The dual variable μ_i can be viewed as the marginal penalty of violating user i 's individual rationality.

Moreover, we observe that the Lagrangian $\mathcal{L}$ represents the total augmented cost of all offloading users minus the system's maximum tolerable congestion cost. Let a convex function $\phi_i(\cdot)$ represent the augmented cost of user $i \in \mathcal{X}$, which incorporates the original offloading cost with congestion costs and a rationality penalty:

$$\begin{aligned} \phi_i(f_i, b_i, \lambda_F, \lambda_B, \mu_i) \triangleq C_i^e(f_i, b_i) + \lambda_F f_i + \lambda_B b_i \\ + \mu_i (C_i^e(f_i, b_i) - C_i^l). \end{aligned} \quad (19)$$

Let the term $\lambda_F F + \lambda_B B$ represent the maximum tolerable congestion cost under the resource capacities. By regrouping the terms in (18), the Lagrangian $\mathcal{L}$ can be rewritten as

$$\mathcal{L} = \sum_{i \in \mathcal{X}} \phi_i(f_i, b_i, \lambda_F, \lambda_B, \mu_i) - (\lambda_F F + \lambda_B B), \quad (20)$$

This structure motivates the design of a distributed primal-dual algorithm to solve the inner problem.

We summarize the proposed distributed primal-dual algorithm in Algorithm 1. The algorithm relies on an iterative process between the SPs and the users. In each iteration, the ESP and the NSP broadcast their respective scalar shadow prices $\lambda_F^{(t)}$ and $\lambda_B^{(t)}$ (Line 4). Given these prices and their marginal penalties $\mu_i^{(t)}$, users in $\mathcal{X}$ independently solve their local cost minimization problems to update their resource demands $(f_i^{(t)}, b_i^{(t)})$ (Lines 6–9). Upon receiving the resource demands from the users, the ESP and the NSP update the shadow prices using gradient ascent to enforce capacity constraints (Lines 10–11), while users update their penalty multipliers locally (Lines 12–14). This distributed architecture preserves the privacy of users' local cost functions and requires minimal communication overhead. Furthermore, it guarantees that the algorithm converges to the unique primal optimal solution $\{(f_i^*, b_i^*)\}_{i \in \mathcal{X}}$ and approaches the set of optimal dual variables $(\lambda_F^*, \lambda_B^*, \{\mu_i^*\}_{i \in \mathcal{X}})$ in polynomial time [35]–[37]. The detailed convergence result and complexity analysis are provided in Appendix F.

Algorithm 1 Distributed Primal–Dual Algorithm for the Inner Problem

Input: Offloading user set $\mathcal{X}$, initial step sizes $\eta_F^{(0)}, \eta_B^{(0)}, \eta_\mu^{(0)}$, tolerance $\varepsilon > 0$.

Output: An optimal resource allocation $\{(f_i^*, b_i^*)\}_{i \in \mathcal{X}}$, and the corresponding dual variables $(\lambda_F^*, \lambda_B^*, \{\mu_i^*\}_{i \in \mathcal{X}})$.

- 1: Initialize $\lambda_F^{(0)} \geq 0$, $\lambda_B^{(0)} \geq 0$, and $\mu_i^{(0)} \geq 0$ for all $i \in \mathcal{X}$.
- 2: Set iteration index $t \leftarrow 0$.
- 3: **repeat**
- 4: The ESP broadcasts $\lambda_F^{(t)}$ and the NSP broadcasts $\lambda_B^{(t)}$ to each user $i \in \mathcal{X}$.
- 5: Set step sizes $\eta_F^{(t)} = \eta_F^{(0)} / \sqrt{t+1}$, $\eta_B^{(t)} = \eta_B^{(0)} / \sqrt{t+1}$, and $\eta_\mu^{(t)} = \eta_\mu^{(0)} / \sqrt{t+1}$.
- 6: **for all** user $i \in \mathcal{X}$ in parallel **do**
- 7: User i computes $(f_i^{(t)}, b_i^{(t)})$ by minimizing $\phi_i(f_i, b_i, \lambda_F^{(t)}, \lambda_B^{(t)}, \mu_i^{(t)})$ over $f_i > 0$ and $b_i > 0$.
- 8: User i sends $f_i^{(t)}$ to the ESP and $b_i^{(t)}$ to the NSP, respectively.
- 9: **end for**
- 10: The ESP updates λ_F according to $\lambda_F^{(t+1)} = \left[\lambda_F^{(t)} + \eta_F^{(t)} \left(\sum_{i \in \mathcal{X}} f_i^{(t)} - F \right) \right]_+$.
- 11: The NSP updates λ_B according to $\lambda_B^{(t+1)} = \left[\lambda_B^{(t)} + \eta_B^{(t)} \left(\sum_{i \in \mathcal{X}} b_i^{(t)} - B \right) \right]_+$.
- 12: **for all** user $i \in \mathcal{X}$ in parallel **do**
- 13: User i updates μ_i according to $\mu_i^{(t+1)} = \left[\mu_i^{(t)} + \eta_\mu^{(t)} (C_i^e(f_i^{(t)}, b_i^{(t)}) - C_i^l) \right]_+$.
- 14: **end for**
- 15: $t \leftarrow t + 1$.
- 16: **until** $\max \left\{ \left[\sum_{i \in \mathcal{X}} f_i^{(t)} - F \right]_+, \left[\sum_{i \in \mathcal{X}} b_i^{(t)} - B \right]_+, \max_{i \in \mathcal{X}} [C_i^e(f_i^{(t)}, b_i^{(t)}) - C_i^l]_+ \right\} \leq \varepsilon$
- 17: **return** $\{(f_i^*, b_i^*)\}_{i \in \mathcal{X}} = \{(f_i^{(t)}, b_i^{(t)})\}_{i \in \mathcal{X}}$, $(\lambda_F^*, \lambda_B^*, \{\mu_i^*\}_{i \in \mathcal{X}}) = (\lambda_F^{(t)}, \lambda_B^{(t)}, \{\mu_i^{(t)}\}_{i \in \mathcal{X}})$.

In the next subsection, we will investigate the offloading user selection in the outer problem and design a heuristic distributed algorithm for the NP-hard SCM problem.

D. Offloading User Selection in the Outer Problem

We aim to determine the offloading user set $\mathcal{X} \subseteq \mathcal{I}$ that minimizes the social cost by addressing the outer problem of the SCM problem:

$$\min_{\mathcal{X} \subseteq \mathcal{I}} V(\mathcal{X}) \triangleq \sum_{i \in \mathcal{X}} C_i^e(f_i^*(\mathcal{X}), b_i^*(\mathcal{X})) + \sum_{j \in \mathcal{I} \setminus \mathcal{X}} C_j^l, \quad (21)$$

s.t. $\{(f_i^*(\mathcal{X}), b_i^*(\mathcal{X}))\}_{i \in \mathcal{X}}$ satisfies (16b) – (16e),

where $V(\mathcal{X})$ denotes the social cost associated with the offloading user set $\mathcal{X}$, and $\{(f_i^*(\mathcal{X}), b_i^*(\mathcal{X}))\}_{i \in \mathcal{X}}$ denotes the optimal solution of the corresponding inner problem in (17).

In fact, such a set minimization problem remains challenging to solve. First, optimizing over the exponentially many offloading user sets is NP-hard, as shown in Proposition 1.

Moreover, we find that the social cost function $V(\cdot)$ is neither submodular nor monotone in general (see Appendix E). To the best of our knowledge, existing approximation frameworks [38]–[40] for set function optimization do not directly apply to minimizing a set function of this kind.

To address these challenges, we develop a greedy approach specific to the problem in (21). The key idea is to iteratively select the user that provides the largest marginal social cost reduction when it offloads. Specifically, for a given offloading user set $\mathcal{X} \subset \mathcal{I}$ and a set of candidate users in $\mathcal{I} \setminus \mathcal{X}$, we select a user $j^* \in \arg \min_{j \in \mathcal{I} \setminus \mathcal{X}} \Delta V(\mathcal{X}, \{j\})$, where $\Delta V(\mathcal{X}, \{j\})$ is the marginal social cost when user j offloads, i.e.,

$$\Delta V(\mathcal{X}, \{j\}) \triangleq V(\mathcal{X} \cup \{j\}) - V(\mathcal{X}). \quad (22)$$

However, evaluating this marginal social cost centrally incurs high computational overhead. It requires repeatedly solving the inner problem in (17) for many different offloading user sets while selecting the users.

Instead, we characterize an approximation of the marginal social cost $\Delta V(\mathcal{X}, \{j\})$ that can be computed efficiently in a distributed way. Recall that in Section IV-C, the shadow prices λ_F and λ_B reflect the congestion level of the computation resources and bandwidth, respectively. Given a pair of shadow prices (λ_F, λ_B) , we define the social-cost impact for a non-offloading user $j \in \mathcal{I} \setminus \mathcal{X}$ as follows:

$$h_j(\lambda_F, \lambda_B) \triangleq \min_{0 < f_j \leq F, 0 < b_j \leq B} \left\{ C_j^e(f_j, b_j) - C_j^l + \lambda_F f_j + \lambda_B b_j \right\}. \quad (23)$$

This social-cost impact h_j can be locally computed by the user j efficiently. It tractably approximates the marginal social cost when user j offloads under the current congestion level.

Proposition 2. *Given a fixed offloading user set $\mathcal{X} \subseteq \mathcal{I}$, let $(\lambda_F^*, \lambda_B^*)$ be the optimal shadow prices in the inner problem (17) associated with $\mathcal{X}$. For any non-offloading user $j \in \mathcal{I} \setminus \mathcal{X}$, if the inner problem associated with $\mathcal{X} \cup \{j\}$ is feasible, then we have*

$$h_j(\lambda_F^*, \lambda_B^*) \leq \Delta V(\mathcal{X}, \{j\}). \quad (24)$$

Proof. The proof is provided in Appendix G. $\square$

Proposition 2 shows that if $h_j(\lambda_F^*, \lambda_B^*) \geq 0$, we have $\Delta V(\mathcal{X}, \{j\}) \geq 0$, which means admitting user j to offload increases the social cost. This implies that we should not include the user with a positive h_j in the offloading set.

We leverage the social-cost impact to guide the offloading-user selection process and propose a distributed offloading-user selection algorithm in Algorithm 2. It requires a coordinator, which can be one of the SPs. The algorithm maintains an offloading user set $\mathcal{X}$. In each iteration, it first evaluates the social-cost impact h_j of each non-offloading user (Lines 4–7). The user with the most negative impact is then temporarily admitted to offload (Lines 8–12). As a negative lower bound of the marginal social cost does not guarantee a reduction in social cost, the algorithm further re-solves the inner problem in (17) and checks the true marginal social cost (Lines 13–14). The admission is accepted only if the true marginal social cost is negative, i.e., the admission indeed reduces the social

cost (Lines 15–16). Otherwise, the user is removed from the candidate set (Line 18).

Algorithm 2 is efficient in both computation and communication. It terminates within $|\mathcal{I}|$ iterations, requiring a total of $O(|\mathcal{I}|)$ inner-problem evaluations. Each iteration involves exchanging a constant number of scalars between the coordinator and users, which implies a per-iteration communication overhead of $O(|\mathcal{I}|)$.

Finally, we analyze the theoretical approximation bound of Algorithm 2. As discussed before, the social cost $V(\cdot)$ is neither submodular nor monotone in general (see Appendix E). To our knowledge, no established approximation bound (e.g., $1/2$ for submodular maximization [39]) is known for minimizing such a function. Nevertheless, we derive an approximation bound relative to the optimal offloading user set $\mathcal{X}^*$ and the total local computation cost $V(\emptyset) = \sum_{i \in \mathcal{I}} C_i^l$. Let $\mathcal{X}^{\text{DG}}$ denote the solution returned by Algorithm 2. For each user $i \in \mathcal{I}$, we define its maximum achievable cost reduction as $\Delta \hat{C}_i \triangleq \max_{0 < f_i \leq F, 0 < b_i \leq B} \{C_i^l - C_i^e(f_i, b_i)\}$. Let $i^* \in \arg \max_{i \in \mathcal{I}} \Delta \hat{C}_i$ be a user with the largest reduction. We present the approximation bound as follows.

Theorem 2. *The approximation bound of Algorithm 2 is*

$$\frac{V(\mathcal{X}^{\text{DG}})}{V(\mathcal{X}^*)} \leq \frac{V(\emptyset) - \Delta \hat{C}_{i^*}}{V(\emptyset) - |\mathcal{X}^*| \Delta \hat{C}_{i^*}}. \quad (25)$$

Proof Sketch. The optimal cost $V(\mathcal{X}^*)$ includes no more than $|\mathcal{X}^*| \Delta \hat{C}_{i^*}$ cost reductions, while Algorithm 2 yields at least one largest reduction, i.e., $V(\mathcal{X}^{\text{DG}}) \leq V(\emptyset) - \Delta \hat{C}_{i^*}$. Combining these, we obtain the approximation bound in (25). The detailed proof is provided in Appendix H. $\square$

Theorem 2 provides a worst-case performance guarantee for Algorithm 2. The bound becomes tight when the maximum individual cost reduction is small relative to the total local-computation cost $V(\emptyset)$, or when the optimal offloading user set contains only a limited number of users. In these cases, the denominator remains close to $V(\emptyset)$, and the approximation ratio approaches one. Hence, the ratio in Theorem 2 ensures that the algorithm's solution cannot be arbitrarily worse than the optimal solution, even though the social cost function is neither submodular nor monotone. We also investigate the empirical approximation bound of Algorithm 2 in Appendix N.

In this section, we analyze the user offloading game in Stage II and compute the GNE. In the next section, we study the SP pricing game in Stage I and approximate the resulting Stackelberg equilibrium between the SPs and users.

V. STAGE I: SERVICE PROVIDER PRICING GAME AND STACKELBERG EQUILIBRIUM COMPUTATION

In this section, we investigate the SPs' pricing strategies in Stage I. In Section V-A, we first formulate the SP pricing game and characterize the piecewise monotone structure of the SPs' revenues. Based on this structure, we establish a necessary switching-price condition for the Nash Equilibrium (NE), showing that any NE with non-empty equilibrium offloading user set in Stage II must occur at a switching price where the offloading user set in Stage II changes. In Section V-B, we estimate the SPs' best responses by approximating the switching

Algorithm 2 Distributed Offloading-User Selection Algorithm for the Outer Problem

Input: User set $\mathcal{I}$, resource capacities (F, B) , prices (p_E, p_N) , a coordinator.

Output: Offloading user set $\mathcal{X}^{\text{DG}}$.

- 1: Initialize $\mathcal{X} \leftarrow \emptyset$, candidate set $\mathcal{A} \leftarrow \mathcal{I}$, and $V_e(\mathcal{X}) \leftarrow 0$.
- 2: **repeat**
- 3: Given the offloading user set $\mathcal{X}$, solve the inner problem in (17) with Algorithm 1 to obtain the optimal total offloading cost $V_e(\mathcal{X}) \triangleq \sum_{i \in \mathcal{X}} C_i^e(f_i^*, b_i^*)$, and shadow prices λ_F and λ_B .
- 4: The ESP and NSP broadcast (λ_F, λ_B) to users in $\mathcal{A} \setminus \mathcal{X}$.
- 5: **for all** users $j \in \mathcal{A} \setminus \mathcal{X}$ in parallel **do**
- 6: User j computes its social-cost impact $h_j(\lambda_F, \lambda_B)$ according to (23) and sends it to the coordinator.
- 7: **end for**
- 8: The coordinator finds $j^* \leftarrow \arg \min_{j \in \mathcal{A} \setminus \mathcal{X}} h_j$.
- 9: **if** $h_{j^*} \geq 0$ **then**
- 10: **break**
- 11: **end if**
- 12: Set $\mathcal{Y} \leftarrow \mathcal{X} \cup \{j^*\}$.
- 13: Solve the inner problem in (17) with Algorithm 1 for $\mathcal{Y}$ and obtain $V_e(\mathcal{Y})$.
- 14: Compute the true marginal social cost $\Delta V_{\text{true}} \leftarrow V_e(\mathcal{Y}) - V_e(\mathcal{X}) - C_{j^*}^l$.
- 15: **if** $\Delta V_{\text{true}} < 0$ and the inner problem in (17) for $\mathcal{Y}$ is feasible **then**
- 16: Accept the admission: $\mathcal{X} \leftarrow \mathcal{Y}$.
- 17: **else**
- 18: Remove j^* from candidate set: $\mathcal{A} \leftarrow \mathcal{A} \setminus \{j^*\}$.
- 19: **end if**
- 20: **until** $\mathcal{A} \setminus \mathcal{X} = \emptyset$
- 21: Set $\mathcal{X}^{\text{DG}} \leftarrow \mathcal{X}$.
- 22: **return** $\mathcal{X}^{\text{DG}}$.

prices with a set of boundary prices, which represent the price thresholds associated with likely offloading user set changes around the current price profile. Finally, in Section V-C, we develop an iterative pricing algorithm to approximate the NE in Stage I and the corresponding Stackelberg equilibrium.

A. SP Pricing Game Formulation and Switching-Price Condition for Nash Equilibrium

In this subsection, we first formally define the SP pricing game and its NE. Then, we reveal the piecewise monotonicity of the SPs' revenues, based on which we characterize a necessary switching-price condition for the NE: any NE with non-empty equilibrium offloading user set in Stage II must lie in a set of switching prices where offloading users in Stage II change.

We first provide the formal definition of the SP pricing game as follows.

Definition 4 (SP Pricing Game). *The SP pricing game in Stage I consists of*

- **Players:** The ESP and the NSP.

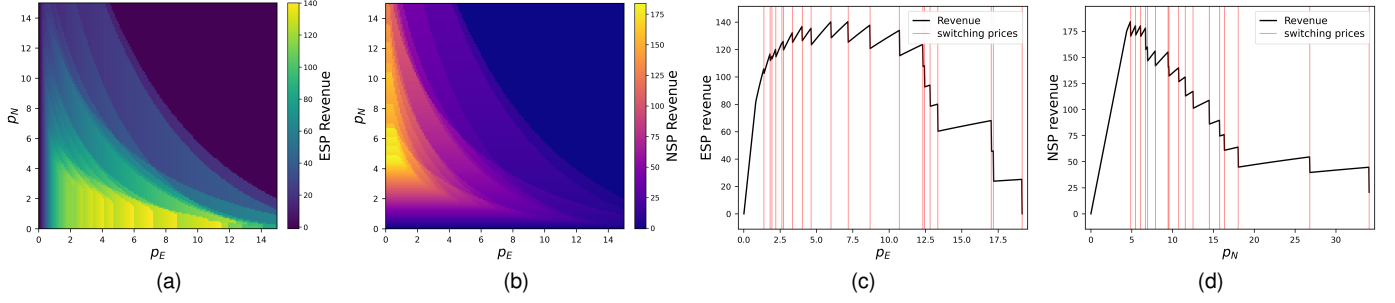


Fig. 1. Revenue structure of the ESP and NSP in the joint pricing space. (a) and (b) show the ESP and NSP revenue heatmaps, respectively, illustrating the layered structure. (c) and (d) show 1D slices of the ESP and NSP revenue curves under a fixed opponent price ($p_N = 0.5$ and $p_E = 0.5$, respectively). The red vertical lines denote the prices where the equilibrium offloading user set in Stage II switches.

- **Strategies:** The ESP and the NSP decide their prices p_E and p_N , respectively, in the feasible pricing space $\mathcal{P} \triangleq \{p = (p_E, p_N) \in \mathbb{R}_+^2 : p_E \geq c_E, p_N \geq c_N\}$.
- **Revenues:** The ESP's and the NSP's revenues are given in (11) and (12), respectively.

We are interested in the NE of the SP pricing game. An NE is a price profile where both SPs simultaneously play their best responses. We formally define it as follows.

Definition 5 (NE of SP Pricing Game). A price profile (p_E^*, p_N^*) is an NE of the SP pricing game if each SP's pricing strategy is the best response to the other's, i.e.,

$$U_E(p_E^*, p_N^*) \geq U_E(p_E, p_N^*), \forall p_E \geq c_E, \quad (26)$$

$$U_N(p_N^*, p_E^*) \geq U_N(p_N, p_E^*), \forall p_N \geq c_N, \quad (27)$$

and $(p_E, p_N^*), (p_E^*, p_N)$ satisfy constraints in (13c) and (14c).

An NE represents a stable outcome because no SP can increase its revenue by unilaterally changing its pricing strategy. Based on this definition, we seek the NE through the SPs' best-response characterization. We define the SPs' best responses as follows.

Definition 6 (Best responses of the SPs). SP $k \in \{E, N\}$'s best response is the price that maximizes its own revenue given the opponent's price p_{-k} , i.e.,

$$p_k^{\text{BR}}(p_{-k}) \in \arg \max_{p_k} U_k(p_k, p_{-k}) \quad \text{s.t. } (p_k, p_{-k}) \in \mathcal{P}. \quad (28)$$

In words, the best response is the most profitable pricing strategy for SP k given the opponent's price.

The main challenge of finding the NE and the best responses is that each SP's revenue depends on the user GNE, which requires solving the NP-hard mixed-integer GNEP in Stage II. As a result, the SPs' revenue functions are implicit and expensive to evaluate. This makes it difficult to analytically derive the NE of the SP pricing game, and also renders direct best-response computation over the strategy space computationally prohibitive.

To address this challenge, we begin by revealing that the SPs' revenues are piecewise monotone functions. Let $\mathcal{X}^*(p)$ denote the equilibrium offloading user set in Stage II under price profile p . In the following analysis, we focus on price profiles $p \in \mathcal{P}$ that induce a nonempty equilibrium offloading

user set in Stage II, i.e., $\mathcal{X}^*(p) \neq \emptyset$. Given the opponent's price p_{-k} , we define SP $k \in \{E, N\}$'s feasible price slice as $\mathcal{P}_k(p_{-k}) \triangleq \{p_k \geq c_k : \mathcal{X}^*((p_k, p_{-k})) \neq \emptyset\}$. Then we have the following lemma.

Lemma 3. For SP $k \in \{E, N\}$, given a fixed price p_{-k} of its opponent, its revenue is piecewise monotone in p_k over $\mathcal{P}_k(p_{-k})$. Specifically, over any pricing interval on which $\mathcal{X}^*((p_k, p_{-k}))$ remains unchanged, SP k 's revenue is strictly increasing in p_k .

Proof. The proof is provided in Appendix J. $\square$

Lemma 3 shows that each SP's revenue is piecewise monotone in its own price. Within any interval where the equilibrium offloading user set in Stage II remains unchanged, SP k 's revenue is strictly increasing in its own price. Changes in the equilibrium offloading user set divide the feasible price slice into such intervals and may cause revenue discontinuities between adjacent intervals.

Fig. 1 validates this piecewise monotonicity using a representative 100-user instance (setup details in Section VI-A). Figs. 1a and 1b show the layered revenue landscapes of the ESP and the NSP in the joint pricing space. The one-dimensional price slices in Figs. 1c and 1d further show that each SP's revenue strictly increases within each interval where the equilibrium offloading user set in Stage II remains fixed. Meanwhile, abrupt drops occur at the prices where the equilibrium offloading user set switches.

Motivated by the piecewise monotonicity of the SPs' revenues, we can study the SPs' best responses by investigating the set of switching prices where the equilibrium offloading user set in Stage II switches. Given a fixed opponent price p_{-k} , we partition SP k 's feasible price slice $\mathcal{P}_k(p_{-k})$ into disjoint intervals over which the equilibrium offloading user set in Stage II remains unchanged. We denote these intervals by $\{\mathcal{J}_{k,m}(p_{-k})\}_{m=1}^{M_k(p_{-k})}$, where $M_k(p_{-k})$ is the number of such intervals. By Lemma 3, SP k 's revenue monotonically increases in p_k over each interval $\mathcal{J}_{k,m}(p_{-k})$. Next, we formally define the set of switching prices as follows.

Definition 7. For SP $k \in \{E, N\}$, given the opponent's price p_{-k} , the switching price associated with interval $\mathcal{J}_{k,m}(p_{-k})$ is defined as its right boundary, i.e., $\bar{p}_{k,m}(p_{-k}) \triangleq \sup \mathcal{J}_{k,m}(p_{-k})$. That is, if SP k increases its price beyond a

switching price, the equilibrium offloading user set in Stage II changes. The switching-price slice of SP k under p_{-k} is defined as

$$\Sigma_k(p_{-k}) \triangleq \{\bar{p}_{k,m}(p_{-k}) : m = 1, \dots, M_k(p_{-k})\}. \quad (29)$$

Furthermore, the set of switching-price profiles is defined as

$$\Sigma \triangleq \{\mathbf{p} \in \mathcal{P} : p_E \in \Sigma_E(p_N), p_N \in \Sigma_N(p_E)\}. \quad (30)$$

We characterize the SPs' best responses and the NE of the SP pricing game through switching prices.

Theorem 3. For SP $k \in \{E, N\}$, given its opponent's price p_{-k} , any best response $p_k^{\text{BR}}(p_{-k})$ must lie in its switching-price slice $\Sigma_k(p_{-k})$ in (29). Furthermore, any NE of the SP pricing game must lie in the switching-price set Σ in (30).

Proof. The proof is provided in Appendix K. $\square$

Theorem 3 indicates that, for each SP k , the best-response search can be restricted to the switching-price slice $\Sigma_k(p_{-k})$. Consequently, the NE search can also be restricted to the switching-price set Σ . The intuition is that, within any interval where the equilibrium offloading user set in Stage II remains fixed, SP k 's revenue is strictly increasing in its own price by Lemma 3. Hence, no interior price of such an interval can be a best response or support an NE.

However, constructing the exact switching-price slice $\Sigma_k(p_{-k})$ requires tracing all changes of the equilibrium offloading user set in Stage II over the entire price slice, which is computationally prohibitive due to the NP-hardness of the SCM problem in Stage II. Therefore, in the next subsection, we construct tractable boundary prices that approximate the switching prices to estimate the SPs' best responses efficiently.

B. Best-Response Estimation via Boundary Prices

In this subsection, we develop a tractable method for estimating each SP's best response by approximating the switching-price slice. We first capture the offloading user set changes that are most relevant to the current price profile by constructing a family of candidate offloading user sets. Then, we compute a boundary price for each of these sets, which is the maximum price where all users in the candidate offloading user set are individually rational. Finally, we estimate each SP's best response by searching for the revenue-maximizing boundary price.

Recall that a switching price is where the equilibrium offloading user set in Stage II changes. Finding the exact switching prices requires enumerating all possible changes of the equilibrium offloading user set over the entire feasible price slice, the number of which grows exponentially with the number of users. Instead, we focus on the set changes that are most relevant to the current price profile and estimate the prices at which such changes may occur.

Specifically, starting from the current equilibrium offloading user set, we identify the marginal users whose offloading decisions are closest to changing. By removing or adding these users, we construct candidate offloading user sets and compute the associated boundary prices. These boundary prices are then used to estimate the SP's best response.

Algorithm 3 Best-Response Estimation via Boundary Prices

Input: Current price profile $\mathbf{p} = (p_k, p_{-k})$, SP index $k \in \{E, N\}$, maximum number Q of considered marginal users.

Output: Estimated best-response price $\hat{p}_k^{\text{BR}}(p_{-k} \mid \mathbf{p})$.

- 1: Run Algorithm 1 and Algorithm 2 at $\mathbf{p}$ to obtain the Stage-II outcome $\Psi^*(\mathbf{p})$.
- 2: Construct the candidate offloading-user-set family $\mathcal{N}_Q(\mathbf{p})$.
- 3: Initialize the boundary-price set $\mathcal{C} \leftarrow \{p_k\}$.
- 4: **for all** non-empty $\mathcal{Y} \in \mathcal{N}_Q(\mathbf{p})$ **do**
- 5: Compute the boundary price $\hat{p}_k(\mathcal{Y} \mid p_{-k})$ by (33) and add to $\mathcal{C}$.
- 6: **end for**
- 7: Set $\mathbf{p}_{\text{best}} \leftarrow \mathbf{p}$.
- 8: **for all** $q_k \in \mathcal{C} \setminus \{p_k\}$ **do**
- 9: Run Algorithm 1 and Algorithm 2 at $\mathbf{q} = (q_k, p_{-k})$ to obtain $\Psi^*(\mathbf{q})$.
- 10: **if** $U_k(\mathbf{q}) > U_k(\mathbf{p}_{\text{best}})$ **then**
- 11: $\mathbf{p}_{\text{best}} \leftarrow \mathbf{q}$.
- 12: **end if**
- 13: **end for**
- 14: **return** $\hat{p}_k^{\text{BR}}(p_{-k} \mid \mathbf{p}) = p_{\text{best},k}$.

We now describe how we construct candidate offloading user sets. Specifically, given a price profile $\mathbf{p} \in \mathcal{P}$, we define the Stage-II outcome as $\Psi^*(\mathbf{p}) \triangleq (\mathcal{X}^*(\mathbf{p}), \{f_i^*(\mathbf{p}), b_i^*(\mathbf{p})\}_{i \in \mathcal{X}^*(\mathbf{p})}, \lambda_F^*(\mathbf{p}), \lambda_B^*(\mathbf{p}))$, which consists of the equilibrium offloading user set, the resource allocation, and the shadow prices at the user GNE under price profile $\mathbf{p}$. We define the leaving incentive of the offloading user $i \in \mathcal{X}^*(\mathbf{p})$ as offloading cost minus local computation cost:

$$\ell_i(\mathbf{p}) \triangleq C_i^e(f_i^*(\mathbf{p}), b_i^*(\mathbf{p}); \mathbf{p}) - C_i^l. \quad (31)$$

A larger $\ell_i(\mathbf{p})$, i.e., a value closer to zero, indicates that user i is closer to leaving the current offloading user set. Conversely, we define the joining incentive of the non-offloading user $j \in \mathcal{I} \setminus \mathcal{X}^*(\mathbf{p})$ as the maximum cost saving from offloading after considering the resource congestion costs:

$$g_j(\mathbf{p}) \triangleq \max_{0 < f_j \leq F, 0 < b_j \leq B} \{C_j^l - C_j^e(f_j, b_j) - \lambda_F^*(\mathbf{p})f_j - \lambda_B^*(\mathbf{p})b_j\}. \quad (32)$$

A larger $g_j(\mathbf{p})$ indicates that user j is more likely to join offloading. Given the current price profile $\mathbf{p}$ and the maximum number Q of considered marginal users, we rank the current offloading users in $\mathcal{X}^*(\mathbf{p})$ in decreasing order of leaving incentive and the non-offloading users in $\mathcal{I} \setminus \mathcal{X}^*(\mathbf{p})$ in decreasing order of joining incentive. For each feasible pair (r, s) with $0 \leq r \leq \min\{Q, |\mathcal{X}^*(\mathbf{p})|\}$ and $0 \leq s \leq \min\{Q, |\mathcal{I} \setminus \mathcal{X}^*(\mathbf{p})|\}$, we construct one candidate offloading user set by removing the top r offloading users from the current offloading user set $\mathcal{X}^*(\mathbf{p})$ and adding the top s non-offloading users to it. Enumerating all feasible pairs (r, s) yields a family of candidate offloading user sets, denoted by $\mathcal{N}_Q(\mathbf{p})$.

The candidate offloading-user-set family $\mathcal{N}_Q(\mathbf{p})$ represents the offloading user sets that are likely to emerge around the current price profile. For a candidate offloading user set

$\mathcal{Y} \in \mathcal{N}_Q(\mathbf{p})$, obtaining the exact switching price involving $\mathcal{Y}$ remains difficult. This would require repeatedly solving the NP-hard SCM problem in Stage II and tracking the price at which the equilibrium offloading user set switches to another set. To avoid such exhaustive tracking, we use an individual-rationality boundary price as a tractable approximation for the switching price. It captures the price at which $\mathcal{Y}$ can no longer be sustained, because at least one user in $\mathcal{Y}$ would prefer local computation.

We next compute the boundary price for each candidate offloading user set in $\mathcal{N}_Q(\mathbf{p})$. Given a non-empty candidate offloading user set $\mathcal{Y} \in \mathcal{N}_Q(\mathbf{p})$, for any price profile $\mathbf{q} = (q_k, q_{-k})$, let $\{f_i^{\mathcal{Y}}(\mathbf{q}), b_i^{\mathcal{Y}}(\mathbf{q})\}_{i \in \mathcal{Y}}$ denote the optimal resource allocation when the offloading user set is fixed as $\mathcal{Y}$, obtained from the inner resource allocation problem in (17). For each user $i \in \mathcal{Y}$, we define its individual-rationality margin under $\mathcal{Y}$ as $\ell_i^{\mathcal{Y}}(\mathbf{q}) \triangleq C_i^e(f_i^{\mathcal{Y}}(\mathbf{q}), b_i^{\mathcal{Y}}(\mathbf{q}); \mathbf{q}) - C_i^l$, similar to (31). Then, given an opponent price p_{-k} , we define the boundary price associated with $\mathcal{Y}$ as

$$\hat{p}_k(\mathcal{Y} | p_{-k}) \triangleq \sup \left\{ q_k \geq c_k : \max_{i \in \mathcal{Y}} \ell_i^{\mathcal{Y}}(q_k, p_{-k}) \leq 0 \right\}. \quad (33)$$

In words, $\hat{p}_k(\mathcal{Y} | p_{-k})$ is the largest price of SP k under which all users in $\mathcal{Y}$ remain individually rational, *i.e.*, their offloading costs are no larger than their local computation costs. Once this boundary is crossed, at least one user in $\mathcal{Y}$ would prefer local computation, so $\mathcal{Y}$ can no longer be sustained as an individually-rational offloading user set. The boundary price thus serves as a tractable approximation of a switching price involving the candidate offloading user set $\mathcal{Y}$. We have derived the closed-form expression for the boundary price $\hat{p}_k(\mathcal{Y} | p_{-k})$ in Appendix L.

By collecting the boundary prices over all candidate offloading user sets in $\mathcal{N}_Q(\mathbf{p})$, we estimate SP k 's best response as the boundary price that yields the highest revenue:

$$\hat{p}_k^{\text{BR}}(p_{-k} | \mathbf{p}) \in \arg \max_{p'_k \in \{p_k\} \cup \{\hat{p}_k(\mathcal{Y} | p_{-k})\}_{\mathcal{Y} \in \mathcal{N}_Q(\mathbf{p})}} U_k(p'_k, p_{-k}). \quad (34)$$

We summarize the procedure of estimating the best response for SP $k \in \{E, N\}$ in Algorithm 3. Given the current price profile $\mathbf{p}$, it computes the Stage-II outcome and then constructs the corresponding candidate offloading-user-set family $\mathcal{N}_Q(\mathbf{p})$ (Lines 1–2). For each candidate set $\mathcal{Y} \in \mathcal{N}_Q(\mathbf{p})$, the algorithm computes the boundary price $\hat{p}_k(\mathcal{Y} | p_{-k})$ (Lines 4–6). Finally, the algorithm evaluates SP k 's revenue at each boundary price and returns the price that yields the highest revenue as the best response (Lines 8–13). The complexity of Algorithm 3 is characterized as follows.

Proposition 3. *The number of candidate offloading user sets in $\mathcal{N}_Q(\mathbf{p})$ satisfies $|\mathcal{N}_Q(\mathbf{p})| \leq (Q+1)^2$. Let $\epsilon_{\text{pd}} > 0$ denote the accuracy tolerance of Algorithm 1. Then Algorithm 3 has time complexity*

$$O\left((Q+1)^2 |\mathcal{I}|^2 (1 + \epsilon_{\text{pd}}^{-2})\right). \quad (35)$$

Proof. The proof is provided in Appendix M. $\square$

In the next subsection, we will develop an iterative price update algorithm for the ESP and the NSP based on the

best-response estimation obtained in this subsection. This algorithm approximates the NE of the SP pricing game and the corresponding Stackelberg equilibrium.

C. Iterative Pricing Algorithm for the SP Pricing Game

In this subsection, we develop an iterative pricing algorithm to approximate the NE of the SP pricing game in Stage I, as presented in Algorithm 4. Algorithm 4 alternates price updates between the ESP and the NSP. In each iteration, both SPs estimate their best responses and corresponding gains, and one SP updates its price. After the iteration stops, the algorithm solves the user offloading game in Stage II at the final price profile to obtain the corresponding user GNE.

To determine whether the price updates have converged, we define SP k 's restricted best-response gain under price profile $\mathbf{p}$ using the boundary-price-based estimated best response:

$$\hat{G}_k(\mathbf{p}) = U_k(\hat{p}_k^{\text{BR}}(p_{-k} | \mathbf{p}), p_{-k}) - U_k(p_k, p_{-k}). \quad (36)$$

We define the corresponding restricted NE gap under price profile $\mathbf{p}$ as the maximum of the restricted best-response gains of the ESP and the NSP:

$$\delta(\mathbf{p}) \triangleq \max\{\hat{G}_E(\mathbf{p}), \hat{G}_N(\mathbf{p})\}. \quad (37)$$

Algorithm 4 stops when this restricted NE gap falls below a prescribed tolerance.

We use the restricted NE gap in (37) as the convergence criterion because evaluating exact best responses requires searching all unilateral deviations in the joint pricing space. To clarify this motivation, given a price profile $\mathbf{p} \in \mathcal{P}$, the exact best-response gain G_k of SP $k \in \{E, N\}$ is

$$G_k(\mathbf{p}) = \max_{p'_k} U_k(p'_k, p_{-k}) - U_k(p_k, p_{-k}) \quad \text{s.t. } (p'_k, p_{-k}) \in \mathcal{P}. \quad (38)$$

The exact NE gap is the maximum of these gains:

$$\delta(\mathbf{p}) \triangleq \max\{G_E(\mathbf{p}), G_N(\mathbf{p})\}. \quad (39)$$

Because $U_k(p'_k, p_{-k})$ is determined by the Stage-II GNE, evaluating the global maximization in $G_k(\mathbf{p})$ entails searching over the feasible unilateral price range and recomputing the corresponding Stage-II GNE at each evaluated price. To avoid such costly evaluations, we thus use the restricted NE gap, which provides a computable approximation by replacing the exact best responses with the boundary-price-based estimates.

Finally, we derive the computational complexity of Algorithm 4 as follows. Each iteration of Algorithm 4 invokes Algorithm 3 once for the ESP and once for the NSP. Therefore, by Proposition 3, the total complexity of Algorithm 4 over at most T_{max} iterations is

$$O\left(T_{\text{max}}(Q+1)^2 |\mathcal{I}|^2 (1 + \epsilon_{\text{pd}}^{-2})\right). \quad (40)$$

In summary, Section V establishes a tractable computational framework for the SP pricing game, whose main difficulty lies in the fact that each SP's revenue depends on the GNE of the NP-hard mixed-integer GNEP in Stage II. We first characterize the piecewise monotone revenue structure and prove that any NE in Stage I with non-empty equilibrium offloading user set in Stage II must lie in the switching-price

Algorithm 4 Iterative Pricing Algorithm for the Stage-I SP Pricing Game

Input: Initial price profile $\mathbf{p}^{(0)} \in \mathcal{P}$, tolerance $\varepsilon > 0$, maximum number of iterations $T_{max} > 0$.

Output: Final price profile $\mathbf{p}^*$, the corresponding Stage-II outcome $\Psi^*(\mathbf{p}^*)$ at user GNE.

- 1: Set $\mathbf{p}^{cur} \leftarrow \mathbf{p}^{(0)}$.
- 2: **for** $t = 0, 1, \dots, T_{max} - 1$ **do**
- 3: Run Algorithm 3 for the ESP at $\mathbf{p}^{cur}$ and obtain its estimated best response $\hat{p}_E^{BR}(\mathbf{p}_N^{cur} | \mathbf{p}^{cur})$.
- 4: Compute the ESP's best-response gain $\hat{G}_E(\mathbf{p}^{cur})$ by (36).
- 5: Run Algorithm 3 for the NSP at $\mathbf{p}^{cur}$ and obtain its estimated best response $\hat{p}_N^{BR}(\mathbf{p}_E^{cur} | \mathbf{p}^{cur})$.
- 6: Compute the NSP's best-response gain $\hat{G}_N(\mathbf{p}^{cur})$ by (36).
- 7: Set $\hat{\delta}(\mathbf{p}^{cur}) \leftarrow \max\{\hat{G}_E(\mathbf{p}^{cur}), \hat{G}_N(\mathbf{p}^{cur})\}$.
- 8: **if** $\hat{\delta}(\mathbf{p}^{cur}) \leq \varepsilon$ **then**
- 9: **break**
- 10: **end if**
- 11: **if** $t \bmod 2 = 0$ **then**
- 12: Set $\mathbf{p}^{next} \leftarrow (\hat{p}_E^{BR}(\mathbf{p}_N^{cur}), \mathbf{p}_N^{cur})$.
- 13: **else**
- 14: Set $\mathbf{p}^{next} \leftarrow (\mathbf{p}_E^{cur}, \hat{p}_N^{BR}(\mathbf{p}_E^{cur}))$.
- 15: **end if**
- 16: Set $\mathbf{p}^{cur} \leftarrow \mathbf{p}^{next}$.
- 17: **end for**
- 18: Set $\mathbf{p}^* \leftarrow \mathbf{p}^{cur}$.
- 19: Run Algorithm 2 and Algorithm 1 at $\mathbf{p}^*$ for the Stage-II outcome $\Psi^*(\mathbf{p}^*)$ at user GNE.
- 20: **return** $\mathbf{p}^*, \Psi^*(\mathbf{p}^*)$.

set. Based on this insight, we estimate the SPs' best responses using boundary prices and further develop an iterative pricing algorithm to approximate the NE of the SP pricing game and the corresponding Stackelberg equilibrium. In the next section, we evaluate the proposed algorithms through numerical experiments.

VI. EXPERIMENTAL RESULTS

A. Experimental Setup

1) *Simulation Settings:* The parameters follow similar settings in existing studies [18], [21], [41]. For each user $i \in \mathcal{I}$, the computation workload w_i is uniformly sampled from $[0.5, 2.5]$ Gcycles, and the input data size d_i from $[1, 10]$ MB. The user i 's local CPU frequency f_i^l is uniformly sampled from $[0.5, 1.2]$ GHz, the transmission power ρ_i from $[0.5, 2]$ W, and the channel gain c_i is drawn from an exponential distribution with unit mean. Background noise is set as $N_0 = -114$ dBm. To capture heterogeneous user preferences, each user's delay sensitivity α_i is uniformly sampled from $[10, 15]$, and the energy cost coefficient β_i from $[0.1, 0.5]$. We set the user i 's hardware-specific energy coefficient $\kappa_i = 10^{-27}$ and transmit antenna power efficiency parameter $\varpi_i = 1/\eta_{PA}$ with

$\eta_{PA} = 0.35$ for all $i \in \mathcal{I}$. The ESP and NSP provide total resources of $F = 100$ GHz and $B = 40$ MHz, respectively, with unit operational costs $c_E = c_N = 0.01$. We conduct 10 independent trials and present the averaged results. We also plot the shaded bands to represent the standard deviations.

2) *Baseline Methods:* We use two groups of baselines to evaluate the effectiveness of the proposed algorithms and the impact of the proposed strategic formulation, respectively.

a) Baselines for Stackelberg equilibrium computation: We compare the proposed algorithms with four baselines. For all baselines, we use a centralized MINLP solver to solve the SCM problem in (16) for the Stage-II user GNE.

- **Grid Search Oracle (GSO):** A brute-force benchmark that exhaustively searches a dense pricing grid to find the price profile minimizing the exact NE gap in (39).
- **Genetic Algorithm (GA):** A population-based genetic search over the joint pricing space. Price profiles are evolved through selection, crossover, and mutation, and are evaluated using a discretized NE-gap.
- **Bayesian Optimization (BO):** A black-box optimization method that models the NE-gap objective with a Gaussian process. At each iteration, it selects and evaluates a next price profile by computing a discretized NE-gap.
- **Multi-Agent Reinforcement Learning (MARL):** A joint-action Q-learning method where the two SPs act as two pricing agents, choose prices as actions, and receive their respective revenues as rewards.

b) Baselines for strategic settings: To study the impact of the ESP–NSP–user strategic interactions, we compare our Stackelberg model with four alternative strategic settings:

- **Market Equilibrium (ME) [11]:** A *non-selfish dual-SP* model with *competitive users*, where the ESP and the NSP compute the market-clearing prices.
- **SingleSP [21]:** A *single selfish SP* model with *non-strategic users*, where the SP controls user offloading, computation resources, and bandwidth and extracts users' cost savings as revenue.
- **Cooperative Pricing (Coop) [9]:** A *cooperative dual-SP* model with *competitive users*, where the ESP and NSP jointly maximize their total revenue.
- **Random Offloading (Rand):** A *selfish dual-SP* model with *non-strategic users*, where users offload and purchase resources randomly.

B. Convergence of the Proposed Algorithms

We first examine the convergence of the proposed Stage-II algorithms, *i.e.*, Algorithm 2 and Algorithm 1, for solving the user GNE in Stage II. Fig. 2a shows the social-cost trajectories under fixed prices $(p_E, p_N) = (0.5, 0.5)$ with varying numbers of users $|\mathcal{I}| \in \{50, 100, 150, 200\}$, together with the corresponding centralized optima. In all four cases, the social cost approaches the centralized optima in 12 iterations. Specifically, the optimality gap is only 7.05% on average, with gaps of 11.82%, 4.90%, 4.83%, and 6.65% for $|\mathcal{I}| = 50, 100, 150$, and 200, respectively. This confirms the *fast convergence and near-optimal performance* of the proposed Stage-II algorithms.

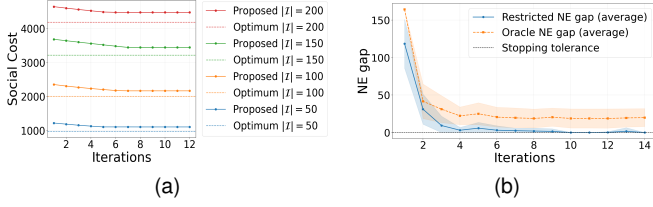


Fig. 2. (a) Social cost in Stage II. (b) NE gap in Stage I.

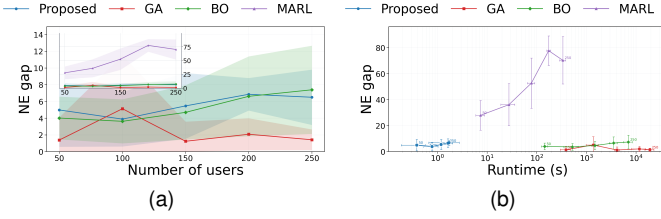


Fig. 3. (a) NE gap versus the number of users (b) NE gap versus runtime.

We next examine the convergence of the proposed iterative pricing algorithm, *i.e.*, Algorithm 4, in Stage I. Fig. 2b shows the boundary-price-restricted NE gap $\hat{\delta}(\mathbf{p})$ in (37), together with the oracle NE gap $\delta(\mathbf{p})$ in (39) measured by the GSO baseline. The restricted NE gap is the objective minimized by Algorithm 4, while the oracle NE gap measures the real gap between the current price profile and the NE in Stage I. As shown in Fig. 2b, the restricted NE gap $\hat{\delta}(\mathbf{p})$ drops sharply in the first 4 iterations and reaches zero within 8 iterations. Meanwhile, the oracle NE gap also decreases substantially, from 164.24 initially to 19.92 at convergence on average, which is an 87.9% reduction. This shows that *Algorithm 4 effectively approximates the NE of the SP pricing game with fast convergence.*

C. Comparison with Baselines for Computing Stackelberg Equilibrium

In this subsection, we evaluate the effectiveness and computational efficiency of the proposed method for computing the Stackelberg equilibrium. We compare it with GA, BO, and MARL across user scales $|I| \in \{50, 100, 150, 200, 250\}$. For each returned price pair, we compute the NE gap using the GSO baseline, and present both the runtime and the number of Stage-II solver calls.

Figs. 3a and 3b show the resulting NE gap and the NE-gap-runtime tradeoff, respectively. Averaged over all tested user scales, the proposed method achieves an average NE gap of 5.54, which is comparable to BO (5.26) and remains much lower than MARL (52.96), yet higher than GA (2.25). However, we demonstrate that the proposed method achieves a favorable accuracy-efficiency tradeoff in Fig. 3b. *The proposed method achieves competitive NE gaps with substantially lower runtime*, while GA and BO require significantly longer computation to obtain comparable or moderately smaller NE gaps.

To further explain the source of this favorable tradeoff, we examine the computational cost of each method in terms

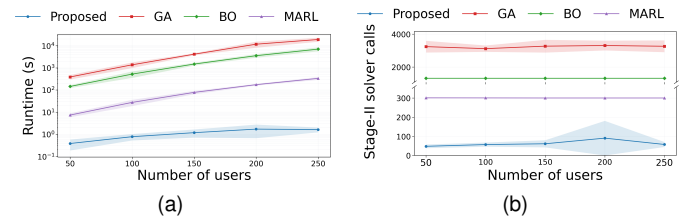
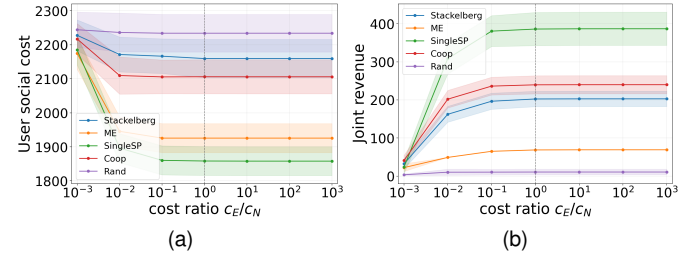


Fig. 4. (a) Wall-clock runtime (b) Number of Stage-II solver calls versus the number of users.

Fig. 5. (a) User social cost (b) ESP-NSP joint revenue versus the cost ratio c_E/c_N .

of runtime and Stage-II solver calls in Fig. 4. *The proposed method consistently achieves the lowest runtime and the fewest Stage-II solver calls across all user scales.* As shown in Fig. 4a, when the number of users increases from 50 to 250, the average runtime of the proposed method remains within 0.38–1.72 seconds, corresponding to average runtime speedups of 6434.2 $\times$, 2231.4 $\times$, and 109.4 $\times$ over GA, BO, and MARL, respectively. Fig. 4b further shows that the proposed method requires only 47.8–90.4 Stage-II solver calls on average. Over all tested user scales, it uses about 51.7 $\times$, 20.9 $\times$, and 4.8 $\times$ fewer calls than GA, BO, and MARL, respectively. These results demonstrate the high computational efficiency of the proposed method, which stems from exploiting the structural properties of the SP pricing game.

Altogether, Figs. 3 and 4 demonstrate that *the proposed method is both effective and efficient in computing the Stackelberg equilibrium.* It closely approximates the Stackelberg equilibrium with remarkably low computational cost.

D. Comparison with Baselines for Strategic Settings

We next compare the proposed Stackelberg model with four strategic-setting baselines under different SP cost ratios. Specifically, we vary the cost ratio c_E/c_N from 10^{-3} to 10^3 while fixing $c_E = 0.01$. This experiment examines the impact of SP strategic pricing and user offloading competition under different relative costs of computation and bandwidth.

Fig. 5a shows that the user social cost first decreases and then remains fixed as c_E/c_N increases. When c_E/c_N is small, bandwidth is much more expensive than computation, which suppresses users' offloading incentives and leads to high user social cost. As c_E/c_N increases, bandwidth becomes less costly, and the user social cost decreases until it remains fixed. The Stackelberg model yields higher user social cost than ME,

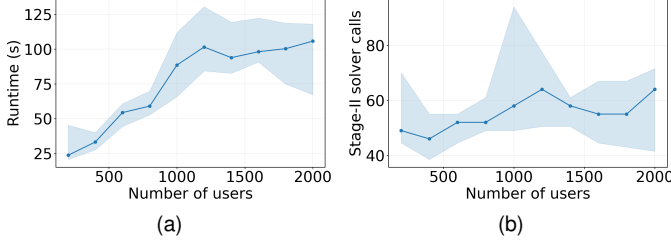


Fig. 6. (a) Runtime (b) Number of Stage-II solver calls versus the number of users.

Coop, and SingleSP, reflecting that *strategic pricing by the SPs imposes additional costs on users*. However, it consistently achieves lower user social cost than Rand, which shows that *competition among users helps them avoid inefficient offloading decisions*.

Fig. 5b shows that the ESP–NSP joint revenue first increases and then remains fixed as c_E/c_N increases. When c_E/c_N is small, an increase in c_E/c_N indicates a lower bandwidth cost, which encourages users to offload and leads to higher joint revenue. When c_E/c_N is large, the bandwidth-cost bottleneck has been relieved, so the joint revenue remains fixed. Further, the Stackelberg model achieves much higher joint revenue than ME and Rand, which indicates that *strategic pricing by the SPs can effectively extract revenue from competing users*. Nevertheless, its joint revenue remains lower than Coop and SingleSP, as *the independent pricing of complementary resources leads to a coordination loss between the ESP and the NSP*.

E. Scalability and Sensitivity Analysis

In this subsection, we evaluate the scalability of the proposed algorithms at substantially larger user populations. We increase the number of users from $|\mathcal{I}| = 200$ to $|\mathcal{I}| = 2000$ and investigate the computational overhead of the proposed methods.

Fig. 6a and Fig. 6b show that the proposed algorithms perform efficiently as the user population scales. The average runtime rises from 32.72 seconds at $|\mathcal{I}| = 200$ to 100.22 seconds at $|\mathcal{I}| = 2000$, i.e., only a $3.06\times$ increase under a $10\times$ growth in the number of users. More importantly, the average number of calls to Algorithms 2 and 1 stays within a narrow range of 49.0 to 71.2. Overall, these results confirm that *the proposed algorithms exhibit strong scalability and efficiently accommodate large-scale user populations with moderate growth in computational overhead*.

Supplementary sensitivity analyses on resource capacities and the maximum number Q of considered marginal users (see Algorithm 3) are provided in Appendices P and Q, respectively.

F. Extensions to a Multiple-SP Market

1) *Experimental Setup*: We further extend to a multiple-SP MEC market with 3 ESPs and 3 NSPs. The experiment contains 50 users, and the user-side parameters follow the experimental setup in Section VI-A. The ESP computation

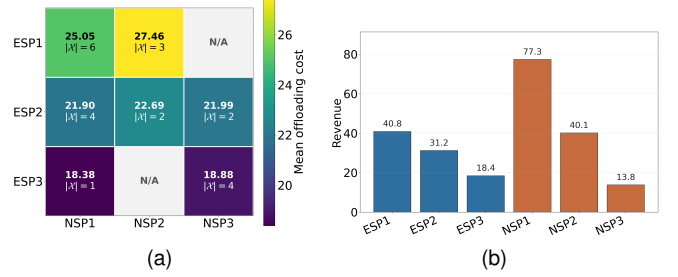


Fig. 7. (a) Mean offloading cost for each ESP–NSP pair. The annotation $|\mathcal{U}|$ denotes the number of offloading users served by the corresponding pair, and N/A indicates that no user offloads through that pair. (b) Revenues of the three ESPs and the three NSPs.

capacities are $[45, 35, 25]$, and the NSP bandwidth capacities are $[45, 35, 25]$, respectively. The unit computation costs of the ESPs are $[0.008, 0.010, 0.013]$, and the unit bandwidth costs of the NSPs are $[0.007, 0.010, 0.014]$. This setting preserves the heterogeneous SP structure. The users select an ESP–NSP pair to offload their tasks. The Stackelberg outcome is computed by iteratively performing SP best-response updates. The detailed formulation and algorithms are described in Appendix O.

2) *Experimental Results*: Among 50 users, 22 users choose to offload. The final prices returned by the algorithm are $p_E = [0.916, 0.901, 0.750]$, $p_N = [1.970, 2.401, 0.620]$, respectively. Fig. 7 illustrates the SP competition in the multi-SP market. Unlike the single-ESP single-NSP setting, users now choose among multiple ESP–NSP pairs. Thus, each SP competes with other SPs through the pairs it can form with the complementary SPs.

Fig. 7a and Fig. 7b together show that *the competitiveness of an ESP–NSP pair is strongly related to the resource capacities of the two SPs*. On the ESP side, ESP1, ESP2, and ESP3 have decreasing computation capacities $[45, 35, 25]$, and they serve 9, 8, and 5 offloading users, respectively. Their revenues follow the same order, with ESP1 achieving the highest revenue (40.8) and ESP3 the lowest (18.4). A similar pattern appears on the NSP side. NSP1 has the largest bandwidth capacity (45) and serves the largest number of users (11), leading to the highest NSP-side revenue (77.3). In contrast, NSP2 and NSP3 have smaller capacities (35 and 25) and serve fewer users (5 and 6, respectively). This suggests that a larger resource capacity allows the ESP–NSP pairs to serve more offloading users and sustain higher revenue.

The equilibrium prices reveal a price–demand tradeoff under ESP–NSP-pair competition. A higher price increases the unit profit but may reduce the demand of all ESP–NSP pairs involving that SP. Conversely, a lower price does not necessarily translate into higher demand or revenue. For example, ESP1 achieves the highest ESP-side revenue because it combines the highest computation price (0.916) with the largest offloading demand (9 users). Meanwhile, ESP3 sets the lowest computation price (0.750) but earns the lowest revenue (18.4) due to limited offloading demand (5 users). Similarly, NSP2 sets the highest bandwidth price (2.401), but its offloading demand is limited (5 users). NSP3 sets the lowest price (0.620), but its revenue is still low (13.8). NSP1 achieves

the highest NSP revenue (77.3) with a moderate price (1.970) and the largest offloading demand (11 users). Thus, *multi-SP competition is governed by the balance between prices and pair-level offloading demand.*

These observations highlight a key feature of MEC markets that arises from the complementarity of computation and bandwidth resources. In the single-ESP single-NSP setting, this complementarity creates strategic interdependence between the ESP and the NSP. In the multiple-SP setting, this complementarity further turns SP strategic interaction into ESP-NSP-pair competition. The competitiveness of an ESP-NSP pair depends on whether the two complementary resources can jointly support users under resource competition.

VII. CONCLUSION

In this paper, we formulate the strategic interactions among an ESP, an NSP, and multiple users in MEC as a two-stage MLMF Stackelberg game. For the user offloading game in Stage II, we prove the existence of a GNE using its exact generalized potential game structure. To compute a GNE, we develop a distributed primal-dual algorithm for resource allocation and a distributed offloading-user selection algorithm. For the SP pricing game in Stage I, we characterize the piecewise monotonicity of SPs' revenues and prove that any NE in Stage I with non-empty equilibrium offloading user set in Stage II must lie in the switching-price set. We propose a boundary-price-based best-response estimation algorithm and an iterative pricing algorithm to approximate the Stackelberg equilibrium. Extensive simulations demonstrate that the proposed framework efficiently and effectively approximates the Stackelberg equilibrium.

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APPENDIX

A. Proof of Lemma 1

We first prove that the user offloading game is an exact generalized potential game with exact potential function $\Phi(\mathbf{s}) = \sum_{i \in \mathcal{I}} C_i(s_i)$. Recall that each user $i \in \mathcal{I}$'s cost C_i is expressed as

$$C_i(s_i) = o_i C_i^e(f_i, b_i) + (1 - o_i) C_i^l, \quad (41)$$

where $s_i = (o_i, f_i, b_i)$ is user i 's strategy, and the computation offloading cost C_i^e and the local computation cost C_i^l depend only on s_i . Hence, for any two feasible strategy profiles $(s_i, \mathbf{s}_{-i}), (s'_i, \mathbf{s}_{-i}) \in \mathcal{S}$ of user i , we have:

$$\begin{aligned} & \Phi(s_i, \mathbf{s}_{-i}) - \Phi(s'_i, \mathbf{s}_{-i}) \\ &= C_i(s_i) + \sum_{j \in \mathcal{I}, j \neq i} C_j(s_j) - C_i(s'_i) - \sum_{j \in \mathcal{I}, j \neq i} C_j(s_j) \\ &= C_i(s_i) - C_i(s'_i). \end{aligned} \quad (42)$$

Therefore, the user offloading game is an exact generalized potential game with exact potential function $\Phi(\mathbf{s}) = \sum_{i \in \mathcal{I}} C_i(s_i)$.

We next prove that the joint strategy space $\mathcal{S}$ of the user offloading game is non-empty and compact. We start by proving the non-emptiness and compactness of the joint strategy space given any fixed offloading user set. For any fixed offloading user set $\mathcal{X} \subseteq \mathcal{I}$, each user $i \in \mathcal{X}$ has $o_i = 1$, and each user $j \notin \mathcal{X}$ has $o_j = 0$. According to constraint (10b), the feasible space for resource usage $\{(f_i, b_i)\}_{i \in \mathcal{X}}$ is

$$\mathcal{S}_{\mathcal{X}} \triangleq \left\{ \{(f_i, b_i)\}_{i \in \mathcal{X}} : f_i \geq 0, b_i \geq 0, \sum_{i \in \mathcal{X}} f_i \leq F, \sum_{i \in \mathcal{X}} b_i \leq B \right\}. \quad (43)$$

This is an intersection of closed half-spaces in a finite-dimensional space, hence a closed, convex set. It is also bounded because we have $0 \leq f_i \leq F$, and $0 \leq b_i \leq B$ for all $i \in \mathcal{X}$. Therefore, the set $\mathcal{S}_{\mathcal{X}}$ is compact. Moreover, it is nonempty because the origin $\{(0, 0)\}_{i \in \mathcal{X}}$ is feasible.

We now prove the compactness of the joint feasible space $\mathcal{S}$. The joint feasible space is the finite union of the feasible space for resource usage of all offloading user sets, i.e.,

$$\mathcal{S} = \bigcup_{\mathcal{X} \subseteq \mathcal{I}} \mathcal{S}_{\mathcal{X}}. \quad (44)$$

To verify this, consider any $\mathbf{s} \in \mathcal{S}$ and let $\mathcal{X}(\mathbf{s})$ be the offloading user set implied by $\mathbf{s}$; then $\mathbf{s} \in \mathcal{S}_{\mathcal{X}(\mathbf{s})}$. If $\mathbf{s} \in \mathcal{S}_{\mathcal{X}}$ for some $\mathcal{X} \subseteq \mathcal{I}$, then it clearly satisfies the constraints in (10b), i.e., $\mathbf{s} \in \mathcal{S}$. Therefore, we have the equation in (44). Each $\mathcal{S}_{\mathcal{X}}$ is non-empty and compact, and a finite union of compact sets is compact. Thus $\mathcal{S}$ is non-empty and compact. This finishes the proof of Lemma 1.

B. Proof of Theorem 1

To prove the existence of GNE in the user offloading game, we prove that the exact potential function $\Phi(\mathbf{s}) = \sum_{i \in \mathcal{I}} C_i(s_i)$ has a global minimizer over the joint feasible strategy space $\mathcal{S}$, and the global minimizer is a GNE of the user offloading game.

As proved in Lemma 1 (see Appendix A), for any offloading user set $\mathcal{X} \subseteq \mathcal{I}$, the feasible space for resource usage $\mathcal{S}_{\mathcal{X}}$ in (43) is non-empty, convex, and compact. The joint feasible strategy space $\mathcal{S}$ is the finite union of $\mathcal{S}_{\mathcal{X}}$ for all $\mathcal{X} \subseteq \mathcal{I}$, i.e., $\mathcal{S} = \bigcup_{\mathcal{X} \subseteq \mathcal{I}} \mathcal{S}_{\mathcal{X}}$.

We now prove that the exact potential function $\Phi(\mathbf{s}) = \sum_{i \in \mathcal{I}} C_i(s_i)$ has a global minimizer over the joint feasible strategy space $\mathcal{S}$. Given any fixed offloading user set $\mathcal{X}$, the offloading decisions $\{o_i\}_{i \in \mathcal{I}}$ are fixed. With fixed $\{o_i\}_{i \in \mathcal{I}}$, the exact potential function Φ is continuous on $\mathcal{S}_{\mathcal{X}}$. Hence, by the Weierstrass extreme value theorem, there exists $\mathbf{s}_{\mathcal{X}}^* \in \mathcal{S}_{\mathcal{X}}$ minimizing Φ over $\mathcal{S}_{\mathcal{X}}$, i.e., $\mathbf{s}_{\mathcal{X}}^* = \arg \min_{\mathbf{s} \in \mathcal{S}_{\mathcal{X}}} \Phi(\mathbf{s})$. Since there are finitely many $\mathcal{X} \subseteq \mathcal{I}$, let $\bar{\mathcal{X}} \in \arg \min_{\mathcal{X} \subseteq \mathcal{I}} \Phi(\mathbf{s}_{\mathcal{X}}^*)$, and let $\mathbf{s}^* = \mathbf{s}_{\bar{\mathcal{X}}}^*$. Then $\mathbf{s}^* \in \mathcal{S}$ and $\Phi(\mathbf{s}^*) = \min_{\mathbf{s} \in \mathcal{S}} \Phi(\mathbf{s})$, i.e., $\mathbf{s}^*$ is a global minimizer of Φ over $\mathcal{S}$.

Next, we prove that the global minimizer $\mathbf{s}^*$ of Φ over $\mathcal{S}$ is a GNE. Suppose, by contradiction, that $\mathbf{s}^*$ is not a GNE of the user offloading game. Then there exists a user $i \in \mathcal{I}$ and a feasible unilateral deviation $s'_i \in \mathcal{S}_i(\mathbf{s}_{-i}^*)$ such that $C_i(s'_i) < C_i(s_i^*)$. By the definition of the exact potential function, we have

$$\Phi(s'_i, \mathbf{s}_{-i}^*) - \Phi(\mathbf{s}^*) = C_i(s'_i) - C_i(s_i^*) < 0, \quad (45)$$

which means that $\Phi(s'_i, \mathbf{s}_{-i}^*) < \Phi(\mathbf{s}^*)$ while $(s'_i, \mathbf{s}_{-i}^*) \in \mathcal{S}$, contradicting the fact that $\mathbf{s}^*$ is a global minimizer of Φ over $\mathcal{S}$. Therefore, $\mathbf{s}^*$ is a GNE of the user offloading game, and the user offloading game admits at least one GNE. This completes the proof.

C. Proof of Proposition 1

We show that the 0–1 knapsack problem can be reduced to an instance of the SCM problem.

0–1 Knapsack Problem. Given a set of items $N = \{1, \dots, n\}$, each item has a weight $\omega_i \in \mathbb{Z}_+$ and a value $v_i \in \mathbb{Z}_+$. A knapsack has a maximum weight capacity $W \in \mathbb{Z}_+$. The 0–1 knapsack problem is modeled as follows.

$$\max_{\{x_i\}_{i=1}^n} \sum_{i=1}^n v_i x_i \quad (46a)$$

$$\text{s.t.} \quad \sum_{i=1}^n \omega_i x_i \leq W, \quad (46b)$$

$$x_i \in \{0, 1\}, \forall i = 1, \dots, n. \quad (46c)$$

We construct a polynomial-time reduction from the 0–1 knapsack problem to the SCM problem. Given an instance of the 0–1 knapsack problem, we construct an instance of the SCM problem by setting $\mathcal{I} = N$, $p_N = 0$, $p_E = 1$, $B = 0$, $F = W$, and, for all $i \in \mathcal{I}$, $\alpha_i = 1$, $\beta_i = 0$, $w_i = \omega_i^2$, $d_i = 0$, $f_i^l = \frac{\omega_i^2}{2\omega_i + \tilde{v}_i}$, where $\tilde{v}_i = v_i/M$ and $M \triangleq 8n^2 W \sum_{i \in \mathcal{I}} v_i$. The complexity of the above reduction is $O(n)$, and hence it is a polynomial-time reduction. In this instance of the SCM problem, user i 's computation offloading cost is $C_i^e(f_i) = \frac{\omega_i^2}{f_i} + f_i$, whereas its local computation cost is

$C_i^l = 2\omega_i + \tilde{v}_i$. This instance of the SCM problem is expressed as follows.

$$\min_{\mathcal{X} \subseteq \mathcal{I}} \min_{\{f_i\}_{i \in \mathcal{X}}} \sum_{i \in \mathcal{X}} \left(\frac{\omega_i^2}{f_i} + f_i \right) + \sum_{j \notin \mathcal{X}} (2\omega_j + \tilde{v}_j) \quad (47a)$$

$$\text{s.t. } \frac{\omega_i^2}{f_i} + f_i \leq 2\omega_i + \tilde{v}_i, \forall i \in \mathcal{X} \quad (47b)$$

$$f_i > 0, \forall i \in \mathcal{X} \quad (47c)$$

$$\sum_{i \in \mathcal{X}} f_i \leq W. \quad (47d)$$

Let $(\mathcal{X}, \{f_i\}_{i \in \mathcal{X}})$ be any feasible solution to the problem in (47). For $i \in \mathcal{X}$, we define

$$\Delta_i \triangleq \frac{\omega_i^2}{f_i} + f_i - 2\omega_i = \frac{(\omega_i - f_i)^2}{f_i} \geq 0. \quad (48)$$

Constraints in (47b) impose that $\frac{\omega_i^2}{f_i} + f_i \leq 2\omega_i + \tilde{v}_i$, hence $\Delta_i \leq \tilde{v}_i$. By summing over $i \in \mathcal{X}$, we have

$$\sum_{i \in \mathcal{X}} \Delta_i \leq \sum_{i \in \mathcal{X}} \tilde{v}_i \leq \frac{1}{M} \sum_{i \in \mathcal{I}} v_i \leq \frac{1}{8n^2 W}. \quad (49)$$

We apply the Cauchy–Schwarz inequality to the nonnegative parts $(\omega_i - f_i)_+$ and have

$$\begin{aligned} \sum_{i \in \mathcal{X}} (\omega_i - f_i)_+ &\leq \sqrt{\left(\sum_{i \in \mathcal{X}} f_i \right) \left(\sum_{i \in \mathcal{X}} \frac{(\omega_i - f_i)_+^2}{f_i} \right)} \\ &\leq \sqrt{W \cdot \sum_{i \in \mathcal{X}} \frac{(\omega_i - f_i)^2}{f_i}} = \sqrt{W \cdot \sum_{i \in \mathcal{X}} \Delta_i}. \end{aligned} \quad (50)$$

Using (49), we can derive the following inequality:

$$\sum_{i \in \mathcal{X}} (\omega_i - f_i) \leq \sum_{i \in \mathcal{X}} (\omega_i - f_i)_+ \leq \sqrt{W \cdot \frac{1}{8n^2 W}} = \sqrt{\frac{1}{8n^2}} < \frac{1}{2}. \quad (51)$$

Therefore, we have

$$\begin{aligned} \sum_{i \in \mathcal{X}} \omega_i &= \sum_{i \in \mathcal{X}} f_i + \sum_{i \in \mathcal{X}} (\omega_i - f_i) \\ &\leq W + \sum_{i \in \mathcal{X}} (\omega_i - f_i)_+ < W + \frac{1}{2}. \end{aligned} \quad (52)$$

Since each ω_i and W are integers, we conclude that $\sum_{i \in \mathcal{X}} \omega_i \leq W$ holds. Hence, any feasible solution $(\mathcal{X}, \{f_i\}_{i \in \mathcal{X}})$ of the problem in (47) has an $\mathcal{X}$ that is feasible for the 0–1 knapsack problem.

Given a fixed $\mathcal{X} \subseteq \mathcal{I}$ with $\sum_{i \in \mathcal{X}} \omega_i \leq W$, we now consider the inner problem of the problem in (47). For each $i \in \mathcal{X}$, the function $\tilde{g}_i(f) \triangleq \omega_i^2/f + f$ is strictly convex on $(0, \infty)$ with unique minimizer $f = \omega_i$ and minimum value $\tilde{g}_i(\omega_i) = 2\omega_i$. Because $\sum_{i \in \mathcal{X}} \omega_i \leq W$ and $2\omega_i \leq 2\omega_i + \tilde{v}_i$, the point $f_i = \omega_i$ is feasible for all constraints. By strict convexity, the unique inner optimum of the inner problem is $f_i^* = \omega_i$, $\forall i \in \mathcal{X}$, and the optimal value is $\sum_{i \in \mathcal{X}} 2\omega_i$.

For any $\mathcal{X} \subseteq \mathcal{I}$ with $\sum_{i \in \mathcal{X}} \omega_i \leq W$, the optimal social cost $V(\mathcal{X})$ of the problem in (47) is

$$\begin{aligned} V(\mathcal{X}) &= \sum_{i \in \mathcal{X}} 2\omega_i + \sum_{j \notin \mathcal{X}} (2\omega_j + \tilde{v}_j) \\ &= \sum_{i \in \mathcal{I}} 2\omega_i + \sum_{j \notin \mathcal{X}} \tilde{v}_j \\ &= \sum_{i \in \mathcal{I}} 2\omega_i + \sum_{i \in \mathcal{I}} \tilde{v}_i - \sum_{i \in \mathcal{X}} \tilde{v}_i. \end{aligned} \quad (53)$$

Let $C_0 \triangleq \sum_{i \in \mathcal{I}} 2\omega_i + \sum_{i \in \mathcal{I}} \tilde{v}_i$, which is a constant independent of $\mathcal{X}$. Then, we have

$$V(\mathcal{X}) = C_0 - \sum_{i \in \mathcal{X}} \tilde{v}_i. \quad (54)$$

We have proved that, for any feasible $\mathcal{X}$ of the problem in (47), $\sum_{i \in \mathcal{X}} \omega_i \leq W$. Therefore, we have

$$\arg \min_{\mathcal{X} \subseteq \mathcal{I}} V(\mathcal{X}) = \arg \max_{\mathcal{X} \subseteq \mathcal{I}} \sum_{i \in \mathcal{X}} \tilde{v}_i \quad \text{s.t.} \quad \sum_{i \in \mathcal{X}} \omega_i \leq W. \quad (55)$$

Since $\tilde{v}_i = v_i/M$ scales v_i by a positive constant M , an optimizer of the problem in (47) coincides with that of the 0–1 knapsack problem. Hence, any optimal offloading user set $\mathcal{X}^*$ for (47) is an optimal solution to the 0–1 knapsack problem, and vice versa.

Therefore, we have proved that the 0–1 knapsack problem can be reduced to an instance of the SCM problem in polynomial time, and the optimal solutions of the two problems are equivalent. We can conclude that the SCM problem is NP-hard.

D. Proof of Lemma 2

For completeness, we first write down the inner problem explicitly. Given a fixed offloading user set $\mathcal{X} \subseteq \mathcal{I}$ and fixed prices (p_E, p_N) , the inner problem in (17) is

$$\min_{\{(f_i, b_i)\}_{i \in \mathcal{X}}} \sum_{i \in \mathcal{X}} C_i^e(f_i, b_i) \quad (56a)$$

$$\text{s.t. } C_i^e(f_i, b_i) \leq C_i^l, \quad \forall i \in \mathcal{X}, \quad (56b)$$

$$f_i > 0, b_i > 0, \quad \forall i \in \mathcal{X}, \quad (56c)$$

$$\sum_{i \in \mathcal{X}} f_i \leq F, \quad \sum_{i \in \mathcal{X}} b_i \leq B, \quad (56d)$$

where, according to the system model,

$$C_i^e(f_i, b_i) = \alpha_i T_i^e(f_i, b_i) + \beta_i E_i^u(b_i) + p_E f_i + p_N b_i, \quad (57)$$

and

$$T_i^e(f_i, b_i) = \frac{w_i}{f_i} + \frac{d_i}{\sigma_i b_i}, \quad E_i^u(b_i) = \frac{d_i}{\sigma_i b_i} \rho_i \varpi_i. \quad (58)$$

By collecting the terms depending on (f_i, b_i) , we can rewrite C_i^e as

$$C_i^e(f_i, b_i) = \alpha_i \frac{w_i}{f_i} + \left(\alpha_i + \beta_i \rho_i \varpi_i \right) \frac{d_i}{\sigma_i b_i} + p_E f_i + p_N b_i. \quad (59)$$

For convenience, we define a positive constant

$$\theta_i \triangleq \frac{d_i}{\sigma_i} (\alpha_i + \beta_i \rho_i \varpi_i) > 0. \quad (60)$$

Then we have

$$C_i^e(f_i, b_i) = \alpha_i \frac{w_i}{f_i} + \frac{\theta_i}{b_i} + p_E f_i + p_N b_i. \quad (61)$$

a) *Strict convexity of the objective:* We first show that C_i^e is strictly convex in (f_i, b_i) on the domain $\{(f_i, b_i) : f_i > 0, b_i > 0\}$, and then extend to the whole inner problem.

From (61), C_i^e is the sum of four functions:

$$g_{i,1}(f_i) = \alpha_i \frac{w_i}{f_i}, \quad g_{i,2}(b_i) = \frac{\theta_i}{b_i}, \quad (62)$$

$$g_{i,3}(f_i) = p_E f_i, \quad g_{i,4}(b_i) = p_N b_i. \quad (63)$$

The second derivatives of the nonlinear terms are

$$\frac{\partial^2 g_{i,1}}{\partial f_i^2} = \frac{2\alpha_i w_i}{f_i^3} > 0, \quad \frac{\partial^2 g_{i,2}}{\partial b_i^2} = \frac{2\theta_i}{b_i^3} > 0, \quad (64)$$

for all $f_i > 0$ and $b_i > 0$. Thus $g_{i,1}$ is strictly convex in $f_i > 0$ and $g_{i,2}$ is strictly convex in $b_i > 0$. The linear functions $g_{i,3}$ and $g_{i,4}$ are convex.

Moreover, C_i^e has no cross term between f_i and b_i , so its Hessian matrix with respect to (f_i, b_i) is diagonal:

$$\nabla^2 C_i^e(f_i, b_i) = \begin{bmatrix} \frac{2\alpha_i w_i}{f_i^3} & 0 \\ 0 & \frac{2\theta_i}{b_i^3} \end{bmatrix}, \quad (65)$$

which is positive definite for all $f_i > 0$ and $b_i > 0$. Therefore C_i^e is strictly convex in (f_i, b_i) on $\{(f_i, b_i) : f_i > 0, b_i > 0\}$.

Since the total objective in (56a) is the sum of C_i^e over $i \in \mathcal{X}$ and the decision variables of different users are decoupled, the Hessian of the total objective is block diagonal with each block positive definite. Hence the objective function $\sum_{i \in \mathcal{X}} C_i^e(f_i, b_i)$ is strictly convex in the vector $\{(f_i, b_i)\}_{i \in \mathcal{X}}$ on the domain where all $f_i > 0$ and $b_i > 0$.

b) *Convexity of the feasible set and existence of an interior optimum:* We now consider the feasible set of the inner problem, which is defined as follows.

$$\begin{aligned} \mathcal{F}_{>0} \triangleq \{ \{(f_i, b_i)\}_{i \in \mathcal{X}} : & \sum_{i \in \mathcal{X}} f_i \leq F, \sum_{i \in \mathcal{X}} b_i \leq B, \\ & f_i > 0, b_i > 0, \forall i \in \mathcal{X}, \\ & C_i^e(f_i, b_i) \leq C_i^l, \forall i \in \mathcal{X} \}. \end{aligned} \quad (66)$$

Each constraint defining $\mathcal{F}_{>0}$ is convex, and the capacity constraints together with $f_i > 0, b_i > 0$ imply that $0 < f_i \leq F$ and $0 < b_i \leq B$ for all $i \in \mathcal{X}$. Hence $\mathcal{F}_{>0}$ is a nonempty bounded convex set. Because of the strict inequalities $f_i > 0$ and $b_i > 0$, $\mathcal{F}_{>0}$ is in general not closed and therefore not compact.

To establish the existence of an optimal solution, we introduce a corresponding closed feasible set

$$\begin{aligned} \bar{\mathcal{F}} \triangleq \{ \{(f_i, b_i)\}_{i \in \mathcal{X}} : & \sum_{i \in \mathcal{X}} f_i \leq F, \sum_{i \in \mathcal{X}} b_i \leq B, \\ & f_i \geq 0, b_i \geq 0, \forall i \in \mathcal{X}, \\ & C_i^e(f_i, b_i) \leq C_i^l, \forall i \in \mathcal{X} \}. \end{aligned} \quad (67)$$

We interpret C_i^e in (67) as its extended-value extension on $\mathbb{R}_+^2$. The spaces defined by the capacity constraints, the non-negativity constraints and the individual rationality constraints are all closed and convex. Thus, their intersection $\bar{\mathcal{F}}$ is a closed, bounded, and convex set. In particular, $\bar{\mathcal{F}}$ is compact, and clearly we have $\mathcal{F}_{>0} \subseteq \bar{\mathcal{F}}$.

We extend the objective function (56a) to $\bar{\mathcal{F}}$ by setting $C_i^e(f_i, b_i) = +\infty$ whenever $f_i = 0$ or $b_i = 0$ for some $i \in \mathcal{X}$, and keeping the original expression in (61) otherwise. The extended objective $\sum_{i \in \mathcal{X}} C_i^e(f_i, b_i)$ is lower semicontinuous on $\bar{\mathcal{F}}$ and finite at every point in $\mathcal{F}_{>0}$. Since $\bar{\mathcal{F}}$ is compact, the extended objective attains its minimum at some $\mathbf{x}^* = \{(f_i^*, b_i^*)\}_{i \in \mathcal{X}} \in \bar{\mathcal{F}}$.

By (61), for any $i \in \mathcal{X}$ we have

$$C_i^e(f_i, b_i) = \alpha_i \frac{w_i}{f_i} + \frac{\theta_i}{b_i} + p_E f_i + p_N b_i, \quad (68)$$

so $C_i^e(f_i, b_i) \rightarrow +\infty$ as $f_i \rightarrow 0^+$ or $b_i \rightarrow 0^+$. Therefore, any minimizer $\mathbf{x}^*$ of the extended objective over $\bar{\mathcal{F}}$ cannot satisfy $f_i^* = 0$ or $b_i^* = 0$ for any $i \in \mathcal{X}$, otherwise the objective value at $\mathbf{x}^*$ would be $+\infty$, which is strictly larger than the finite objective value at a strictly feasible point in $\mathcal{F}_{>0}$. It follows that $\mathbf{x}^*$ must satisfy $f_i^* > 0$ and $b_i^* > 0$ for all $i \in \mathcal{X}$, i.e., $\mathbf{x}^* \in \mathcal{F}_{>0}$. Together with the strict convexity of the objective established above, this implies that $\mathbf{x}^*$ is the unique optimal solution of the inner problem. This completes the proof.

E. Non-monotonicity and Non-submodularity of the Social Cost Function $V(\mathcal{X})$

In this appendix, we show that the social cost function $V(\mathcal{X})$ in (21) is neither monotone nor submodular. Before we present this result, we introduce the formal definition of monotonicity and submodularity of set functions.

Definition 8 (Monotonicity). *Let U be a finite ground set and $V : 2^U \rightarrow \mathbb{R}$ be a set function.*

- V is monotone increasing if for any $\mathcal{A} \subseteq \mathcal{B} \subseteq U$,

$$V(\mathcal{A}) \leq V(\mathcal{B}). \quad (69)$$

- V is monotone decreasing if for any $\mathcal{A} \subseteq \mathcal{B} \subseteq U$,

$$V(\mathcal{A}) \geq V(\mathcal{B}). \quad (70)$$

If V is either monotone decreasing or monotone increasing, we simply say that V is monotone.

Definition 9 (Submodularity). *Let U be a finite ground set and $V : 2^U \rightarrow \mathbb{R}$ be a set function. V is submodular if it satisfies the diminishing-returns property, i.e., for any $\mathcal{A} \subseteq \mathcal{B} \subseteq U$ and any $i \in U \setminus \mathcal{B}$,*

$$V(\mathcal{A} \cup \{i\}) - V(\mathcal{A}) \geq V(\mathcal{B} \cup \{i\}) - V(\mathcal{B}). \quad (71)$$

Equivalently, V is submodular if for any $\mathcal{A}, \mathcal{B} \subseteq U$,

$$V(\mathcal{A}) + V(\mathcal{B}) \geq V(\mathcal{A} \cup \mathcal{B}) + V(\mathcal{A} \cap \mathcal{B}). \quad (72)$$

Next, we show that the social cost function $V(\mathcal{X})$ is neither monotone nor submodular in the following proposition.

Proposition 4 (Non-monotonicity and non-submodularity of $V(\mathcal{X})$). *The social cost function $V(\mathcal{X})$ defined in (21) is neither monotone nor submodular in general.*

Proof. Recall that for any offloading user set $\mathcal{X} \subseteq \mathcal{I}$, the social cost $V(\mathcal{X})$ is obtained by solving the inner problem of (17) associated with $\mathcal{X}$, and then adding the local computation costs of users in $\mathcal{I} \setminus \mathcal{X}$.

a) *Counterexample instance setup:* Our goal is to prove that the social cost function $V(\mathcal{X})$ is neither monotone nor submodular in general. To this end, it suffices to construct a valid instance of the system model where these properties are violated.

We consider an instance in which the bandwidth constraint does not bind for all offloading user sets involved in our comparisons. Specifically, we ignore the bandwidth capacity constraint $\sum_{i \in \mathcal{X}} b_i \leq B$ in constraints (16d). Since the bandwidth-related term in each user i 's offloading cost C_i^e in (8) is convex in b_i , there exists a unique bandwidth allocation $\bar{b}_i > 0$ that minimizes the bandwidth-related terms in user i 's offloading cost C_i^e , that is,

$$\bar{b}_i \triangleq \arg \min_{b_i > 0} \alpha_i T_i^u(b_i) + \beta_i E_i^u(b_i) + p_N b_i, \quad (73)$$

where $T_i^u(\bar{b}_i)$ is the uplink transmission delay defined in (3), $E_i^u(\bar{b}_i)$ is the energy consumption of data transmission defined in (4), and α_i, β_i are user i 's marginal monetary costs of delay and energy consumption, respectively. Now pick a finite bandwidth capacity

$$B \geq \sum_{i \in \mathcal{S}} \bar{b}_i, \quad (74)$$

where $\mathcal{S}$ is the *largest offloading user set* appearing in the counterexamples below (e.g., $\mathcal{S} = \{1, 2\}$ for monotonicity and $\mathcal{S} = \{1, 2, 3\}$ for submodularity). Then the coupled bandwidth constraint is inactive for every set $\mathcal{X} \subseteq \mathcal{S}$ considered in the proof.

We define K_i as the optimal bandwidth-related cost in C_i^e for each user i , that is,

$$K_i \triangleq \alpha_i T_i^u(\bar{b}_i) + \beta_i E_i^u(\bar{b}_i) + p_N \bar{b}_i. \quad (75)$$

When the bandwidth capacity constraint is inactive, the social cost $V(\mathcal{X})$ reduces to

$$V(\mathcal{X}) = \sum_{i \in \mathcal{X}} \left(\alpha_i T_i^c(f_i^{\mathcal{X}}) + p_E f_i^{\mathcal{X}} \right) + \sum_{i \in \mathcal{X}} K_i + \sum_{j \in \mathcal{I} \setminus \mathcal{X}} C_j^l, \quad (76)$$

where $T_i^c(f_i^{\mathcal{X}})$ is the remote computation delay defined in (5), and $f_i^{\mathcal{X}}$ is the optimal computation resource usage derived from the reduced inner problem we will introduce later in (77). Note that $\sum_{i \in \mathcal{X}} K_i$ is a modular set function, which does not affect submodularity comparisons, and will be explicitly accounted for in the monotonicity counterexample by our choice of local costs. Thus, the inner problem reduces to the convex program

$$\begin{aligned} \min_{\{f_i > 0\}_{i \in \mathcal{X}}} \quad & \sum_{i \in \mathcal{X}} \left(\frac{\alpha_i w_i}{f_i} + p_E f_i \right) \\ \text{s.t.} \quad & \sum_{i \in \mathcal{X}} f_i \leq F. \end{aligned} \quad (77)$$

By KKT conditions, there exists a multiplier $\lambda^{\mathcal{X}} \geq 0$ such that

$$f_i^{\mathcal{X}} = \sqrt{\frac{\alpha_i w_i}{p_E + \lambda^{\mathcal{X}}}}, \quad \forall i \in \mathcal{X}, \quad (78a)$$

$$\lambda^{\mathcal{X}} \left(\sum_{i \in \mathcal{X}} f_i^{\mathcal{X}} - F \right) = 0. \quad (78b)$$

The computing-related offloading cost of user i at this optimum equals

$$\frac{\alpha_i w_i}{f_i^{\mathcal{X}}} + p_E f_i^{\mathcal{X}} = \sqrt{\alpha_i w_i} \left(\sqrt{p_E + \lambda^{\mathcal{X}}} + \frac{p_E}{\sqrt{p_E + \lambda^{\mathcal{X}}}} \right), \quad \forall i \in \mathcal{X}. \quad (79)$$

When $\lambda^{\mathcal{X}} = 0$, (79) reduces to $2\sqrt{\alpha_i w_i p_E}$.

b) $V(\mathcal{X})$ is not monotone in general: We construct one fixed instance in which $V(\cdot)$ is neither monotone increasing nor monotone decreasing under Definition 8.

Let $\mathcal{I} = \{1, 2\}$ and choose

$$\alpha_1 = \alpha_2 = 1, \quad p_E = 1, \quad F = 9.5, \quad (w_1, w_2) = (81, 1). \quad (80)$$

Pick B satisfying (74) with $\mathcal{S} = \{1, 2\}$ so that the bandwidth constraint is inactive for $\emptyset, \{1\}, \{1, 2\}$. Let the (set-independent) optimal bandwidth-related costs be $K_1, K_2 \geq 0$.

Choose the local cost of user 1 as

$$C_1^l \geq \max \left\{ 18 + K_1, \frac{6849}{380} + K_1 \right\} = \frac{6849}{380} + K_1, \quad (81)$$

and choose

$$C_2^l = \frac{761}{380} + K_2 + \varepsilon, \quad 0 < \varepsilon < \frac{9}{380}. \quad (82)$$

This construction ensures feasibility of the inner problem for the offloader sets considered. In particular, the rationality constraints can be satisfied.

We first compute $V(\{1\})$. Let $\mathcal{X} = \{1\}$. Since $\sqrt{w_1} = 9 \leq F$, the computing capacity constraint is inactive and thus $\lambda^{\{1\}} = 0$. Hence $f_1^{\{1\}} = \sqrt{w_1} = 9$, and the computing-related offloading cost of user 1 is

$$\frac{w_1}{f_1^{\{1\}}} + p_E f_1^{\{1\}} = \frac{81}{9} + 9 = 18. \quad (83)$$

Therefore,

$$V(\{1\}) = (18 + K_1) + C_2^l. \quad (84)$$

We next show that $V(\cdot)$ is not monotone increasing in this case. Consider $\emptyset \subset \{1\}$. We have

$$V(\emptyset) = C_1^l + C_2^l > (18 + K_1) + C_2^l = V(\{1\}), \quad (85)$$

where the strict inequality follows from (81). This violates (69) in Definition 8, so $V(\cdot)$ is not monotone increasing.

Then, we compute $V(\{1, 2\})$ and show that $V(\cdot)$ is not monotone decreasing. Let $\mathcal{Y} = \{1, 2\}$. Now $\sqrt{w_1} + \sqrt{w_2} =$

$10 > F$, so the computing constraint is tight and $\lambda^{\{1,2\}} > 0$. From the KKT condition,

$$\begin{aligned} \frac{\sqrt{w_1}}{\sqrt{1+\lambda^{\{1,2\}}}} + \frac{\sqrt{w_2}}{\sqrt{1+\lambda^{\{1,2\}}}} &= F \\ \Rightarrow \sqrt{1+\lambda^{\{1,2\}}} &= \frac{10}{9.5} = \frac{20}{19}. \end{aligned} \quad (86)$$

Thus, we have

$$f_1^{\{1,2\}} = \frac{9}{20/19} = \frac{171}{20}, \quad f_2^{\{1,2\}} = \frac{1}{20/19} = \frac{19}{20}. \quad (87)$$

Substituting back into $\frac{w_i}{f_i} + p_E f_i$ gives

$$\frac{w_1}{f_1^{\{1,2\}}} + p_E f_1^{\{1,2\}} = \frac{81}{171/20} + \frac{171}{20} = \frac{6849}{380}, \quad (88)$$

$$\frac{w_2}{f_2^{\{1,2\}}} + p_E f_2^{\{1,2\}} = \frac{1}{19/20} + \frac{19}{20} = \frac{761}{380}. \quad (89)$$

Therefore,

$$V(\{1,2\}) = \left(\frac{6849}{380} + K_1 \right) + \left(\frac{761}{380} + K_2 \right). \quad (90)$$

Comparing $V(\{1,2\})$ and $V(\{1\})$ and using (82)–(84), we have

$$\begin{aligned} &V(\{1,2\}) - V(\{1\}) \\ &= \left(\frac{6849}{380} + K_1 \right) + \left(\frac{761}{380} + K_2 \right) - (18 + K_1 + C_2^l) \\ &= \frac{9}{380} - \varepsilon > 0. \end{aligned} \quad (91)$$

Hence $V(\{1,2\}) > V(\{1\})$ even though $\{1\} \subset \{1,2\}$, which violates (70) in Definition 8. Therefore, $V(\cdot)$ is *not monotone decreasing*. Together with (85), $V(\cdot)$ is *not monotone in general*.

c) $V(\mathcal{X})$ is *not submodular in general*: Let $\mathcal{I} = \{1,2,3\}$ and choose

$$\alpha_1 = \alpha_2 = \alpha_3 = 1, \quad p_E = 1, \quad F = 1, \quad (92)$$

$$(w_1, w_2, w_3) = (1, 0.25, 0.01), \quad (93)$$

so that $(\sqrt{w_1}, \sqrt{w_2}, \sqrt{w_3}) = (1, 0.5, 0.1)$. Pick B satisfying (74) with $\mathcal{S} = \{1,2,3\}$ so that the bandwidth constraint is inactive for all sets appearing below. Let the (set-independent) optimal bandwidth-related costs be $K_1, K_2, K_3 \geq 0$, and choose C_i^l large enough so that the inner problem is feasible for all sets considered.

For convenience, for any offloader set $\mathcal{X} \subseteq \mathcal{I}$, we define

$$\begin{aligned} \tilde{V}(\mathcal{X}) &\triangleq \min_{\{(f_i, b_i)\}_{i \in \mathcal{X}}} \sum_{i \in \mathcal{X}} C_i^e(f_i, b_i) \\ \text{s.t.} \quad &\sum_{i \in \mathcal{X}} f_i \leq F, \quad \sum_{i \in \mathcal{X}} b_i \leq B, \quad C_i^e(f_i, b_i) \leq C_i^l, \quad \forall i \in \mathcal{X}, \end{aligned} \quad (94)$$

i.e., $\tilde{V}(\mathcal{X})$ is the minimum total *offloading cost* attained by users in $\mathcal{X}$ in the inner problem, excluding the local costs of users in $\mathcal{I} \setminus \mathcal{X}$. Under the special instance constructed in the proof, the coupled bandwidth constraint is inactive for all sets considered, and the individual rationality constraints are also inactive (by choosing sufficiently large $\{C_i^l\}$). In this case, the minimization over $\{b_i\}_{i \in \mathcal{X}}$ decouples across users and yields a

set-independent constant K_i for each offloading user i . Hence, for every $\mathcal{X}$ involved in the counterexamples, (94) reduces to

$$\begin{aligned} \tilde{V}(\mathcal{X}) &= \min_{\{f_i > 0\}_{i \in \mathcal{X}}} \sum_{i \in \mathcal{X}} \left(\frac{\alpha_i w_i}{f_i} + p_E f_i \right) + \sum_{i \in \mathcal{X}} K_i \\ \text{s.t.} \quad &\sum_{i \in \mathcal{X}} f_i \leq F, \end{aligned} \quad (95)$$

which is exactly the reduced computing-only inner problem (77) plus the modular term $\sum_{i \in \mathcal{X}} K_i$.

Accordingly, the social cost can be decomposed as

$$V(\mathcal{X}) = \tilde{V}(\mathcal{X}) + \sum_{j \in \mathcal{I} \setminus \mathcal{X}} C_j^l. \quad (96)$$

For $\mathcal{X} \subseteq \mathcal{I}$ and $i \notin \mathcal{X}$,

$$V(\mathcal{X} \cup \{i\}) - V(\mathcal{X}) = (\tilde{V}(\mathcal{X} \cup \{i\}) - \tilde{V}(\mathcal{X})) - C_i^l, \quad (97)$$

so the term $-C_i^l$ cancels when comparing marginal social cost increment for different supersets, and submodularity can be checked via $\tilde{V}(\cdot)$.

Consider $\mathcal{X} = \{1\} \subset \mathcal{Y} = \{1,2\}$ and $i = 3 \notin \mathcal{Y}$. Let $S(\mathcal{A}) \triangleq \sum_{j \in \mathcal{A}} \sqrt{w_j}$ for any offloading user set $\mathcal{A}$. When the computation resource capacity constraint $\sum_{i \in \mathcal{A}} f_i \leq F$ is tight for $\mathcal{A}$, from (78) with $p_E = 1$ and $F = 1$ we have

$$\sqrt{1+\lambda^{\mathcal{A}}} = S(\mathcal{A}), \quad (98)$$

and the optimal total computing cost equals

$$\begin{aligned} \tilde{V}(\mathcal{A}) &= S(\mathcal{A}) \left(S(\mathcal{A}) + \frac{1}{S(\mathcal{A})} \right) + \sum_{j \in \mathcal{A}} K_j \\ &= S(\mathcal{A})^2 + 1 + \sum_{j \in \mathcal{A}} K_j. \end{aligned} \quad (99)$$

Now, we have

$$\tilde{V}(\{1\}) = 2\sqrt{w_1} + K_1 = 2 + K_1, \quad (100)$$

$$\tilde{V}(\{1,3\}) = 1.1^2 + 1 + K_1 + K_3 = 2.21 + K_1 + K_3, \quad (101)$$

so

$$\tilde{V}(\{1,3\}) - \tilde{V}(\{1\}) = 0.21 + K_3. \quad (102)$$

Similarly,

$$\tilde{V}(\{1,2\}) = 1.5^2 + 1 + K_1 + K_2 = 3.25 + K_1 + K_2, \quad (103)$$

$$\tilde{V}(\{1,2,3\}) = 1.6^2 + 1 + K_1 + K_2 + K_3 = 3.56 + K_1 + K_2 + K_3, \quad (104)$$

so

$$\tilde{V}(\{1,2,3\}) - \tilde{V}(\{1,2\}) = 0.31 + K_3. \quad (105)$$

Therefore,

$$\begin{aligned} &\tilde{V}(\mathcal{X} \cup \{3\}) - \tilde{V}(\mathcal{X}) = 0.21 + K_3 \\ &< 0.31 + K_3 = \tilde{V}(\mathcal{Y} \cup \{3\}) - \tilde{V}(\mathcal{Y}), \end{aligned} \quad (106)$$

which violates the definition of submodularity in Definition 9. By (97), the same violation holds for $V(\cdot)$ as well. Hence $V(\cdot)$ is *not submodular in general*. This finishes the proof. $\square$

F. Convergence Result and Complexity Analysis of Algorithm 1

In this appendix, we present the convergence result and the complexity analysis of Algorithm 1. Given a fixed offloading user set $\mathcal{X}$, the inner problem in (17) is strictly convex with a unique primal optimal solution $\{(f_i^*, b_i^*)\}_{i \in \mathcal{X}}$ and admits a nonempty set of optimal dual variables $(\lambda_F^*, \lambda_B^*, \{\mu_i^*\}_{i \in \mathcal{X}})$ (see Lemma 2).

Let $\{(f_i^{(t)}, b_i^{(t)})\}_{i \in \mathcal{X}}$ and $(\lambda_F^{(t)}, \lambda_B^{(t)}, \{\mu_i^{(t)}\}_{i \in \mathcal{X}})$ be the primal and dual sequences generated by Algorithm 1. Let $\eta_F^{(t)}$ and $\eta_B^{(t)}$ be the step sizes in the t -th primal update. We define the running averages of the primal iterates:

$$\bar{f}_i^{(T)} \triangleq \frac{\sum_{t=0}^{T-1} \eta_F^{(t)} f_i^{(t)}}{\sum_{t=0}^{T-1} \eta_F^{(t)}}, \quad \bar{b}_i^{(T)} \triangleq \frac{\sum_{t=0}^{T-1} \eta_B^{(t)} b_i^{(t)}}{\sum_{t=0}^{T-1} \eta_B^{(t)}}, \quad \forall i \in \mathcal{X}. \quad (107)$$

The convergence result of Algorithm 1 is presented below.

Proposition 5 (Convergence of Algorithm 1). *Algorithm 1 has the following properties:*

- 1) **(Dual optimality in value)** Let $g(\cdot)$ be the dual function of the inner problem (17) and g^* be its optimal value. Then

$$\lim_{t \rightarrow \infty} g(\lambda_F^{(t)}, \lambda_B^{(t)}, \{\mu_i^{(t)}\}_{i \in \mathcal{X}}) = g^*. \quad (108)$$

- 2) **(Dual iterates approach the optimal set)** Let $\mathcal{D}^*$ denote the (nonempty) set of optimal dual solutions. Then

$$\text{dist}\left((\lambda_F^{(t)}, \lambda_B^{(t)}, \{\mu_i^{(t)}\}_{i \in \mathcal{X}}), \mathcal{D}^*\right) \rightarrow 0. \quad (109)$$

- 3) **(Primal convergence via running averages)** The running averages in (107) are asymptotically feasible and optimal, i.e., the constraint violations vanish, and the objective value achieved by the running-average allocation converges to the optimal value.

$$\sum_{i \in \mathcal{X}} C_i^e(\bar{f}_i^{(T)}, \bar{b}_i^{(T)}) \rightarrow \sum_{i \in \mathcal{X}} C_i^e(f_i^*, b_i^*). \quad (110)$$

Moreover, since the primal optimal solution is unique (Lemma 2), it follows that

$$(\bar{f}_i^{(T)}, \bar{b}_i^{(T)}) \rightarrow (f_i^*, b_i^*), \quad \forall i \in \mathcal{X}. \quad (111)$$

Proof. Algorithm 1 is a projected primal-dual subgradient method applied to the dual problem of the inner problem in (17). At each iteration, the primal update computes an exact minimizer of the Lagrangian given the current multipliers, and the dual update performs a projected ascent step using the corresponding constraint residuals as a subgradient of the concave dual function. This is the standard setup studied in the primal-dual literature [35]–[37], [42].

Lemma 2 guarantees strict convexity and existence of primal and dual optimal solutions. Therefore, the convergence of dual values and the approach of dual iterates to the optimal set follow from standard results on projected subgradient methods for saddle-point problems and dual subgradient methods (e.g., [36], [37]).

Furthermore, the asymptotic feasibility and optimality of the running averages in (107) follow from classical ergodic

primal convergence results for dual subgradient schemes (e.g., [42], [43]). Finally, since the primal optimum is unique, convergence in objective value together with asymptotic feasibility implies the averaged primal sequence converges to $\{(f_i^*, b_i^*)\}_{i \in \mathcal{X}}$. $\square$

The complexity analysis of Algorithm 1 is presented below.

Proposition 6 (Complexity Analysis of Algorithm 1). *Suppose Algorithm 1 returns the running averages $\{(\bar{f}_i^{(T)}, \bar{b}_i^{(T)})\}_{i \in \mathcal{X}}$ after T iterations.*

- 1) **(Rate and iteration complexity)** There exists a constant $c_1 > 0$ such that the objective gap decays as

$$\sum_{i \in \mathcal{X}} C_i^e(\bar{f}_i^{(T)}, \bar{b}_i^{(T)}) - \sum_{i \in \mathcal{X}} C_i^e(f_i^*, b_i^*) \leq \frac{c_1}{\sqrt{T}}. \quad (112)$$

Consequently, to obtain an ϵ -accurate solution in objective gap, it suffices to take $T = O(1/\epsilon^2)$ iterations.

- 2) **(Per-iteration and total cost)** In each iteration, each user $i \in \mathcal{X}$ solves a constant-size (two-variable) convex minimization and communicates $O(1)$ scalars to the service providers; the service providers broadcast $O(1)$ scalars back. Hence, the per-iteration computation and communication costs are both $O(|\mathcal{X}|)$. With $T = O(1/\epsilon^2)$, the total computation and communication complexities scale as $O(|\mathcal{X}|/\epsilon^2)$.

Proof. The $O(1/\sqrt{T})$ rate for projected primal-dual subgradient methods is classical. It follows directly from standard subgradient complexity analyses for primal-dual subgradient schemes (see, e.g., [36], [37], [42]). This yields the iteration complexity $T = O(1/\epsilon^2)$.

The per-iteration and total computation/communication costs follow from the fact that Algorithm 1 performs one constant-size local minimization per user and exchanges only $O(1)$ scalars per user per iteration. Combining this observation with $T = O(1/\epsilon^2)$ completes the proof. $\square$

G. Proof of Proposition 2

In this subsection, we prove Proposition 2. We define the optimal value of the inner problem in (17) associated with $\mathcal{X}$ as follows.

$$V_e(\mathcal{X}) \triangleq \min_{\{(f_i, b_i)\}_{i \in \mathcal{X}}} \sum_{i \in \mathcal{X}} C_i^e \quad \text{s.t. (16b) – (16e)}. \quad (113)$$

Let $g(\lambda_F, \lambda_B, \{\mu_i\}_{i \in \mathcal{X}})$ be the dual function of the inner problem (17) for $\mathcal{X}$, which is given by:

$$\begin{aligned} g(\lambda_F, \lambda_B, \{\mu_i\}_{i \in \mathcal{X}}) &\triangleq \min_{\{f_i, b_i\}_{i \in \mathcal{X}}} \sum_{i \in \mathcal{X}} \left[C_i^e(f_i, b_i) + \lambda_F f_i + \lambda_B b_i \right. \\ &\quad \left. + \mu_i (C_i^e(f_i, b_i) - C_i^l) \right] - (\lambda_F F + \lambda_B B). \end{aligned} \quad (114)$$

For the sake of clarity, we use $g_{\mathcal{Y}}(\lambda_F, \lambda_B, \{\mu_i\}_{i \in \mathcal{Y}})$ to denote the dual function associated with any offloading user set $\mathcal{Y}$. According to Lemma 2, there exists an optimal dual solution $(\lambda_F^*, \lambda_B^*, \{\mu_i^*\}_{i \in \mathcal{X}})$ such that

$$V_e(\mathcal{X}) = g_{\mathcal{X}}(\lambda_F^*, \lambda_B^*, \{\mu_i^*\}_{i \in \mathcal{X}}). \quad (115)$$

Now consider the inner problem for $\mathcal{X} \cup \{j\}$, and pick a dual-feasible point by keeping the same multipliers for users in $\mathcal{X}$ and setting $\mu_j = 0$, i.e., $(\lambda_F^*, \lambda_B^*, \{\mu_i^*\}_{i \in \mathcal{X}}, \mu_j = 0)$. By weak duality, we have

$$V_e(\mathcal{X} \cup \{j\}) \geq g_{\mathcal{X} \cup \{j\}}(\lambda_F^*, \lambda_B^*, \{\mu_i^*\}_{i \in \mathcal{X}}, 0). \quad (116)$$

Moreover, by definition of the dual function in (114), we have

$$g_{\mathcal{X} \cup \{j\}}(\lambda_F^*, \lambda_B^*, \{\mu_i^*\}_{i \in \mathcal{X}}, 0) = g_{\mathcal{X}}(\lambda_F^*, \lambda_B^*, \{\mu_i^*\}_{i \in \mathcal{X}}) + \min_{f_j > 0, b_j > 0} \{C_j^e(f_j, b_j) + \lambda_F^* f_j + \lambda_B^* b_j\}. \quad (117)$$

Combining (115)–(117) yields

$$V_e(\mathcal{X} \cup \{j\}) \geq V_e(\mathcal{X}) + \min_{f_j > 0, b_j > 0} \{C_j^e(f_j, b_j) + \lambda_F^* f_j + \lambda_B^* b_j\}. \quad (118)$$

Subtracting $V_e(\mathcal{X})$ from both sides of (118) and then using (22) and (23), we obtain

$$\Delta V(\mathcal{X}, \{j\}) = V_e(\mathcal{X} \cup \{j\}) - V_e(\mathcal{X}) - C_j^l \geq h_j(\lambda_F^*, \lambda_B^*), \quad (119)$$

which proves (24).

H. Proof of Theorem 2

We prove the claimed approximation bound in Theorem 2 by upper bounding $V(\mathcal{X}^{\text{DG}})$ and lower bounding $V(\mathcal{X}^*)$.

a) *Upper bound on $V(\mathcal{X}^{\text{DG}})$* : In the first iteration with $\mathcal{X}^{(0)} = \emptyset$, Algorithm 2 sets $\lambda_F^{(0)} = \lambda_B^{(0)} = 0$. With the score definition in (23) (restricted to $0 < f_i \leq F$, $0 < b_i \leq B$), we have for each $i \in \mathcal{I}$,

$$h_i^{(0)} = \min_{0 < f_i \leq F, 0 < b_i \leq B} \{C_i^e(f_i, b_i) - C_i^l\} = - \max_{0 < f_i \leq F, 0 < b_i \leq B} \{C_i^l - C_i^e(f_i, b_i)\} = -\Delta \hat{C}_i. \quad (120)$$

Hence the coordinator selects $i^* \in \arg \min_i h_i^{(0)} = \arg \max_i \Delta \hat{C}_i$. If $\Delta \hat{C}_{i^*} = 0$, then $h_{i^*}^{(0)} = 0$ and the algorithm terminates with $\mathcal{X}^{\text{DG}} = \emptyset$, which implies $V(\mathcal{X}^{\text{DG}}) = V(\emptyset)$ and the desired bound holds trivially. Below we consider the case $\Delta \hat{C}_{i^*} > 0$, so that $h_{i^*}^{(0)} < 0$ and i^* is admitted.

For the single-user set $\{i^*\}$, the inner problem optimally chooses $(f_{i^*}^*, b_{i^*}^*)$ to minimize $C_{i^*}^e(f_{i^*}^*, b_{i^*}^*)$ subject to $0 < f_{i^*}^* \leq F$ and $0 < b_{i^*}^* \leq B$, which yields

$$V(\{i^*\}) = V(\emptyset) - \Delta \hat{C}_{i^*}. \quad (121)$$

Moreover, any admitted user is kept only if it strictly decreases the social cost, thus the social cost along the accepted iterations is non-increasing. Therefore,

$$V(\mathcal{X}^{\text{DG}}) \leq V(\{i^*\}) = V(\emptyset) - \Delta \hat{C}_{i^*}. \quad (122)$$

b) *Lower bound on $V(\mathcal{X}^*)$* : Let $\{(f_i^*, b_i^*)\}_{i \in \mathcal{X}^*}$ be an optimal solution of the inner problem for $\mathcal{X}^*$. Since $\sum_{i \in \mathcal{X}^*} f_i^* \leq F$ and $\sum_{i \in \mathcal{X}^*} b_i^* \leq B$, we have $0 < f_i^* \leq F$ and $0 < b_i^* \leq B$ for each $i \in \mathcal{X}^*$. By the definition of $\Delta \hat{C}_i$, it follows that

$$C_i^l - C_i^e(f_i^*, b_i^*) \leq \Delta \hat{C}_i \leq \Delta \hat{C}_{i^*}, \quad \forall i \in \mathcal{X}^*. \quad (123)$$

Summing over $i \in \mathcal{X}^*$ gives

$$V(\emptyset) - V(\mathcal{X}^*) = \sum_{i \in \mathcal{X}^*} (C_i^l - C_i^e(f_i^*, b_i^*)) \leq |\mathcal{X}^*| \Delta \hat{C}_{i^*}, \quad (124)$$

i.e.,

$$V(\mathcal{X}^*) \geq V(\emptyset) - |\mathcal{X}^*| \Delta \hat{C}_{i^*}. \quad (125)$$

Combining the results above, we have

$$\frac{V(\mathcal{X}^{\text{DG}})}{V(\mathcal{X}^*)} \leq \frac{V(\emptyset) - \Delta \hat{C}_{i^*}}{V(\emptyset) - |\mathcal{X}^*| \Delta \hat{C}_{i^*}}, \quad (126)$$

which completes the proof.

I. Closed-form Expressions for Fixed-Offloading-User-Set Resource Allocation

In this appendix, we provide closed-form expressions for the users' optimal resource allocations and the associated shadow prices in Stage II given a fixed offloading user set. For a fixed offloading user set $\mathcal{X} \subseteq \mathcal{I}$, we define the price region associated with $\mathcal{X}$ as

$$\mathcal{P}_{\mathcal{X}} \triangleq \{\mathbf{p} \in \mathcal{P} : \mathcal{X}^*(\mathbf{p}) = \mathcal{X}\}, \quad (127)$$

where $\mathcal{X}^*(\mathbf{p})$ is the equilibrium offloading user set in Stage II under price profile $\mathbf{p}$. Thus, if $\mathbf{p} \in \mathcal{P}_{\mathcal{X}}$, then $\mathcal{X}$ attains the minimum user social cost among all offloading user sets under $\mathbf{p}$.

We define two price thresholds $\bar{p}_E(\mathcal{X}) \triangleq (\sum_{i \in \mathcal{X}} \sqrt{\alpha_i w_i} / F)^2$ and $\bar{p}_N(\mathcal{X}) \triangleq (\sum_{i \in \mathcal{X}} \sqrt{d_i(\alpha_i + \beta_i \rho_i \varpi_i) / \sigma_i} / B)^2$. Given a pair of prices in $\mathcal{P}_{\mathcal{X}}$, let $\tilde{p}_E(\mathbf{p}, \mathcal{X}) \triangleq \max\{p_E, \bar{p}_E(\mathcal{X})\}$ and $\tilde{p}_N(\mathbf{p}, \mathcal{X}) \triangleq \max\{p_N, \bar{p}_N(\mathcal{X})\}$ denote the prices capped from below. Then we have the following lemma.

Lemma 4. *Given a non-empty fixed offloading user set $\mathcal{X} \subseteq \mathcal{I}$ and any pricing strategy $\mathbf{p} = (p_E, p_N) \in \mathcal{P}_{\mathcal{X}}$, the optimal resource allocations $f_i^*(\mathbf{p}, \mathcal{X}), b_i^*(\mathbf{p}, \mathcal{X})$ for all $i \in \mathcal{X}$ and shadow prices $\lambda_F^*(\mathbf{p}, \mathcal{X}), \lambda_B^*(\mathbf{p}, \mathcal{X})$ of the users' inner problem in (17) admit the following closed-form expressions:*

$$f_i^*(\mathbf{p}, \mathcal{X}) = \sqrt{\frac{\alpha_i w_i}{\tilde{p}_E(\mathbf{p}, \mathcal{X})}}, \quad b_i^*(\mathbf{p}, \mathcal{X}) = \sqrt{\frac{d_i(\alpha_i + \beta_i \rho_i \varpi_i) / \sigma_i}{\tilde{p}_N(\mathbf{p}, \mathcal{X})}}, \quad (128a)$$

$$\lambda_F^*(\mathbf{p}, \mathcal{X}) = \tilde{p}_E(\mathbf{p}, \mathcal{X}) - p_E, \quad \lambda_B^*(\mathbf{p}, \mathcal{X}) = \tilde{p}_N(\mathbf{p}, \mathcal{X}) - p_N. \quad (128b)$$

Proof. We prove the closed-form expressions in (128) by solving the users' inner problem in (17) via its KKT conditions. We first relax the individual-rationality constraints in (16b) and derive the closed-form expressions in (128). We then show that, for any $\mathbf{p} \in \mathcal{P}_{\mathcal{X}}$, the optimizer of the relaxed problem automatically satisfies all individual-rationality constraints. Hence the relaxed problem and the original fixed- $\mathcal{X}$ inner problem coincide on $\mathcal{P}_{\mathcal{X}}$.

For a fixed offloading user set $\mathcal{X}$ and a price pair $\mathbf{p} = (p_E, p_N) \in \mathcal{P}_{\mathcal{X}}$, we relax the individual-rationality constraints in (16b) and rewrite the inner problem as follows:

$$\begin{aligned} \min_{\{f_i > 0, b_i > 0\}_{i \in \mathcal{X}}} \quad & \sum_{i \in \mathcal{X}} \left(\frac{\alpha_i w_i}{f_i} + \frac{\theta_i}{b_i} + p_E f_i + p_N b_i \right) \\ \text{s.t.} \quad & \sum_{i \in \mathcal{X}} f_i \leq F, \quad \sum_{i \in \mathcal{X}} b_i \leq B, \end{aligned} \quad (129)$$

where $\theta_i \triangleq \frac{d_i}{\sigma_i}(\alpha_i + \beta_i \rho_i \varpi_i) > 0$. Recall that $\lambda_F \geq 0$ and $\lambda_B \geq 0$ are the Lagrange multipliers associated with the computation and bandwidth resource capacity constraints, respectively. The Lagrangian of (129) is

$$\begin{aligned} \mathcal{L} = \sum_{i \in \mathcal{X}} \left(\frac{\alpha_i w_i}{f_i} + \frac{\theta_i}{b_i} + p_E f_i + p_N b_i \right) + \lambda_F \left(\sum_{i \in \mathcal{X}} f_i - F \right) \\ + \lambda_B \left(\sum_{i \in \mathcal{X}} b_i - B \right). \end{aligned} \quad (130)$$

At the optimum, the KKT stationarity conditions yield, for each $i \in \mathcal{X}$,

$$\frac{\partial \mathcal{L}}{\partial f_i} = -\frac{\alpha_i w_i}{f_i^2} + p_E + \lambda_F = 0 \implies f_i^* = \sqrt{\frac{\alpha_i w_i}{p_E + \lambda_F^*}}, \quad (131)$$

$$\frac{\partial \mathcal{L}}{\partial b_i} = -\frac{\theta_i}{b_i^2} + p_N + \lambda_B = 0 \implies b_i^* = \sqrt{\frac{\theta_i}{p_N + \lambda_B^*}}. \quad (132)$$

The remaining KKT conditions are the primal feasibility, dual feasibility, and complementary slackness:

$$\sum_{i \in \mathcal{X}} f_i^* \leq F, \quad \sum_{i \in \mathcal{X}} b_i^* \leq B, \quad (133)$$

$$\lambda_F^* \geq 0, \quad \lambda_B^* \geq 0, \quad (134)$$

$$\lambda_F^* \left(\sum_{i \in \mathcal{X}} f_i^* - F \right) = 0, \quad \lambda_B^* \left(\sum_{i \in \mathcal{X}} b_i^* - B \right) = 0. \quad (135)$$

We define $t_E \triangleq p_E + \lambda_F^*$ and $t_N \triangleq p_N + \lambda_B^*$. Using (131), we have

$$\sum_{i \in \mathcal{X}} f_i^* = \sum_{i \in \mathcal{X}} \sqrt{\frac{\alpha_i w_i}{t_E}} = \frac{\sum_{i \in \mathcal{X}} \sqrt{\alpha_i w_i}}{\sqrt{t_E}}. \quad (136)$$

If the computation resource capacity constraint is not tight, then we have $\lambda_F^* = 0$ by (135), so we have $t_E = p_E$ and

$$\sum_{i \in \mathcal{X}} f_i^* = \frac{\sum_{i \in \mathcal{X}} \sqrt{\alpha_i w_i}}{\sqrt{p_E}} \leq F \iff p_E \geq \left(\frac{\sum_{i \in \mathcal{X}} \sqrt{\alpha_i w_i}}{F} \right)^2. \quad (137)$$

If the computation resource capacity constraint is tight, then $\sum_{i \in \mathcal{X}} f_i^* = F$ and $\lambda_F^* > 0$, hence we have

$$\frac{\sum_{i \in \mathcal{X}} \sqrt{\alpha_i w_i}}{\sqrt{t_E}} = F \iff t_E = \left(\frac{\sum_{i \in \mathcal{X}} \sqrt{\alpha_i w_i}}{F} \right)^2. \quad (138)$$

Combining these two cases yields

$$t_E = \max \left\{ p_E, \left(\frac{\sum_{i \in \mathcal{X}} \sqrt{\alpha_i w_i}}{F} \right)^2 \right\} \triangleq \tilde{p}_E(\mathbf{p}, \mathcal{X}), \quad (139)$$

$$\lambda_F^* = t_E - p_E = \tilde{p}_E(\mathbf{p}, \mathcal{X}) - p_E. \quad (140)$$

Substituting $t_E = \tilde{p}_E(\mathbf{p}, \mathcal{X})$ into (131) gives

$$f_i^*(\mathbf{p}, \mathcal{X}) = \sqrt{\frac{\alpha_i w_i}{\tilde{p}_E(\mathbf{p}, \mathcal{X})}}, \quad \forall i \in \mathcal{X}. \quad (141)$$

The bandwidth part follows in the same way. We obtain

$$t_N = \max \left\{ p_N, \left(\frac{\sum_{i \in \mathcal{X}} \sqrt{\theta_i}}{B} \right)^2 \right\} \triangleq \tilde{p}_N(\mathbf{p}, \mathcal{X}), \quad (142)$$

$$\lambda_B^*(\mathbf{p}, \mathcal{X}) = \tilde{p}_N(\mathbf{p}, \mathcal{X}) - p_N, \quad (143)$$

and thus we have

$$b_i^*(\mathbf{p}, \mathcal{X}) = \sqrt{\frac{\theta_i}{\tilde{p}_N(\mathbf{p}, \mathcal{X})}}, \quad \forall i \in \mathcal{X}. \quad (144)$$

This gives (128).

Since the objective function in (129) is strictly convex in $\{f_i, b_i\}_{i \in \mathcal{X}}$ and the feasible set is convex, the KKT point derived above is the unique optimizer of the relaxed problem. It remains to show that, for any $\mathbf{p} \in \mathcal{P}_{\mathcal{X}}$, this relaxed optimizer automatically satisfies all individual-rationality constraints.

For each $i \in \mathcal{X}$, we define the achieved offloading cost at the relaxed optimum by

$$C_i^{e,*}(\mathbf{p}, \mathcal{X}) \triangleq \frac{\alpha_i w_i}{f_i^*(\mathbf{p}, \mathcal{X})} + \frac{\theta_i}{b_i^*(\mathbf{p}, \mathcal{X})} + p_E f_i^*(\mathbf{p}, \mathcal{X}) + p_N b_i^*(\mathbf{p}, \mathcal{X}). \quad (145)$$

We also define

$$\tilde{V}(\mathbf{p}, \mathcal{X}) \triangleq \sum_{i \in \mathcal{X}} C_i^{e,*}(\mathbf{p}, \mathcal{X}) + \sum_{j \notin \mathcal{X}} C_j^l, \quad (146)$$

which is exactly the optimal objective value of the relaxed fixed- $\mathcal{X}$ problem augmented by the local costs of users in $\mathcal{I} \setminus \mathcal{X}$.

Suppose, to the contrary, that the relaxed optimizer violates at least one individual-rationality constraint. Define the violating-user set

$$\mathcal{V} \triangleq \{i \in \mathcal{X} : C_i^{e,*}(\mathbf{p}, \mathcal{X}) > C_i^l\}, \quad (147)$$

and assume that $\mathcal{V} \neq \emptyset$. Let

$$\mathcal{Y} \triangleq \mathcal{X} \setminus \mathcal{V}. \quad (148)$$

Consider a feasible solution to the SCM problem in (16), in which each user $i \in \mathcal{Y}$ offloads with the same resource allocations $(f_i^*(\mathbf{p}, \mathcal{X}), b_i^*(\mathbf{p}, \mathcal{X}))$, while every user in $\mathcal{I} \setminus \mathcal{Y}$ executes locally. This solution is feasible because

$$\sum_{i \in \mathcal{Y}} f_i^*(\mathbf{p}, \mathcal{X}) \leq \sum_{i \in \mathcal{X}} f_i^*(\mathbf{p}, \mathcal{X}) \leq F, \quad (149)$$

$$\sum_{i \in \mathcal{Y}} b_i^*(\mathbf{p}, \mathcal{X}) \leq \sum_{i \in \mathcal{X}} b_i^*(\mathbf{p}, \mathcal{X}) \leq B, \quad (150)$$

and, by the definition of $\mathcal{Y}$,

$$C_i^{e,*}(\mathbf{p}, \mathcal{X}) \leq C_i^l, \quad \forall i \in \mathcal{Y}. \quad (151)$$

Hence the social cost of this feasible solution is

$$\begin{aligned} & \sum_{i \in \mathcal{V}} C_i^{e,*}(\mathbf{p}, \mathcal{X}) + \sum_{j \notin \mathcal{V}} C_j^l \\ &= \sum_{i \in \mathcal{X}} C_i^{e,*}(\mathbf{p}, \mathcal{X}) + \sum_{j \notin \mathcal{X}} C_j^l - \sum_{i \in \mathcal{V}} (C_i^{e,*}(\mathbf{p}, \mathcal{X}) - C_i^l) \\ &= \tilde{V}(\mathbf{p}, \mathcal{X}) - \sum_{i \in \mathcal{V}} (C_i^{e,*}(\mathbf{p}, \mathcal{X}) - C_i^l) < \tilde{V}(\mathbf{p}, \mathcal{X}), \end{aligned} \quad (152)$$

where the strict inequality follows from $\mathcal{V} \neq \emptyset$ and $C_i^{e,*}(\mathbf{p}, \mathcal{X}) > C_i^l$ for all $i \in \mathcal{V}$.

On the other hand, $\tilde{V}(\mathbf{p}, \mathcal{X})$ is the optimal value of the relaxed problem, and hence it is a lower bound on the social cost of any feasible SCM solution whose offloading user set is $\mathcal{X}$. Therefore, the above feasible solution with offloading user set $\mathcal{V}$ has strictly smaller social cost than any feasible SCM solution with offloading user set $\mathcal{X}$. This contradicts $\mathbf{p} \in \mathcal{P}_{\mathcal{X}}$, because by definition of $\mathcal{P}_{\mathcal{X}}$ in (127), the set $\mathcal{X}$ is the offloading user set of the user GNE under prices $\mathbf{p}$, *i.e.*, it attains the minimum user social cost in Stage II compared to other offloading user sets.

Therefore, $\mathcal{V} = \emptyset$, which implies that

$$C_i^{e,*}(\mathbf{p}, \mathcal{X}) \leq C_i^l, \quad \forall i \in \mathcal{X}. \quad (153)$$

That is, the relaxed optimizer automatically satisfies all individual-rationality constraints. Hence, on $\mathcal{P}_{\mathcal{X}}$, the relaxed problem (129) and the original fixed- $\mathcal{X}$ inner problem coincide. Therefore, the closed-form solution in (128) is also the optimizer of the original fixed- $\mathcal{X}$ inner problem. This completes the proof. $\square$

J. Proof of Lemma 3

In this subsection, we prove Lemma 3. We prove the result for the ESP. The proof for the NSP is analogous.

We fix the NSP's price at $\bar{p}_N$. For any non-empty offloading user set $\mathcal{X} \subseteq \mathcal{I}$, we define

$$\mathcal{R}_{\mathcal{X}}(\bar{p}_N) \triangleq \{p_E \in \mathcal{P}_E(\bar{p}_N) : \mathcal{X}^*(p_E, \bar{p}_N) = \mathcal{X}\}. \quad (154)$$

The collection of connected components of $\{\mathcal{R}_{\mathcal{X}}(\bar{p}_N)\}_{\mathcal{X} \subseteq \mathcal{I}}$ partitions the one-dimensional price slice $\mathcal{P}_E(\bar{p}_N)$ into intervals on which the equilibrium offloading user set in Stage II remains unchanged. It remains to show that the ESP's revenue is monotone on each such interval.

Consider any such interval $\mathcal{J}$, and let $\mathcal{X}_{\mathcal{J}}$ be the corresponding non-empty equilibrium offloading user set, *i.e.*,

$$\mathcal{X}^*(p_E, \bar{p}_N) = \mathcal{X}_{\mathcal{J}}, \quad \forall p_E \in \mathcal{J}. \quad (155)$$

For notational simplicity, we write $\mathcal{X} = \mathcal{X}_{\mathcal{J}}$. Since the offloading user set is fixed as $\mathcal{X}$ on $\mathcal{J}$, the ESP's revenue is

$$U_E(p_E, \bar{p}_N) = (p_E - c_E) \sum_{i \in \mathcal{X}} f_i^*(p_E, \bar{p}_N), \quad \forall p_E \in \mathcal{J}, \quad (156)$$

where $\{f_i^*(p_E, \bar{p}_N)\}_{i \in \mathcal{X}}$ is the optimal computation-resource allocation in the inner problem in (17).

We define

$$S_E(\mathcal{X}) \triangleq \sum_{i \in \mathcal{X}} \sqrt{\alpha_i w_i}, \quad \bar{p}_E(\mathcal{X}) \triangleq \left(\frac{S_E(\mathcal{X})}{F} \right)^2. \quad (157)$$

For the fixed offloading user set $\mathcal{X}$, the optimal computation-resource allocation satisfies

$$f_i^*(p_E, \bar{p}_N) = \sqrt{\frac{\alpha_i w_i}{\tilde{p}_E(p_E, \mathcal{X})}}, \quad \tilde{p}_E(p_E, \mathcal{X}) = \max\{p_E, \bar{p}_E(\mathcal{X})\}. \quad (158)$$

Thus, we have

$$\sum_{i \in \mathcal{X}} f_i^*(p_E, \bar{p}_N) = \frac{S_E(\mathcal{X})}{\sqrt{\tilde{p}_E(p_E, \mathcal{X})}}, \quad (159)$$

and hence

$$U_E(p_E, \bar{p}_N) = (p_E - c_E) \frac{S_E(\mathcal{X})}{\sqrt{\tilde{p}_E(p_E, \mathcal{X})}}. \quad (160)$$

We now show that $U_E(p_E, \bar{p}_N)$ is strictly increasing in p_E . There are two cases.

First, if $p_E \leq \bar{p}_E(\mathcal{X})$, then we have $\tilde{p}_E(p_E, \mathcal{X}) = \bar{p}_E(\mathcal{X})$. Therefore, we have

$$U_E(p_E, \bar{p}_N) = (p_E - c_E) \frac{S_E(\mathcal{X})}{\sqrt{\bar{p}_E(\mathcal{X})}}, \quad (161)$$

which is strictly increasing in p_E , since $S_E(\mathcal{X}) > 0$.

Second, if $p_E > \bar{p}_E(\mathcal{X})$, then $\tilde{p}_E(p_E, \mathcal{X}) = p_E$. Therefore, we have

$$U_E(p_E, \bar{p}_N) = S_E(\mathcal{X}) \left(\sqrt{p_E} - \frac{c_E}{\sqrt{p_E}} \right). \quad (162)$$

Differentiating it with respect to p_E , we obtain

$$\frac{\partial U_E(p_E, \bar{p}_N)}{\partial p_E} = S_E(\mathcal{X}) \left(\frac{1}{2\sqrt{p_E}} + \frac{c_E}{2p_E^{3/2}} \right) > 0. \quad (163)$$

Hence, $U_E(p_E, \bar{p}_N)$ is strictly increasing in p_E in this case as well.

Combining the two cases, the ESP's revenue is strictly increasing in p_E on every interval where the equilibrium of offloading user set in Stage II is fixed and nonempty. Combining the two cases proves that $U_E(p_E, \bar{p}_N)$ is piecewise monotone over $\mathcal{P}_E(\bar{p}_N)$.

The proof for the NSP follows the same argument. Specifically, for a fixed offloading user set $\mathcal{X}$, we define

$$S_N(\mathcal{X}) \triangleq \sum_{i \in \mathcal{X}} \sqrt{\theta_i}, \quad \bar{p}_N(\mathcal{X}) \triangleq \left(\frac{S_N(\mathcal{X})}{B} \right)^2, \quad (164)$$

where $\theta_i \triangleq d_i(\alpha_i + \beta_i \rho_i \varpi_i) / \sigma_i$. The optimal bandwidth allocation satisfies the same form as the computation-resource allocation. Therefore, $U_N(p_N, \bar{p}_E)$ is strictly increasing in p_N on every interval with a fixed nonempty offloading user set. This completes the proof.

K. Proof of Theorem 3

In this subsection, we prove Theorem 3.

We first prove that SP k 's best response must lie in its switching-price slice $\Sigma_k(p_{-k})$ in (29). Recall that in Section V-A, we have partitioned the feasible price slice of SP k , $\mathcal{P}_k(p_{-k})$, into monotone intervals $\{\mathcal{J}_{k,m}(p_{-k})\}_{m=1}^{M_k(p_{-k})}$. By Lemma 3, for any fixed p_{-k} , the revenue of SP k is strictly increasing in p_k on every interval $\mathcal{J}_{k,m}(p_{-k})$. Therefore, an attained maximizer cannot lie in the interior of any such interval. Hence any exact best response must lie on the boundary between adjacent intervals, *i.e.*, in $\Sigma_k(p_{-k})$.

We next prove that any Nash equilibrium of the SP pricing game in Stage I that induces a nonempty equilibrium offloading user set in Stage II must lie in the switching-price set Σ . Let $\mathcal{P}^+$ denote the set of all price profiles that induce a nonempty equilibrium offloading user set in Stage II. We show that there is no Nash equilibrium at any price pair in $\mathcal{P}^+ \setminus \Sigma$. Suppose, by contradiction, that there exists an NE $\mathbf{p}^* = (p_E^*, p_N^*) \in \mathcal{P}^+ \setminus \Sigma$. Since $\mathbf{p}^* \notin \Sigma$, there exists at least one SP $k \in \{E, N\}$ such that $p_k^* \notin \Sigma_k(p_{-k}^*)$. Hence, p_k^* lies in the interior of a monotone interval of the price slice $\mathcal{P}_k(p_{-k}^*)$. Because $\mathbf{p}^* \in \mathcal{P}^+$, the corresponding equilibrium offloading user set is nonempty. By Lemma 3, SP k 's revenue is strictly increasing in p_k within this interval. Thus, SP k can slightly increase its price while keeping p_{-k}^* fixed and obtain a strictly larger revenue. This profitable unilateral deviation contradicts the NE condition. Therefore, no price profile in $\mathcal{P}^+ \setminus \Sigma$ can be an NE. Equivalently, any NE of the Stage-I SP pricing game on $\mathcal{P}^+$ must lie in the switching-price set Σ .

L. Closed-Form Expression of the Boundary Price

In this appendix, we derive the closed-form expression for the boundary price $\hat{p}_k(\mathcal{Y} \mid p_{-k})$ defined in (33).

For notational brevity, given a fixed non-empty candidate offloading user set $\mathcal{Y} \subseteq \mathcal{I}$, we define

$$a_i \triangleq \sqrt{\alpha_i w_i}, \quad s_i \triangleq \sqrt{\theta_i}, \quad \theta_i \triangleq \frac{d_i}{\sigma_i} (\alpha_i + \beta_i \rho_i \varpi_i), \quad i \in \mathcal{Y}. \quad (165)$$

Also define

$$S_E(\mathcal{Y}) \triangleq \sum_{j \in \mathcal{Y}} a_j, \quad S_N(\mathcal{Y}) \triangleq \sum_{j \in \mathcal{Y}} s_j, \quad (166)$$

and recall the two fixed-set price thresholds

$$\bar{p}_E(\mathcal{Y}) \triangleq \left(\frac{S_E(\mathcal{Y})}{F} \right)^2, \quad \bar{p}_N(\mathcal{Y}) \triangleq \left(\frac{S_N(\mathcal{Y})}{B} \right)^2. \quad (167)$$

Before presenting the closed-form characterization, we introduce some auxiliary notation. For the ESP ($k = E$) with opponent price $p_N \geq c_N$, we define

$$\chi_i^N(p_N, \mathcal{Y}) \triangleq \begin{cases} s_i \left(\sqrt{\bar{p}_N(\mathcal{Y})} + \frac{p_N}{\sqrt{\bar{p}_N(\mathcal{Y})}} \right), & p_N \leq \bar{p}_N(\mathcal{Y}), \\ 2s_i \sqrt{\bar{p}_N(\mathcal{Y})}, & p_N > \bar{p}_N(\mathcal{Y}), \end{cases} \quad (168)$$

and

$$R_i^E(p_N, \mathcal{Y}) \triangleq C_i^l - \chi_i^N(p_N, \mathcal{Y}). \quad (169)$$

For the NSP ($k = N$) with opponent price $p_E \geq c_E$, we define symmetrically

$$\chi_i^E(p_E, \mathcal{Y}) \triangleq \begin{cases} a_i \left(\sqrt{\bar{p}_E(\mathcal{Y})} + \frac{p_E}{\sqrt{\bar{p}_E(\mathcal{Y})}} \right), & p_E \leq \bar{p}_E(\mathcal{Y}), \\ 2a_i \sqrt{\bar{p}_E(\mathcal{Y})}, & p_E > \bar{p}_E(\mathcal{Y}), \end{cases} \quad (170)$$

and

$$R_i^N(p_E, \mathcal{Y}) \triangleq C_i^l - \chi_i^E(p_E, \mathcal{Y}). \quad (171)$$

For any price profile $q = (q_E, q_N)$ and any fixed candidate offloading user set $\mathcal{Y}$, let

$$\tilde{q}_E(q, \mathcal{Y}) \triangleq \max\{q_E, \bar{p}_E(\mathcal{Y})\}, \quad \tilde{q}_N(q, \mathcal{Y}) \triangleq \max\{q_N, \bar{p}_N(\mathcal{Y})\}. \quad (172)$$

For each user $i \in \mathcal{Y}$, we define its fixed-set individual-rationality margin under $\mathcal{Y}$ as

$$m_i(q, \mathcal{Y}) \triangleq C_i^l - C_i^{e,*}(q, \mathcal{Y}), \quad (173)$$

where

$$C_i^{e,*}(q, \mathcal{Y}) \triangleq a_i \left(\sqrt{\tilde{q}_E(q, \mathcal{Y})} + \frac{q_E}{\sqrt{\tilde{q}_E(q, \mathcal{Y})}} \right) + s_i \left(\sqrt{\tilde{q}_N(q, \mathcal{Y})} + \frac{q_N}{\sqrt{\tilde{q}_N(q, \mathcal{Y})}} \right). \quad (174)$$

Thus, $m_i(q, \mathcal{Y}) \geq 0$ means that user i 's offloading cost under the fixed offloading set $\mathcal{Y}$ is no larger than its local computation cost.

Finally, we define the threshold functions for each user $i \in \mathcal{Y}$. For brevity, let $\bar{p}_E^* = \bar{p}_E(\mathcal{Y})$, $\bar{p}_N^* = \bar{p}_N(\mathcal{Y})$, $R_E^* = R_E^E(p_N, \mathcal{Y})$, $R_N^* = R_N^N(p_E, \mathcal{Y})$, $S_E^* = S_E(\mathcal{Y})$, $S_N^* = S_N(\mathcal{Y})$. Then

$$\tau_{E,i}(p_N, \mathcal{Y}) \triangleq \begin{cases} \frac{S_E^*}{F} \frac{R_E^*}{a_i} - \bar{p}_E^*, & R_E^* \leq 2a_i \sqrt{\bar{p}_E^*}, \\ \frac{(R_E^*)^2}{4a_i^2}, & R_E^* > 2a_i \sqrt{\bar{p}_E^*}, \end{cases} \quad (175)$$

and

$$\tau_{N,i}(p_E, \mathcal{Y}) \triangleq \begin{cases} \frac{S_N^*}{B} \frac{R_N^*}{s_i} - \bar{p}_N^*, & R_N^* \leq 2s_i \sqrt{\bar{p}_N^*}, \\ \frac{(R_N^*)^2}{4s_i^2}, & R_N^* > 2s_i \sqrt{\bar{p}_N^*}. \end{cases} \quad (176)$$

Lemma 5 (Closed-form boundary price). *Given a fixed non-empty candidate offloading user set $\mathcal{Y} \subseteq \mathcal{I}$ and the opponent's price p_{-k} , let the auxiliary functions $\chi_i^{k,*}$, R_i^k , and $\tau_{k,i}$ be as defined in Appendix L. Then the boundary price $\hat{p}_k(\mathcal{Y} \mid p_{-k})$ defined in (33) is characterized as follows.*

(i) **ESP boundary price.** For a fixed opponent price $p_N \geq c_N$, suppose that

$$\min_{i \in \mathcal{Y}} m_i((c_E, p_N), \mathcal{Y}) \geq 0, \quad (177)$$

so that $\hat{p}_E(\mathcal{Y} \mid p_N)$ exists on $[c_E, \infty)$. Then

$$\hat{p}_E(\mathcal{Y} \mid p_N) = \min_{i \in \mathcal{Y}} \tau_{E,i}(p_N, \mathcal{Y}). \quad (178)$$

(ii) **NSP boundary price.** For a fixed opponent price $p_E \geq c_E$, suppose that

$$\min_{i \in \mathcal{Y}} m_i((p_E, c_N), \mathcal{Y}) \geq 0, \quad (179)$$

so that $\hat{p}_N(\mathcal{Y} | p_E)$ exists on $[c_N, \infty)$. Then

$$\hat{p}_N(\mathcal{Y} | p_E) = \min_{i \in \mathcal{Y}} \tau_{N,i}(p_E, \mathcal{Y}). \quad (180)$$

Proof. We prove part (i). The proof of part (ii) is completely analogous.

Fix $\mathcal{Y} \neq \emptyset$ and $p_N \geq c_N$. By Lemma 4 in Appendix I, for each $i \in \mathcal{Y}$ we have

$$f_i^*((p_E, p_N), \mathcal{Y}) = \sqrt{\frac{\alpha_i w_i}{\tilde{p}_E}}, \quad b_i^*((p_E, p_N), \mathcal{Y}) = \sqrt{\frac{\theta_i}{\tilde{p}_N}}, \quad (181)$$

where

$$\tilde{p}_E \triangleq \max\{p_E, \bar{p}_E(\mathcal{Y})\}, \quad \tilde{p}_N \triangleq \max\{p_N, \bar{p}_N(\mathcal{Y})\}. \quad (182)$$

Substituting these expressions into the offloading cost

$$C_i^e(f_i, b_i) = \frac{\alpha_i w_i}{f_i} + \frac{\theta_i}{b_i} + p_E f_i + p_N b_i, \quad (183)$$

yields

$$C_i^{e,*}((p_E, p_N), \mathcal{Y}) = a_i \left(\sqrt{\tilde{p}_E} + \frac{p_E}{\sqrt{\tilde{p}_E}} \right) + s_i \left(\sqrt{\tilde{p}_N} + \frac{p_N}{\sqrt{\tilde{p}_N}} \right) \quad (184)$$

Since p_N is fixed, the second term in (184) is a constant with respect to p_E . By the definition of $\chi_i^N(p_N, \mathcal{Y})$, the fixed-set margin of user i can be rewritten as

$$m_i((p_E, p_N), \mathcal{Y}) = C_i^l - \chi_i^N(p_N, \mathcal{Y}) - \phi_i^E(p_E, \mathcal{Y}), \quad (185)$$

where

$$\begin{aligned} \phi_i^E(p_E, \mathcal{Y}) &\triangleq a_i \left(\sqrt{\tilde{p}_E} + \frac{p_E}{\sqrt{\tilde{p}_E}} \right) \\ &= \begin{cases} a_i \left(\sqrt{\bar{p}_E(\mathcal{Y})} + \frac{p_E}{\sqrt{\bar{p}_E(\mathcal{Y})}} \right), & p_E \leq \bar{p}_E(\mathcal{Y}), \\ 2a_i \sqrt{p_E}, & p_E > \bar{p}_E(\mathcal{Y}). \end{cases} \end{aligned} \quad (186)$$

Hence

$$m_i((p_E, p_N), \mathcal{Y}) = R_i^E(p_N, \mathcal{Y}) - \phi_i^E(p_E, \mathcal{Y}). \quad (187)$$

Because $\phi_i^E(p_E, \mathcal{Y})$ is continuous and strictly increasing in p_E on $[c_E, \infty)$, the function $m_i((p_E, p_N), \mathcal{Y})$ is continuous and strictly decreasing in p_E .

We next solve the individual equation

$$m_i((p_E, p_N), \mathcal{Y}) = 0 \iff \phi_i^E(p_E, \mathcal{Y}) = R_i^E(p_N, \mathcal{Y}). \quad (188)$$

There are two cases.

Case 1: the solution lies in the congested regime $p_E \leq \bar{p}_E(\mathcal{Y})$. Then

$$a_i \left(\sqrt{\bar{p}_E(\mathcal{Y})} + \frac{p_E}{\sqrt{\bar{p}_E(\mathcal{Y})}} \right) = R_i^E(p_N, \mathcal{Y}), \quad (189)$$

and thus

$$\begin{aligned} p_E &= \sqrt{\bar{p}_E(\mathcal{Y})} \frac{R_i^E(p_N, \mathcal{Y})}{a_i} - \bar{p}_E(\mathcal{Y}) \\ &= \frac{S_E(\mathcal{Y})}{F} \frac{R_i^E(p_N, \mathcal{Y})}{a_i} - \bar{p}_E(\mathcal{Y}). \end{aligned} \quad (190)$$

This solution indeed belongs to the congested regime iff

$$R_i^E(p_N, \mathcal{Y}) \leq 2a_i \sqrt{\bar{p}_E(\mathcal{Y})}. \quad (191)$$

Case 2: the solution lies in the non-congested regime $p_E > \bar{p}_E(\mathcal{Y})$. Then

$$2a_i \sqrt{p_E} = R_i^E(p_N, \mathcal{Y}), \quad (192)$$

which gives

$$p_E = \frac{(R_i^E(p_N, \mathcal{Y}))^2}{4a_i^2}. \quad (193)$$

This solution indeed belongs to the non-congested regime iff

$$R_i^E(p_N, \mathcal{Y}) > 2a_i \sqrt{\bar{p}_E(\mathcal{Y})}. \quad (194)$$

Therefore, for each $i \in \mathcal{Y}$, the individual zero-margin price is exactly $\tau_{E,i}(p_N, \mathcal{Y})$ defined in the statement.

Finally, let

$$\tau_E^{\min}(p_N, \mathcal{Y}) \triangleq \min_{i \in \mathcal{Y}} \tau_{E,i}(p_N, \mathcal{Y}). \quad (195)$$

Since each $m_i((p_E, p_N), \mathcal{Y})$ is strictly decreasing, we have

$$m_i((p_E, p_N), \mathcal{Y}) \geq 0 \iff p_E \leq \tau_{E,i}(p_N, \mathcal{Y}). \quad (196)$$

Hence

$$\begin{aligned} \min_{i \in \mathcal{Y}} m_i((p_E, p_N), \mathcal{Y}) &\geq 0 \\ \iff p_E &\leq \min_{i \in \mathcal{Y}} \tau_{E,i}(p_N, \mathcal{Y}) = \tau_E^{\min}(p_N, \mathcal{Y}). \end{aligned} \quad (197)$$

At $p_E = \tau_E^{\min}(p_N, \mathcal{Y})$, at least one user reaches zero margin and all the others have nonnegative margins; for $p_E > \tau_E^{\min}(p_N, \mathcal{Y})$, the minimum margin becomes negative. Therefore, the unique solution of

$$\min_{i \in \mathcal{Y}} m_i((p_E, p_N), \mathcal{Y}) = 0 \quad (198)$$

over $p_E \geq c_E$ is

$$\hat{p}_E(\mathcal{Y} | p_N) = \min_{i \in \mathcal{Y}} \tau_{E,i}(p_N, \mathcal{Y}). \quad (199)$$

This proves part (i). Part (ii) follows by the same argument after exchanging $(a_i, S_E, F, \bar{p}_E, p_E)$ with $(s_i, S_N, B, \bar{p}_N, p_N)$. $\square$

M. Proof of Proposition 3

In this subsection, we provide the proof of Proposition 3.

We first prove that the size of the set of candidate offloading user sets $\mathcal{N}_Q(\mathbf{p})$ satisfies $|\mathcal{N}_Q(\mathbf{p})| \leq O((Q+1)^2)$. As described in Section V-B, the candidate offloading user sets are generated by choosing

$$0 \leq r \leq \min\{Q, |\mathcal{X}^*(\mathbf{p})|\}, \quad 0 \leq s \leq \min\{Q, |\mathcal{I} \setminus \mathcal{X}^*(\mathbf{p})|\}. \quad (200)$$

Hence, we have

$$\begin{aligned} |\mathcal{N}_Q(\mathbf{p})| &\leq (\min\{Q, |\mathcal{X}^*(\mathbf{p})|\} + 1)(\min\{Q, |\mathcal{I}| - |\mathcal{X}^*(\mathbf{p})|\} + 1) \\ &\leq (Q + 1)^2. \end{aligned} \quad (201)$$

We next prove the time complexity of Algorithm 3. Algorithm 3 performs one Stage-II evaluation at the current price profile $\mathbf{p}$, and then at most one additional Stage-II evaluation for each boundary price generated from $\mathcal{N}_Q(\mathbf{p})$. Therefore, the total number of Stage-II evaluations is at most

$$|\mathcal{N}_Q(\mathbf{p})| + 1 = O((Q + 1)^2). \quad (202)$$

Now consider one Stage-II evaluation. Algorithm 2 has iteration complexity $O(|\mathcal{I}|)$ and invokes Algorithm 1 at most $2|\mathcal{I}| + 1$ times. By Proposition 6 in Appendix F, one call of Algorithm 1 on a user set $\mathcal{X}$ has complexity

$$O\left(\frac{|\mathcal{X}|}{\epsilon_{\text{pd}}^2}\right) \leq O\left(\frac{|\mathcal{I}|}{\epsilon_{\text{pd}}^2}\right). \quad (203)$$

Therefore, all calls to Algorithm 1 within one execution of Algorithm 2 cost

$$O\left((2|\mathcal{I}| + 1) \frac{|\mathcal{I}|}{\epsilon_{\text{pd}}^2}\right) = O\left(\frac{|\mathcal{I}|^2}{\epsilon_{\text{pd}}^2}\right). \quad (204)$$

Besides these inner solves, each iteration of Algorithm 2 requires $O(|\mathcal{I}|)$ additional operations, including the rollback check, score collection, minimum-score selection, and set update. Since the number of iterations is $O(|\mathcal{I}|)$, these additional operations contribute

$$O(|\mathcal{I}|^2). \quad (205)$$

Hence, one Stage-II evaluation costs

$$O(|\mathcal{I}|^2(1 + \epsilon_{\text{pd}}^{-2})). \quad (206)$$

The remaining operations of Algorithm 3, including constructing $\mathcal{N}_Q(\mathbf{p})$, computing the boundary prices, and comparing the corresponding revenues, require at most

$$O(|\mathcal{I}| \log |\mathcal{I}| + |\mathcal{N}_Q(\mathbf{p})| |\mathcal{I}|), \quad (207)$$

which is absorbed by the repeated Stage-II evaluations. Therefore, we have

$$T_{\text{Alg3}} = O\left((|\mathcal{N}_Q(\mathbf{p})| + 1) |\mathcal{I}|^2 (1 + \epsilon_{\text{pd}}^{-2})\right). \quad (208)$$

Using $|\mathcal{N}_Q(\mathbf{p})| \leq (Q + 1)^2$, we obtain

$$T_{\text{Alg3}} = O\left((Q + 1)^2 |\mathcal{I}|^2 (1 + \epsilon_{\text{pd}}^{-2})\right). \quad (209)$$

In particular, if ϵ_{pd} is treated as a fixed constant, the above bound reduces to

$$O((Q + 1)^2 |\mathcal{I}|^2). \quad (210)$$

This completes the proof.

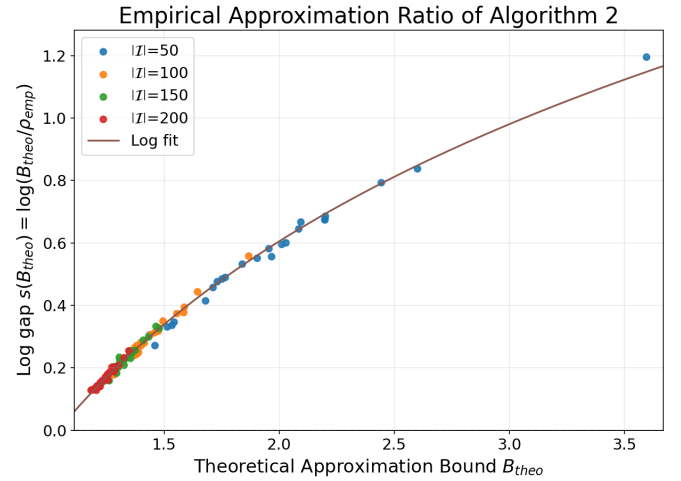


Fig. 8. Approximation bound versus empirical approximation ratio of Algorithm 2. Each marker corresponds to one valid instance and is color-coded by $|\mathcal{I}|$. The brown line is the logarithmic fit of $s(B_{\text{theo}}) = \log(B_{\text{theo}}/\rho_{\text{emp}})$ against $\ln B_{\text{theo}}$.

N. Empirical Approximation Bound of Algorithm 2

We investigate the empirical approximation bound of Algorithm 2 in Fig. 8. Recall that we derived a theoretical approximation bound (25) in Theorem 2. For each $|\mathcal{I}| = 50, 100, 150, 200$, we run 10 trials and compute the empirical approximation ratio $\rho_{\text{emp}} \triangleq V(\mathcal{X}^{\text{DG}})/V(\mathcal{X}^*)$ from the experimental results. We then compute

$$s(B_{\text{theo}}) = \log(B_{\text{theo}}/\rho_{\text{emp}}), \quad (211)$$

where $B_{\text{theo}} \triangleq (V(\emptyset) - \Delta \hat{C}_{i^*}) / (V(\emptyset) - |\mathcal{X}^*| \Delta \hat{C}_{i^*})$ is the theoretical approximation bound in (25). We then plot $s(B_{\text{theo}})$ against B_{theo} for each instance. For all instances, we observe $s(B_{\text{theo}}) \geq 0$, confirming that the theoretical bound in (25) provides a valid upper bound on the empirical approximation ratio. Moreover, the empirical ratio is consistently much tighter than the theoretical one. Fitting $s(B_{\text{theo}})$ against $\ln B_{\text{theo}}$ over the computed instances yields the logarithmic relationship

$$s(B_{\text{theo}}) \approx -0.0358 + 0.9249 \ln B_{\text{theo}}, \quad (212)$$

which implies the following empirical approximation bound

$$V(\mathcal{X}^{\text{DG}})/V(\mathcal{X}^*) \approx e^{0.0358} \left(\frac{V(\emptyset) - \Delta \hat{C}_{i^*}}{V(\emptyset) - |\mathcal{X}^*| \Delta \hat{C}_{i^*}} \right)^{0.0751}. \quad (213)$$

This indicates that Algorithm 2 closely approximates the optimal solution of the NP-hard SCM problem in Stage II.

O. Multiple-SP Market Formulation and Algorithms

This appendix describes the multiple-SP extension used in Section VI-F. The extension considers a market with multiple ESPs, multiple NSPs, and multiple users. We experimentally extend the proposed method to this multiple-SP setting. The strategic interactions follow the same two-stage structure. In Stage I, the SPs set their resource prices. In Stage II, users decide whether to compute locally or offload through one ESP-NSP pair.

1) *Model*: Let $\mathcal{E} = \{1, 2, 3\}$, $\mathcal{N} = \{1, 2, 3\}$ denote the sets of ESPs and NSPs, respectively. Each ESP $e \in \mathcal{E}$ has computation capacity F_e , unit operational cost c_e^E , and unit price p_e^E . Each NSP $n \in \mathcal{N}$ has bandwidth capacity B_n , unit operational cost c_n^N , and unit price p_n^N . Each user can either compute locally or select one ESP–NSP pair (e, n) for offloading.

For user i , we define the following notation for brevity:

$$a_i \triangleq \alpha_i w_i, \quad \theta_{i,n} \triangleq \frac{d_i(\alpha_i + \beta_i \rho_i \varpi_i)}{\sigma_{i,n}}, \quad (214)$$

where $\sigma_{i,n}$ is the spectral efficiency from user i to NSP n . Let C_i^l denote the local computation cost. If user i offloads through pair (e, n) , purchases computation resource f_i , and purchases bandwidth b_i , its offloading cost is

$$C_{i,e,n}^o(f_i, b_i) = \frac{a_i}{f_i} + \frac{\theta_{i,n}}{b_i} + p_e^E f_i + p_n^N b_i. \quad (215)$$

Let $x_{i,e,n} \in \{0, 1\}$ indicate whether user i offloads through pair (e, n) . Each user can select at most one ESP–NSP pair:

$$\sum_{e \in \mathcal{E}} \sum_{n \in \mathcal{N}} x_{i,e,n} \leq 1. \quad (216)$$

When $\sum_{e,n} x_{i,e,n} = 0$, user i computes locally.

2) *Stage-II User GNE*: Given provider prices $\mathbf{p} = (p_1^E, p_2^E, p_3^E, p_1^N, p_2^N, p_3^N)$, the Stage-II user GNE is formulated as the solution to the following social cost minimization problem:

$$\min_{\mathbf{x}, \mathbf{f}, \mathbf{b}} \sum_i \left(1 - \sum_{e,n} x_{i,e,n} \right) C_i^l + \sum_i \sum_{e,n} x_{i,e,n} C_{i,e,n}^o(f_i, b_i) \quad (217)$$

$$\text{s.t.} \quad \sum_i \sum_{n \in \mathcal{N}} x_{i,e,n} f_i \leq F_e, \quad \forall e \in \mathcal{E}, \quad (218)$$

$$\sum_i \sum_{e \in \mathcal{E}} x_{i,e,n} b_i \leq B_n, \quad \forall n \in \mathcal{N}, \quad (219)$$

$$C_{i,e,n}^o(f_i, b_i) \leq C_i^l, \quad \forall i, e, n \text{ with } x_{i,e,n} = 1. \quad (220)$$

This problem is complex to solve because it jointly optimizes users' binary offloading decisions and continuous resource allocations. Therefore, we solve this problem in two steps. First, we compute the user best response under a fixed offloading assignment. Second, we adopt a greedy algorithm to approximate the optimal offloading assignment.

a) *User Best Response under Fixed Offloading Assignment*: Consider a fixed offloading assignment A , where $A_i = (e_i, n_i)$ if user i offloads and $A_i = \emptyset$ otherwise. We define

$$\mathcal{I}_e(A) = \{i : A_i = (e, n) \text{ for some } n \in \mathcal{N}\}, \quad (221)$$

$$\mathcal{I}_n(A) = \{i : A_i = (e, n) \text{ for some } e \in \mathcal{E}\}. \quad (222)$$

For each ESP and NSP, we define the congestion threshold prices

$$\bar{p}_e^E(A) = \left(\frac{\sum_{i \in \mathcal{I}_e(A)} \sqrt{a_i}}{F_e} \right)^2, \quad \bar{p}_n^N(A) = \left(\frac{\sum_{i \in \mathcal{I}_n(A)} \sqrt{\theta_{i,n}}}{B_n} \right)^2 \quad (223)$$

Algorithm 5 Multiple-SP Stage-II Greedy Algorithm

Input: Prices $\mathbf{p}$, capacities $\{F_e\}, \{B_n\}$

Output: Offloading assignment, resources, shadow prices, social cost

- 1: Initialize an empty offloading assignment A and the local-user set.
- 2: **repeat**
- 3: Compute the user best response using (224)–(225).
- 4: Roll back the last admitted user if the realized social cost does not decrease.
- 5: Remove assigned users violating $m_i(A) \geq 0$.
- 6: Compute $h_{i,e,n}(A)$ for every local user and every pair (e, n) .
- 7: **if** the minimum score is negative **then**
- 8: Temporarily assign the corresponding user to the corresponding pair.
- 9: **else**
- 10: Stop.
- 11: **end if**
- 12: **until** the candidate user set is empty or the iteration limit is reached
- 13: Recompute the final user best response and remove infeasible users.
- 14: **return** Final offloading assignment, resources, shadow prices, social cost.

The corresponding effective prices are

$$\tilde{p}_e^E(A) = \max\{p_e^E, \bar{p}_e^E(A)\}, \quad \tilde{p}_n^N(A) = \max\{p_n^N, \bar{p}_n^N(A)\}. \quad (224)$$

For any offloading user assigned to pair (e, n) , the best response of user i is

$$f_i^*(A) = \sqrt{\frac{a_i}{\tilde{p}_e^E(A)}}, \quad b_i^*(A) = \sqrt{\frac{\theta_{i,n}}{\tilde{p}_n^N(A)}}. \quad (225)$$

The associated capacity shadow prices are

$$\lambda_e(A) = \tilde{p}_e^E(A) - p_e^E, \quad \nu_n(A) = \tilde{p}_n^N(A) - p_n^N. \quad (226)$$

The individual-rationality margin of an offloading user is

$$m_i(A) = C_i^l - C_{i,e_i,n_i}^o(f_i^*(A), b_i^*(A)). \quad (227)$$

An assigned user is feasible only when $m_i(A) \geq 0$.

b) *Greedy Algorithm for Offloading Assignment*: We use a greedy algorithm for offloading assignment, which is summarized in Algorithm 5. Starting from the empty offloading assignment, we admit users one by one, similar to Algorithm 2. Given the current assignment A , each local user j evaluates every ESP–NSP pair using the social-cost impact similar to (23):

$$h_{j,e,n}(A) = \min_{0 < f \leq F_e} \left\{ \frac{a_j}{f} + (p_e^E + \lambda_e(A))f \right\} + \min_{0 < b \leq B_n} \left\{ \frac{\theta_{j,n}}{b} + (p_n^N + \nu_n(A))b \right\} - C_j^l. \quad (228)$$

Each one-dimensional minimization in (228) admits a closed-form expression. The algorithm selects the user-SP triple with the smallest social-cost reduction:

$$(j^*, e^*, n^*) \in \arg \min_{j, e, n} h_{j, e, n}(A). \quad (229)$$

If the minimum score is non-negative, the algorithm terminates. Otherwise, user j^* is temporarily assigned to pair (e^*, n^*) . The fixed-assignment user best response is then recomputed, and a rollback check is applied. The temporary admission is removed if it increases the real social cost or if any offloading user violates individual rationality. Users that fail the rollback check are excluded from the candidate users.

3) *SP Revenues*: After the Stage-II user offloading decisions and resource allocations are computed, each SP's demand and revenue are evaluated from the assigned users:

$$U_e^E(\mathbf{p}) = (p_e^E - c_e^E) \sum_{i \in \mathcal{I}_e(A^*(\mathbf{p}))} f_i^*, \quad e \in \mathcal{E}, \quad (230)$$

$$U_n^N(\mathbf{p}) = (p_n^N - c_n^N) \sum_{i \in \mathcal{I}_n(A^*(\mathbf{p}))} b_i^*, \quad n \in \mathcal{N}. \quad (231)$$

Here $A^*(\mathbf{p})$ denotes the offloading assignment returned by the Stage-II greedy algorithm under prices $\mathbf{p}$.

4) *Stage-I SP Pricing Game*: The Stage-I SP pricing game contains six players, i.e., three ESPs and three NSPs. Since exhaustive search in the six-dimensional pricing space is impractical, we use a boundary-price best-response heuristic inspired by the proposed method in Section V-B.

a) *Candidate Assignment Set*: At a current price profile $\mathbf{p}$, we first compute the Stage-II assignment $A^*(\mathbf{p})$. We then construct a set of candidate assignments. Current offloading users are sorted by increasing individual-rationality margin m_i , so that users with smaller margins are treated as more likely to leave. Current local users are sorted by their best social-cost-impact score $\min_{e, n} h_{i, e, n}$, so that users with smaller scores are treated as more likely to enter.

For a small integer Q , candidate assignments are generated by removing the first r fragile offloading users and adding the first s promising local users, where $0 \leq r \leq Q$, $0 \leq s \leq Q$. Each newly added local user is assigned to its currently best ESP-NSP pair. Duplicate and empty assignments are discarded. The number of candidate offloading user sets is at most $(Q + 1)^2$.

5) *Algorithm Summary*: We summarize the Stage-II algorithm in Algorithm 5 and the Stage-I algorithm in Algorithm 6.

a) *Boundary price and restricted best response*: For a SP $k \in \mathcal{E} \cup \mathcal{N}$, all other SPs' prices are fixed. For each candidate assignment A , we estimate the right boundary of the feasible price interval for SP k . This boundary is the first price where one user served by SP k becomes indifferent between offloading and local computation:

$$\min_{i \in \mathcal{I}_k(A)} m_i(A; p_k, p_{-k}) = 0, \quad (232)$$

where $\mathcal{I}_k(A)$ denotes the set of users served by SP k in assignment A . This boundary is computed by bracketing and bisection. If the candidate assignment is already infeasible at the SP's unit cost, it is skipped. If no such boundary price is

Algorithm 6 Multiple-SP Boundary-Price Iterative Pricing Algorithm

Input: Initial prices $\mathbf{p}^{(0)}$, candidate-assignment-set size Q , tolerance ε , maximum number of iterations $T_{\max}$.

Output: Final prices, offloading assignment, revenues, resource utilization

```

1: for  $t = 0, 1, \dots, T_{\max} - 1$  do
2:   Run Algorithm 5 at  $\mathbf{p}^{(t)}$ .
3:   for all SP  $k \in \mathcal{E} \cup \mathcal{N}$  do
4:     Construct the candidate assignment set.
5:     Compute boundary prices using (232).
6:     Evaluate each boundary price by rerunning Stage II.
7:     Obtain  $\hat{p}_k^{\text{BR}}$  and  $\hat{G}_k$ .
8:   end for
9:   Set  $\hat{\delta}(\mathbf{p}^{(t)}) = \max_k \hat{G}_k$ .
10:  if  $\hat{\delta}(\mathbf{p}^{(t)}) \leq \varepsilon$  then
11:    Stop.
12:  end if
13:  Update only the SP  $k^* \in \arg \max_k \hat{G}_k$ .
14: end for
15: Run Stage II at the final price profile.
16: return Final prices, offloading assignment, revenues,
    resource utilization.
```

found, the maximum reachable price is used as the candidate boundary.

For each SP k , the evaluated price set contains the current price and all valid boundary prices from the candidate assignment set. Each boundary price is evaluated by rerunning the Stage-II greedy algorithm under the corresponding unilateral deviation. The SP's estimated best response is

$$\hat{p}_k^{\text{BR}} \in \arg \max_{q_k \in \hat{\mathcal{C}}_k(\mathbf{p})} U_k(q_k, p_{-k}), \quad (233)$$

where $\hat{\mathcal{C}}_k(\mathbf{p})$ denotes the candidate price set of SP k . The restricted best-response gain is

$$\hat{G}_k(\mathbf{p}) = U_k(\hat{p}_k^{\text{BR}}, p_{-k}) - U_k(p_k, p_{-k}). \quad (234)$$

b) *Iterative Pricing Algorithm*: The restricted NE gap is defined as

$$\hat{\delta}(\mathbf{p}) = \max_{k \in \mathcal{E} \cup \mathcal{N}} \hat{G}_k(\mathbf{p}). \quad (235)$$

At each Stage-I iteration, all six SPs compute their estimated best-response gains. If

$$\hat{\delta}(\mathbf{p}) \leq \varepsilon, \quad (236)$$

the pricing algorithm terminates. Otherwise, the algorithm updates the single SP with the largest restricted gain:

$$k^* \in \arg \max_{k \in \mathcal{E} \cup \mathcal{N}} \hat{G}_k(\mathbf{p}), \quad p_{k^*} \leftarrow \hat{p}_{k^*}^{\text{BR}}. \quad (237)$$

All other SP prices remain unchanged. After the final Stage-I iteration, Stage II is solved one more time and the final assignment, SP revenues, restricted NE gap, and resource utilization are presented.

6) *Implementation Settings and Remarks:* We run the experiment with 50 users. The parameter settings follow the description in Section VI-A and Section VI-F. The presented runs use $Q = 2$ and a maximum price of 6.0 for both resource types. The method is heuristic. It preserves the intuition of the main paper, and its outcome is an approximation of the multi-SP Stackelberg equilibrium.

P. Sensitivity Analysis on Resource Capacities

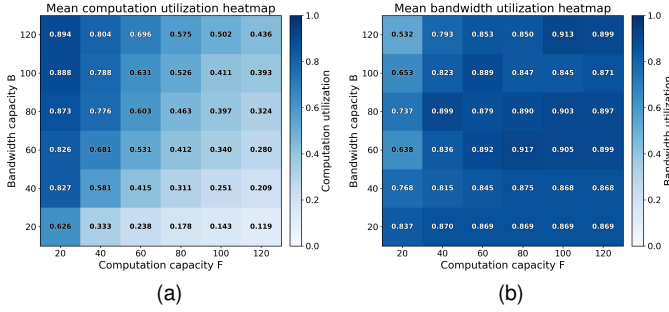


Fig. 9. Two-dimensional heatmaps of resource utilization under different computation capacity F (GHz) and bandwidth capacity B (MHz). (a) Average computation resource utilization. (b) Average bandwidth utilization.

We next examine the impact of computation capacity F and bandwidth capacity B on the system performance. Figs. 9–11 show the resource utilization, user-side outcomes, and SP-side outcomes under different (F, B) configurations. Overall, the results show that the system performance is shaped by the relative scarcity of the two complementary resources. Expanding only one resource often leads to rapidly diminishing returns, whereas jointly increasing both resources can more effectively improve the market outcome.

Figs. 9a and 9b show the utilization of computation and bandwidth resources under different capacity configurations. As expected, the utilization of a resource decreases when its capacity becomes larger. For example, when $(F, B) = (20, 120)$, the computation utilization reaches 0.894, while the bandwidth utilization is only 0.532. In contrast, when $(F, B) = (120, 20)$, the bandwidth utilization remains high at 0.869, whereas the computation utilization drops to 0.119. More importantly, even when the two resources are increased simultaneously along the diagonal, bandwidth utilization is still much higher than computation utilization in most configurations. Specifically, when (F, B) is $(40, 40)$, $(80, 80)$, and $(120, 120)$, the computation and bandwidth utilizations are $(0.581, 0.815)$, $(0.463, 0.890)$, and $(0.436, 0.899)$, respectively. On average, bandwidth utilization is 35.5% higher than computation utilization along the diagonal. This indicates that bandwidth is consumed more intensively in the considered system setting and is more likely to become the dominant bottleneck.

Fig. 10 shows the user-side outcomes under different capacity configurations. Figs. 10a and 10b show that both the offloading ratio and the number of offloading users increase as F and B become larger. However, this increase is neither symmetric nor unlimited.

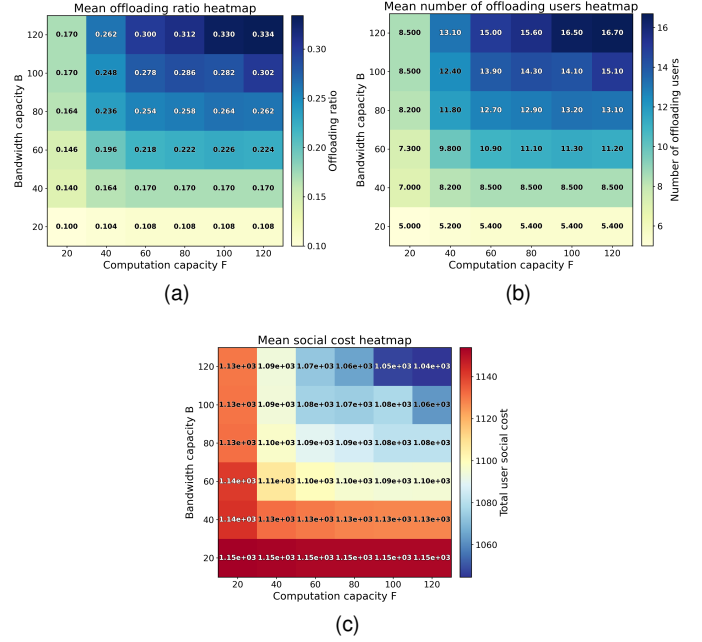


Fig. 10. Two-dimensional heatmaps of user offloading decisions and total system cost under different computation capacity F (GHz) and bandwidth capacity B (MHz). (a) Average offloading ratio. (b) Average number of offloading users. (c) Average total user social cost.

When the system operates under a low-resource regime, increasing only one type of resource leads to significantly different effects. When bandwidth is tightly limited at $B = 20$, increasing computation capacity from $F = 20$ to $F = 120$ barely expands the offloading scale: the average offloading ratio only increases from 0.100 to 0.108, and the average number of offloading users increases from 5.0 to 5.4, corresponding to only an 8% improvement. In contrast, when computation capacity is fixed at a low level $F = 20$, increasing bandwidth capacity from $B = 20$ to $B = 120$ significantly improves user participation. The average offloading ratio increases from 0.100 to 0.170, and the average number of offloading users increases from 5.0 to 8.5, both representing a 70% increase.

The user social cost in Fig. 10c further confirms this asymmetry. When $B = 20$, increasing F from 20 to 120 keeps the social cost almost unchanged around 1.15×10^3 . By contrast, when $F = 20$, increasing B to 120 reduces the social cost from about 1.15×10^3 to about 1.13×10^3 , corresponding to a reduction of approximately 1.7%. These results indicate that, in the low-resource regime, increasing bandwidth capacity brings a much larger improvement than increasing computation capacity. Therefore, bandwidth is the more critical limiting resource in the considered setting.

Moreover, the benefit of increasing computation capacity becomes significant only when bandwidth is no longer the dominant bottleneck. For example, when $B = 120$, increasing F from 20 to 120 raises the offloading ratio from 0.170 to 0.334, an increase of 96.5%, and the average number of offloading users increases from 8.5 to 16.7. Meanwhile, the user social cost decreases from about 1.13×10^3 to about 1.04×10^3 . This shows that expanding computation capacity

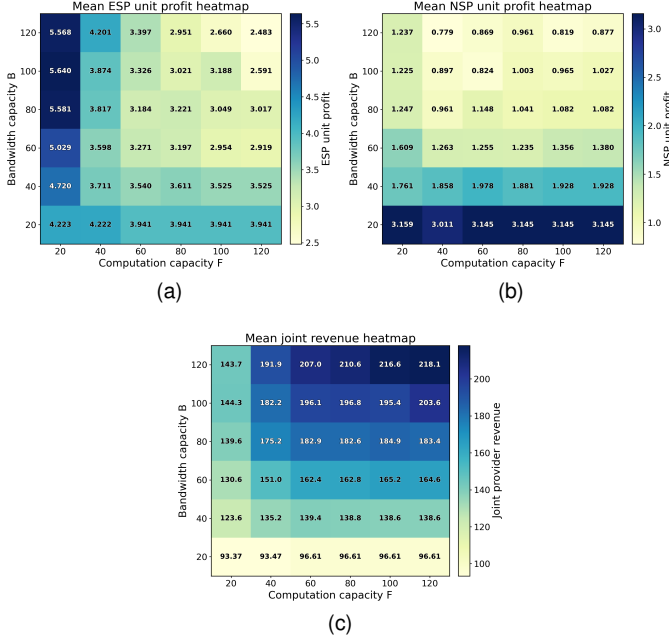


Fig. 11. Service-provider-side economic metrics under different computation capacity F and bandwidth capacity B . (a) Average ESP unit profit. (b) Average NSP unit profit. (c) Average joint revenue of the ESP and NSP.

alone is ineffective if bandwidth is not expanded accordingly. Only coordinated resource expansion can substantially improve user-side offloading participation.

The results also reveal a diminishing-return effect. Taking the average number of offloading users as an example, when $B = 120$, increasing F from 20 to 40 raises the number of offloading users from 8.50 to 13.10, an increment of 4.60. However, when F is already large, increasing it from 100 to 120 only increases the number of offloading users from 16.50 to 16.70, an increment of only 0.20. A similar pattern appears when only bandwidth is expanded. With $F = 120$, increasing B from 20 to 40 increases the number of offloading users by 3.10, whereas increasing B from 100 to 120 increases it by only 1.60. Therefore, as resource capacities grow, the marginal improvement in offloading participation gradually decreases.

Fig. 11 shows the SP-side economic outcomes under different resource capacity configurations. Figs. 11a and 11b show a clear scarcity-premium effect. The relatively scarcer resource obtains stronger pricing power and higher unit profit. Specifically, the ESP unit profit is maximized in the region where computation is scarce and bandwidth is abundant. For example, when $(F, B) = (20, 100)$, the ESP unit profit reaches 5.640. When F increases to 120, the ESP unit profit drops to 2.591, a reduction of 54%. Conversely, the NSP unit profit is high when bandwidth is scarce. When $B = 20$, the NSP unit profit remains in the high range of 3.01–3.16 across different values of F . However, when bandwidth capacity increases to $B = 120$, the NSP unit profit drops to 0.877 at $F = 120$, a reduction of 72%.

Unlike unit profit, the joint revenue in Fig. 11c improves substantially only when the two resources are expanded in a coordinated manner. When bandwidth is fixed at $B = 20$,

increasing computation capacity from $F = 20$ to $F = 120$ only raises joint revenue from 93.37 to 96.61, an increase of merely 3.5%. By contrast, when computation capacity is fixed at $F = 20$, increasing bandwidth capacity from $B = 20$ to $B = 120$ raises joint revenue from 93.37 to 143.7, an increase of 54%. This again confirms that bandwidth is the more critical bottleneck in the considered setting. When the bandwidth bottleneck is relaxed, increasing computation capacity becomes much more effective. Under $B = 120$, increasing F from 20 to 120 raises joint revenue from 143.7 to 218.1, an increase of nearly 52%.

The joint revenue also exhibits diminishing returns. Under abundant bandwidth ($B = 120$), increasing F from 20 to 40 increases joint revenue by 48.2, from 143.7 to 191.9. However, increasing F from 100 to 120 only raises joint revenue from 216.6 to 218.1, an increment of 1.5. A similar trend appears when bandwidth is expanded alone. With $F = 120$, increasing B from 20 to 40 increases joint revenue by 41.99, whereas increasing B from 100 to 120 only increases joint revenue by 14.5. These observations show that the marginal revenue gain from resource expansion shrinks rapidly as the system becomes sufficiently provisioned.

In summary, the two-dimensional capacity-sensitivity results reveal three main findings. First, improving system performance requires coordinated expansion of both computation and bandwidth resources, rather than isolated expansion of a single resource. Second, under the considered parameter setting, bandwidth is more likely to become the dominant bottleneck that limits market expansion. Third, both user-side improvement and SP-side revenue growth exhibit clear diminishing returns as resource capacities increase. Therefore, MEC resource provisioning should consider not only the total amount of computation and bandwidth resources, but also the balance between these two complementary resources and the marginal benefit of further capacity expansion.

Q. Sensitivity Analysis on the Maximum Number Q of Considered Marginal Users in Algorithm 3

We further evaluate the sensitivity of the proposed method to the maximum number Q of considered marginal users in constructing the candidate offloading user sets $\mathcal{N}_Q(\mathbf{p})$. A larger Q allows the algorithm to consider more possible leaving and joining users, thereby enlarging the candidate sets. However, it also increases the number of Stage-II evaluations required in each best-response computation. Therefore, this experiment examines the tradeoff between equilibrium quality and computational cost as Q varies.

Fig. 12 shows the resulting equilibrium-quality and computational-cost metrics. As shown in Figs. 12a and 12b, increasing Q from 1 to a moderate value significantly improves the quality of the returned pricing outcome. When $Q = 1$, the remaining best-response gains are large, and the oracle NE gap is 9.77. When $Q = 2$ or $Q = 3$, the NE gap decreases to 3.09, and the NSP-side best-response gain becomes zero. When $Q = 4$, both ESP and NSP best-response gains vanish, and the oracle NE gap reaches zero. This confirms that enlarging the candidate offloading user sets helps the boundary-price

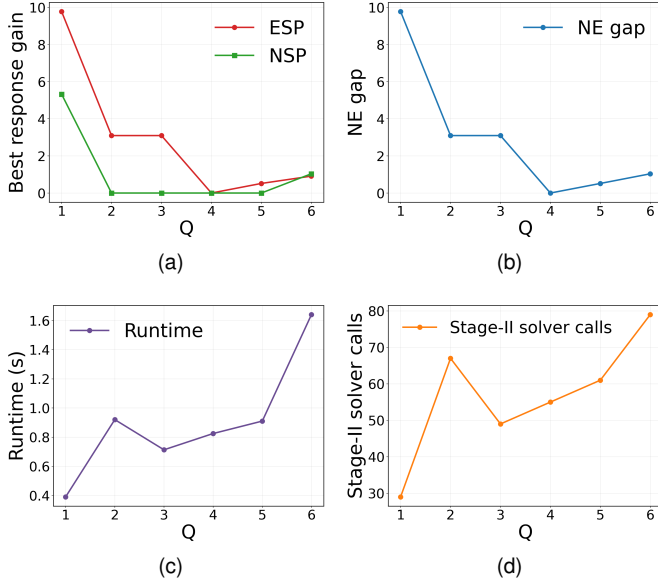


Fig. 12. Sensitivity to the maximum number Q of considered marginal users. (a) Best-response gains of the ESP and NSP under different Q . (b) Oracle NE gap under different Q . (c) Runtime under different Q . (d) Number of Stage-II solver calls under different Q .

search capture more relevant switching-price candidates and therefore improves the pricing equilibrium approximation.

However, the benefit of increasing Q is not monotone. When Q further increases to 5 and 6, the direct-grid oracle NE gap becomes 0.52 and 1.04, respectively. This might be because Q changes the candidate boundary-price set and leads the best-response dynamics to a different trajectory. Hence, a larger number of candidate offloading user sets does not necessarily guarantee a better final pricing outcome under the heuristic boundary-price dynamics. Instead, a moderate Q is sufficient to capture the most important local switching structures.

Figs. 12c and 12d show the corresponding computational cost. The runtime increases from 0.39 seconds at $Q = 1$ to 1.64 seconds at $Q = 6$, while the number of Stage-II solver calls increases from 29 to 79. This increase remains moderate in the tested range. Even when Q increases from 1 to 6, the runtime stays at or below 1.64 seconds and the number of Stage-II solver calls remains at or below 79. Overall, Fig. 12 demonstrates that Q controls a clear quality-cost tradeoff. Small Q may miss important boundary-price candidates, whereas very large Q increases computation and may not further improve the final outcome. In this experiment, moderate values of Q , especially around $Q = 4$, achieve strong equilibrium quality with limited computational overhead.